

Migration with Costly Information

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Abstract

Information is critical for migration decisions. Yet, depending on where individuals reside and who they interact with, they may face different costs of accessing information about employment opportunities. How does this imperfect and heterogeneous information structure affect the spatial allocation of economic activity and welfare? To address this question, I develop a quantitative dynamic model of migration with costly information acquisition and local information sharing. Rationally inattentive agents optimally acquire more information about nearby locations and learn about payoffs in other locations from the migrants around them. I apply this model to internal migration in Brazil and estimate it using migration flows between regions. To illustrate its quantitative implications, I evaluate the counterfactual effects of the roll-out of broadband internet in Brazil. By allowing workers to make better mobility choices, expanding internet access increases average welfare by 1.6%, reduces migration flows by 1.2% and reduces the cross-sectional dispersion in earnings by 4%.

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1 Introduction

Migration decisions shape nearly every aspect of workers' lives, from jobs and housing to schools and leisure opportunities. Evaluating these opportunities requires information about many features of potential destinations, yet collecting and processing such information involves costs. Despite the central role of information in these choices, economic research provides little quantitative evidence on the nature, magnitude, and consequences of these costs. Are workers forgoing attractive opportunities because they lack information about them? Does unequal access to information contribute to inequality in incomes? How have recent improvements in information technologies altered workers' response to local shocks, and what scope remains for policies to improve the informational environment in which migration decisions are made?

In this paper, I address these questions by treating what migrants know about opportunities elsewhere not as an exogenous feature of the environment, but as an equilibrium outcome that co-evolves with migration. Information availability depends on where people live and who they interact with: some locations provide better infrastructure for collecting information, and migrants share what they learn with those around them. Because migration reshapes local networks and relocates workers across space, it also transforms the structure of information, creating feedback between migration patterns and information flows.

The paper makes four main contributions. First, it develops a dynamic theory of spatial equilibrium with migration and costly information acquisition, in which the equilibrium structure of workers' information is determined jointly with migration patterns. This framework endogenizes the distribution of information and highlights how local information networks shape spatial adjustment. Second, I estimate the model using detailed data on internal migration in Brazil to quantify the magnitude and spatial heterogeneity of information costs. Third, I show that the model replicates key empirical features of migration patterns—such as the stronger responsiveness of nearby flows and the delayed reaction to local shocks—that standard models with perfect information cannot explain. Finally, I use the estimated model to evaluate the welfare effects of Brazil's broadband expansion, showing how improved access to information raises average welfare and reshapes regional inequality and migration flows.

In the first part of the paper, I develop a dynamic model of migration with costly information acquisition and local information sharing. Each period, unobserved location-specific productivity shocks change local earnings. Agents are rationally inattentive: they can acquire information about these shocks only by incurring a location-specific cost. They rely on these beliefs to make migration decisions, facing bilateral costs of moving between any two locations. Within each location, agents differ in both their beliefs about payoffs and their preferences for destinations. While migration based purely on preferences reflects utility maximization, migration under incomplete information leads to systematic errors in location choices.

The model admits a tractable steady-state solution and yields three main results. First, bilateral migration shares take a closed-form multinomial logit expression, consistent with standard models of migration based on preference heterogeneity. When information costs vanish, the model collapses to that benchmark. Second, agents optimally choose to be better informed about locations offering higher expected payoffs net of migration costs, creating a bias toward nearby destinations, which typically involve

lower migration costs. This bias effectively generates endogenous bilateral frictions that increase with distance, as in gravity models of migration.¹ Third, under a distributional assumption that permits closed-form aggregation, the model disentangles the effects of idiosyncratic preferences from those of imperfect information. While both dampen the responsiveness of migration flows to earnings differences, information frictions specifically weaken responses to unobserved shocks, making it possible to identify them separately from preference heterogeneity.

Despite its analytical tractability, solving the model remains computationally demanding because migration decisions depend on many state variables—productivity shocks, migration flows, and the distribution of local beliefs. To make the problem tractable, I show that in the stochastic steady state, beliefs can be summarized as a function of migration flows, which substantially reduces the dimensionality of the state space. I then solve the resulting dynamic programming problem using standard methods from approximate dynamic programming, combining functional approximations of the value function with a sampling-based solution algorithm.² I assess the accuracy of this approach by comparing it to exact or nearly exact solutions in small-scale economies and discrete-state environments.

In the second part of the paper, I apply the model to internal migration in Brazil to assess the quantitative importance of information frictions. Brazil provides an ideal setting, with large regional disparities and a staggered diffusion of broadband internet. The closed-form expression for migration probabilities allows for a transparent estimation strategy. I estimate the model using administrative matched employer–employee data covering virtually the universe of formal-sector workers, tracking their location and earnings annually over fifteen years. To account for changes in broadband availability and local shocks, I estimate the model separately for 2000–2007 and 2008–2014, treating each as a steady state. Exploiting the gravity structure of migration flows implied by the model, I recover information costs, preference heterogeneity, and migration costs separately. The key idea is that migration flows respond differently to payoff variation depending on whether these payoffs are observed or must be learned. Agents are assumed to observe slow-moving regional population levels but not the partly unpredictable location-specific productivity shocks; comparing migration elasticities to wages and to inferred productivity shocks therefore isolates the contribution of information frictions.

The estimated information costs average about 3% of annual earnings. They are lower in regions with greater internet access. Instrumenting for internet connectivity with proximity to backbone internet cables, I find that a 1% increase in the share of residents with internet access reduces local information costs by 0.83%. The annualized average bilateral migration cost is also close to 3% of annual earnings.³ Hence, collecting information about where to move appears nearly as costly as moving itself. Ignoring information frictions would lead to overestimating migration costs by roughly 40%, because poor information amplifies apparent immobility.⁴ In the presence of information frictions, even moderate migration

¹See Monte et al. (2018), Morten and Oliveira (2024), Tombe and Zhu (2019), Fan (2019), Caliendo et al. (2019).

²See Powell (2011) for an overview of approximate dynamic programming techniques.

³I estimate the average migration cost to be about 75% of annual earnings. However, the median worker migrates only once in a lifetime. With an annual discount factor of 0.96, paying 75% of earnings once is equivalent to paying about 3% of earnings every year.

⁴The idea that part of the large estimated migration costs could be attributed to information frictions can be traced back at least to Sjaastad (1962), who noted that “one is strongly tempted to appeal to market imperfections such as the lack of information to explain the apparently high distance cost of migration.”

costs can rationalize the low observed migration flows: when migration costs to a region are large, agents acquire little information about that region, further reducing their likelihood of moving there.

In the third part of the paper, I show that the estimated model replicates two salient features of observed migration patterns that models with complete information cannot match. First, allowing for flexible bilateral migration costs, I find empirically that migration elasticities are larger between geographically proximate regions, between regions with stronger past migration links, and from origins with greater internet access. The model generates the same patterns, with magnitudes comparable to those in the data. In the simulated equilibrium, agents' beliefs are more tightly correlated with true payoffs in regions with which they interact more frequently, making migration flows more responsive along those links. Second, I analyze local labor demand shocks and find that migration responses to a positive regional shock are slower from more distant origins and from those with lower internet access. The model reproduces these delayed responses through the gradual updating of beliefs transmitted via local information networks. Together, these two validation exercises demonstrate that the model captures the observed heterogeneity and sluggishness of migration responses arising from the spatial diffusion of information rather than from ad hoc mobility frictions.

The final part of the paper quantifies the welfare implications of information frictions through two counterfactual exercises. First, eliminating information frictions raises average welfare by 5.5 percent and reduces wage dispersion by 15 percent, reflecting a more efficient spatial allocation of workers. In the steady state with complete information, gross migration flows become more concentrated toward high-payoff regions yet 4.1 percent lower overall, consistent with more directed but less frequent migration. Second, I evaluate the impact of Brazil's broadband expansion during the early 2010s.⁵ I compare a steady-state economy with no broadband to one in which information costs fall according to the observed increase in internet access between 2008 and 2014. Average welfare rises by 1.6 percent. These gains arise through several channels: lower individual information-acquisition costs; more effective information sharing within local networks; improved sorting of workers toward high-productivity regions; and, indirectly, wage adjustments following the new spatial allocation of labor. The average welfare gains conceal substantial regional heterogeneity. Several remote regions experience small welfare losses despite improved connectivity. Their residents benefit less from broadband expansion—both because their internet access grew more slowly than elsewhere, and because faster information diffusion in other regions allows workers there to capture new opportunities more quickly. Moreover, as fewer well-informed workers move to relatively unattractive regions, local information sharing weakens, further limiting spatial learning and amplifying regional disparities.

This paper is related to several existing literatures. A growing body of empirical work emphasizes the role of information frictions in migration decisions. First, there is increasing evidence that migration choices can be shaped by the availability of information. This has been shown by exploiting variation in migrants' exposure to information about opportunities across locations, either through differential media access (Farré and Fasani, 2013; Wilson, 2021), or through randomized interventions that provide

⁵I abstract from the direct productivity effects of broadband access and focus on its informational role. Broadband still affects aggregate productivity indirectly, through the reallocation of labor across space.

information about average earnings elsewhere (Baseler, 2023; Bryan et al., 2014).⁶ Second, a recent literature documents that migrants often hold inaccurate beliefs about the returns to migration. In international settings, several papers directly elicit migrants’ expectations in surveys, lab experiments, and field studies, and find that, for many, these beliefs are misaligned with realized outcomes (McKenzie et al., 2013; Bah et al., 2023; Shrestha, 2019). In the context of internal migration in Brazil, Porcher, Morales and Fujiwara (2024) use a revealed-preference approach to infer the structure of migrants’ information consistent with a model that allows for unobserved migration costs. They find that the information needed to rationalize observed flows is concentrated on neighboring regions and large cities, with limited knowledge of regions beyond several hundred kilometers. My paper contributes to this literature by developing a general-equilibrium theory of optimal information acquisition that endogenizes the spatial structure of information—also concentrated on nearby regions and large cities—and quantifies the welfare implications of these frictions.

This paper also contributes to a rapidly growing empirical literature on information transmission through social networks. Classic work (e.g., Granovetter (1973)) showed that weak and strong ties help individuals learn about opportunities from others. More recent studies document that workers obtain job- or migration-relevant information from coworkers (Dustmann et al., 2015; Glitz and Vejlin, 2021; Saygin et al., 2021; Caldwell and Harmon, 2019), family members (Kramarz and Skans, 2014), and neighbors (Bayer et al., 2008; Schmutte, 2015). More broadly, variation in social network structure is strongly predictive of migration (Blumenstock et al., 2023). This paper provides suggestive evidence that workers also rely on their networks to learn about migration opportunities across regions.

This paper is related to the emerging theoretical literature on rational inattention in discrete choice settings. Following the seminal contribution by Sims (2003), Matějka and McKay (2015) showed that static rational inattention problems lead to a multinomial-logit decision rule.⁷ Steiner et al. (2017) extended this property to single-agent dynamic problems, opening the way to incorporating rational inattention into richer dynamic environments. I combine these insights with properties of social learning in networks derived by Molavi et al. (2018) to show that, in steady state with a continuum of agents, heterogeneous preferences, and endogenous payoffs, a dynamic logit structure survives. Empirical applications that embed rational inattention into structural models remain scarce. Exceptions include work in industrial organization such as Brown and Jeon (2024) and Joo (2023), and in international trade Dasgupta and Mondria (2018), which provides a microfoundation for Eaton and Kortum (2002). In the context of migration, Bertoli et al. (2020) document heterogeneous elasticities consistent with rational inattention. I provide the first estimation of a dynamic rational-inattention model with preference heterogeneity.

My paper also contributes to the quantitative economic geography literature. A large body of research

⁶In the context of seasonal migration in Bangladesh, Bryan et al. (2014) find that providing information about average wages and job availability across four broad regions did not significantly increase migration. By contrast, Baseler (2023) shows that revealing otherwise hidden destination earnings can substantially raise migration rates. Consistent with the interpretation that some information is already known at broad geographic levels, Porcher, Morales and Fujiwara (2024) cannot reject that internal migrants in Brazil are aware of average wages at the regional level but appear to lack information about labor market outcomes at finer geographic scales.

⁷Other important contributions include Caplin and Dean (2015); Caplin et al. (2022), and Fosgerau et al. (2020) establish a general equivalence between additive random-utility models and rational inattention with generalized entropy. For an overview of this literature, see the recent survey by Wiederholt (2023).

documents that local labor markets adjust slowly to shocks, in part because migration responses are limited.⁸ Analyzing U.S. internal migration, Kennan and Walker (2011) show that large migration costs are necessary to rationalize the coexistence of sizable spatial income disparities, a measurable elasticity of migration with respect to income, and overall limited population movements.⁹ Such large migration costs have significant implications for aggregate productivity and welfare.¹⁰ My analysis highlights how information frictions affect both migration elasticities and measured migration costs, suggesting that as much as 40% of the migration costs estimated under full-information assumptions may actually reflect informational barriers. Recent research further emphasizes how spatial linkages through trade and labor mobility mediate regional adjustment to shocks (Monte et al., 2018; Tombe and Zhu, 2019; Caliendo et al., 2019; Albert and Monras, 2022). By incorporating information frictions into a dynamic spatial-equilibrium framework, this paper shows how the endogenous structure of information interacts with the spatial allocation of economic activity, adding a new adjustment channel with important welfare implications.

Section 2 introduces the model. Section 3.1 describes the data used for estimation, and Section 3.2 presents the estimation procedure and main results. Section 4 compares the model’s predictions to the data, and Section 5 reports the counterfactual exercises.

2 Model

2.1 Overview

I consider an infinite-horizon economy with a finite set of locations $\mathcal{J} = \{1, \dots, J\}$. For any set X , let ΔX denote the set of probability distributions over X , and for any sequence $(x_\tau)_\tau$, define $x^t = (x_1, \dots, x_t)$. Each location j has three time-invariant attributes: a baseline productivity A_j , an amenity level B_j , and a cost of acquiring information λ_j . In each period t , locations experience an exogenous stochastic productivity shock $\theta_{jt} \in \Theta$, capturing time-varying migration opportunities.¹¹ Let $\Gamma(\cdot | \theta^{t-1})$ denote the conditional distribution of the productivity vector $\theta_t = (\theta_{1t}, \dots, \theta_{Jt})$ given its past realizations θ^{t-1} . Geography is summarized by a matrix of fixed bilateral migration costs $\{\kappa_{jk}\}_{j,k=1,\dots,J}$ that agents must pay to move from location j to location k .

The total population in the economy, $\bar{L}$, is constant over time. In each period t , the distribution of population across destination locations k and origin locations j is denoted by $\{L_{jkt}\}_{j,k=1,\dots,J}$, where L_{jkt}

⁸Following early analyses of local labor-demand shocks (Bartik, 1991; Blanchard and Katz, 1992), modest migration responses have been observed after international trade shocks (Topalova, 2010; Kovak, 2013; Autor et al., 2013; Dix-Carneiro and Kovak, 2017; Pierce and Schott, 2020; Dix-Carneiro and Kovak, 2019), technological change (Bustos et al., 2016; Acemoglu and Restrepo, 2020), fluctuations in housing wealth (Mian and Sufi, 2014), differential exposure to business cycles (Yagan, 2019; Beraja et al., 2019), place-based policies (Busso et al., 2013), natural resource discoveries (Bartik et al., 2019), and natural disasters (Nakamura et al., 2021).

⁹Kennan and Walker (2011) estimate average state-to-state moving costs above \$300,000 (2010 dollars), while Diamond et al. (2019) find fixed costs of roughly \$40,000 for moves between San Francisco neighborhoods.

¹⁰See, among others, Redding (2016); Diamond (2016); Morten and Oliveira (2024); Bryan and Morten (2019); Caliendo et al. (2021, 2019) for quantitative assessments of the welfare effects of limited mobility.

¹¹As discussed in Footnote 8, such local shocks may stem from differences in exposure to international trade, technological change, government spending, business cycles, firm entry or exit, or natural resource discoveries. Section 4 examines shocks of this type in Brazil between 2000 and 2014—including dam construction, mining and oil booms, and expansions in tourism.

represents the number of individuals residing in location k at time t who lived in location j at time $t - 1$. The vector of all such bilateral population stocks belongs to the set $\mathcal{L} = \{\{L_{jkt}\}_{j,k} : \sum_{j,k} L_{jkt} = \bar{L}, L_{jkt} \geq 0, j, k = 1, \dots, J\}$. Each period, every agent i draws idiosyncratic taste shocks $\{\varepsilon_{ikt}\}_{k=1,\dots,J}$ for locations, which are independently and identically distributed across individuals, locations, and periods.

The remainder of this section proceeds as follows. I begin by outlining a simple production environment and specifying the assumptions on unobserved payoffs. Next, I formalize the worker's optimization problem and describe the laws of motion governing population and beliefs, which together determine the aggregate dynamics of the economy. The section concludes by characterizing the steady-state equilibrium.

2.2 Production

There is a single homogeneous good that is freely traded across locations and chosen as the numeraire. Production in each location follows a Cobb–Douglas technology using labor as the only input. Output in location k at time t is therefore given by

$$y_{kt} = \exp(A_k + \theta_{kt}) L_{kt}^{1-\alpha}, \quad \alpha \geq 0, \quad (1)$$

where θ_{kt} denotes the current productivity shock in location k , and L_{kt} is the labor input in that location, equal to the population since each agent supplies one unit of human capital.

The parameter α governs the extent of decreasing returns to labor and therefore introduces congestion: as more workers move into a location, its attractiveness falls.¹² A positive α implies that equilibrium earnings in each location depend on, and in turn influence, the population level determined by migration choices.

Profit maximization yields the first-order condition for the wage in location k at time t :

$$\log w_{kt} = A_k + \theta_{kt} - \alpha \log L_{kt} + \log(1 - \alpha). \quad (2)$$

Because firms earn positive profits, these are collected and redistributed to workers through a uniform negative tax rate on local wages, τ_t , so that earnings in location k at time t are $w_{kt}(1 + \tau_t)$.¹³

2.3 Assumptions on Unobserved Payoffs

The stochastic productivity vector θ_t is the only variable that agents do not observe directly. Assumption 1 specifies which variables are observed by agents and at what points in time, and makes explicit that the stochastic productivity vector is never observed by any agent.¹⁴

Assumption 1. *In every period t , each agent i :*

¹²Congestion could also arise from a fixed or imperfectly elastic supply of housing or land.

¹³The transfer rate is constant over time and equals $\tau = \alpha/(1-\alpha)$, implying that net-of-transfer earnings are $w_{kt}/(1-\alpha)$. To see this, note that profits in location k equal $\alpha \exp(A_k + \theta_{kt}) L_{kt}^{1-\alpha}$. The transfer rate τ_t solves $\sum_k \tau_t w_{kt} L_{kt} = \sum_k \alpha \exp(A_k + \theta_{kt}) L_{kt}^{1-\alpha}$, yielding $\tau = \alpha/(1-\alpha)$ after substituting wages from (2).

¹⁴One justification is that time-invariant characteristics such as amenities and baseline productivities are assumed to have been gradually learned until they became common knowledge. Although the population distribution varies over time, it is arguably easier for agents to observe population levels than to know the underlying productivity shocks in each location.

- (i) observes all time-invariant payoff-relevant location characteristics, $\{A_j, B_j, \lambda_j\}_{j=1,\dots,J}$;
- (ii) observes realizations of time-varying payoff-relevant variables up to and including period t , $\{L_{jk}^t\}_{j,k=1,\dots,J}$ and $\{\varepsilon_{ik}^t\}_{k=1,\dots,J}$, but not their future realizations;
- (iii) does not observe any past, current, or future realizations of the productivity vector $\{\theta_t\}_t$.

In this economy, agents never observe the realized productivity vector θ_t . In particular, they do not use the wage they receive in their current location k to update their beliefs about θ_{kt} . If productivity shocks are independent over time, learning θ_{kt} after moving provides no information relevant for future migration decisions. However, when the persistence parameter ρ is positive, observing θ_{kt} helps agents form more precise beliefs about future productivity θ_{kt+1} . In Appendix A.3, I extend the model to a setting in which agents can infer local productivity from observed wages. If no additional unobserved shock affects wages, workers in location k can perfectly recover θ_{kt} once they move there. In a simulation with a small number of locations, I show that, for a reasonable range of parameters, allowing workers to observe θ_{kt} has only modest effects on mobility decisions.¹⁵

I assume that the productivity vector follows a first-order Markov process.

Assumption 2. For each location j , the productivity shock θ_{jt} follows an $AR(1)$ process with persistence ρ and innovation variance σ_ξ^2 :

$$\theta_{jt} = \rho \theta_{jt-1} + \xi_{jt}, \quad \xi_{jt} \sim \mathcal{N}(0, \sigma_\xi^2), \quad (3)$$

where the innovations ξ_{jt} are independent across locations and over time.

The assumption that productivity shocks follow an $AR(1)$ process implies that the current shock depends on past realizations only through the most recent one. Consequently, agents form beliefs about the current productivity vector using their beliefs about last period's productivity. Let π_{it} denote the prior belief of agent i at the beginning of period t regarding the vector θ_t :

$$\pi_{it}(\theta) \equiv \Pr(\theta_t = \theta \mid i, t), \quad \forall \theta \in \mathbb{R}^J. \quad (4)$$

2.4 Problem of the Worker

2.4.1 Timeline of Actions

Each period t unfolds in four stages. First, the vectors of productivity shocks θ_t and idiosyncratic preferences ε_{it} are realized. Second, agents who resided in the same location l_{it-1} in period $t-1$ share information. This process aggregates their individual prior beliefs π_{it} into a common prior $\bar{\pi}_{l_{it-1}t}$ shared among all agents in that location. Third, each agent refines this shared prior by acquiring and processing additional information. This decision problem follows the rational inattention framework and results in

¹⁵This insensitivity arises because agents optimally adjust their information strategies. When they can form more precise priors about θ_{kt} , they optimally devote less attention to learning about it. As a result, costless observation of wages has a limited effect on migration choices. Even so, workers are better off, since they save on information acquisition costs. The crowding out of costly attention by free signals was first highlighted by Steiner et al. (2017).

posterior beliefs about θ_t , denoted $\pi_{it|s}$, where s represents the signal received by agent i . Fourth, agents use their posterior beliefs to compute expected payoffs in every location and choose the destination offering the highest expected utility. The resulting location choices l_{it} , once realized for all agents, determine the new population distribution L_t . Finally, agents' posterior beliefs about θ_t at the end of period t become their prior beliefs $\pi_{i,t+1}$ for the next period's productivity vector θ_{t+1} .¹⁶

2.4.2 Preferences

Let $u(c_{kt})$ denote the utility derived from consuming c_{kt} units of the numeraire good in location k at time t . The gross per-period utility of an agent i who moves from location j to location k in period t is then

$$u_{ijkt} = u(c_{kt}) + B_k - \kappa_{jk} + \nu \varepsilon_{ikt}. \quad (5)$$

This expression captures the utility gain from consumption and local amenities in the destination, net of migration costs. The idiosyncratic taste shock ε_{ikt} represents agent i 's individual preference for location k in period t , scaled by the parameter ν , which governs the overall dispersion of preferences across destinations.

Because the utility function in equation (5) is log-linear, rebating profits does not affect migration decisions—earnings rise proportionally in every location. Moreover, since goods are freely traded, local prices are equalized across space and thus do not influence migration choices. Hence, indirect utility—net of information costs—can be written as

$$u_{ijkt} = A_k + \theta_{kt} - \alpha \log L_{kt} + B_k - \kappa_{jk} + \nu \varepsilon_{ikt}. \quad (6)$$

2.4.3 Individual Information Acquisition

At the beginning of each period t , every individual i holds a prior belief $\bar{\pi}_{l_{it-1}t} \in \Delta\Theta$, which assigns probabilities to all possible productivity vectors θ_t and is shared among all agents residing in location l_{it-1} . Each agent then decides how much information to acquire about the realized productivity vector θ_t in order to form posterior beliefs about the payoffs in each location. Following Matějka and McKay (2015) and Steiner et al. (2017), agents can choose to receive signals $s \in S$ about the current productivity vector. These signals enable them to refine their beliefs but are themselves stochastic. While agents cannot control the realizations of the signals they observe, they can optimally choose the conditional distribution from which these signals are drawn.

Let the vector of observed state variables for agent i in period t be $\omega_{it} = (L_{t-1}, \bar{\pi}_{l_{it-1}t}, l_{it-1}, \varepsilon_{it})$, and let $\Omega = \mathcal{L} \times \Delta\Theta \times \mathcal{J} \times \mathbb{R}^J$ denote the set of all possible realizations of this state vector. Given ω_{it} , the agent chooses an information acquisition strategy represented by a conditional distribution $f_t : \Theta \times \Omega \rightarrow \Delta S$, where $f_t(s | \theta_t, \omega_{it})$ is the probability of receiving signal s conditional on the true productivity vector being θ_t and the observed state being ω_{it} .

¹⁶See Figure A.1 in Appendix A.1 for a graphical illustration of this timeline.

These strategies—mappings from observed states ω_{it} to signal distributions f_t —are unrestricted, reflecting the idea that agents can gather information through a wide variety of channels. Although agents are free to design any signal structure, acquiring more informative signals entails higher costs. Consequently, individuals may optimally choose to remain only partially informed about locations, receiving a vector of signals that conveys limited information.

Once the agent has chosen their signal structure for the period, nature draws a signal realization s according to the chosen distribution f_t . Upon observing s , the agent updates their prior using Bayes' rule to obtain a posterior belief $\pi_{it|s}$. Let $\bar{f}_t(s | \omega) = \int_{\theta} f_t(s | \theta, \omega) \bar{\pi}_{l_{it-1}t}(\theta) d\theta$ denote the marginal probability of observing signal s given state ω . Then the posterior distribution is given by

$$\pi_{it|s}(\theta_t) = \frac{f_t(s | \theta_t, \omega_{it}) \bar{\pi}_{l_{it-1}t}(\theta_t)}{\bar{f}_t(s | \omega_{it})}. \quad (7)$$

To model the cost of acquiring information, I follow the rational inattention literature and measure uncertainty using the entropy of beliefs.¹⁷ For a random variable X with continuous support S and probability distribution $p \in \Delta(S)$, its (Shannon) entropy is defined as

$$H(p) \equiv \tilde{H}(X) = - \int_{x \in S} p(x) \log p(x) dx. \quad (8)$$

Entropy quantifies uncertainty about X .¹⁸ Starting from a prior belief distribution π_{it} , any signal distribution that lowers the expected entropy of θ_t is more costly to acquire. To capture this idea, I assume that the cost of a signal structure f_t is proportional to the reduction in expected entropy—that is, to the difference between the entropy of the prior and the expected entropy of the posterior.¹⁹

Before receiving the signal s , the uncertainty held by agent i about θ_t is $H(\bar{\pi}_{l_{it-1}t})$; after observing s , it becomes $H(\pi_{it|s})$. The cost of implementing a signal strategy f_t for an agent with state ω_{it} is therefore defined as

$$I(f_t | \omega_{it}) = \lambda_{l_{it-1}} [H(\bar{\pi}_{l_{it-1}t}) - \mathbb{E}_s[H(\pi_{it|s})]], \quad (9)$$

where the location-specific parameter $\lambda_{l_{it-1}}$ scales the information cost, reflecting that some locations may have more efficient technologies for gathering and processing information.

2.4.4 Dynamic Rational Inattention Migration Problem

The dynamic optimization problem faced by each agent is defined as follows.

Definition 1. *Each agent chooses a pair of information and migration strategies $(f, \sigma) = \{f_t, \sigma_t\}_{t \geq 1}$, where $f_t : \Theta \times \Omega \rightarrow \Delta S$ specifies a signal distribution $f_t(\cdot | \theta_t, \omega_{it})$, and $\sigma_t : S \times \Omega \rightarrow \mathcal{J}$ specifies a location*

¹⁷Specifically, I use Shannon entropy, first introduced in this context by Sims (2003). More recent work, such as Fosgerau et al. (2020), extends this framework to a broader class of entropy-based information costs.

¹⁸For example, entropy equals zero for a Dirac distribution that assigns probability one to a single value and zero to all others, under the convention that $0 \log 0 = 0$. For a normal distribution, entropy is $\log \sqrt{2\pi e \sigma^2}$ and increases with the variance σ^2 .

¹⁹This difference is known as the *conditional mutual information* between s_t and θ_t .

choice $\sigma_t(s_{it}, \omega_{it})$. Given these mappings, the agent solves

$$\max_{f, \sigma} \mathbb{E} \left[\sum_{t=1}^{\infty} \delta^t \left(u_{l_{it-1}l_{it}}(\theta_t, L_t, \varepsilon_{it}) - I(f_t | \omega_{it}) \right) \right] \quad \text{s.t.} \quad l_{it} = \sigma_t(s_{it}, \omega_{it}), \quad \delta \in (0, 1), \quad (10)$$

where l_{it} is the location optimally chosen at time t ; $u_{l_{it-1}l_{it}}(\cdot)$ denotes the flow utility defined in (6); $I(f_t | \omega_{it})$ is the information cost defined in (9); and expectations are taken over all future realizations of productivity shocks and individual states under current beliefs.

2.5 Population Dynamics

Individual migration decisions $l_{it} = \sigma_t(s_{it}, \omega_{it})$ jointly determine the reallocation of workers across locations. The resulting population distribution at time t , denoted $L_t = \{L_{jkt}\}_{j,k=1,\dots,J}$, records the measure of individuals moving from location j to location k . This new distribution evolves as a function of the previous period's allocation L_{t-1} .

Let $p_{jkt}(s, \theta_t, \omega_{it})$ denote the probability that an agent initially in location $l_{it-1} = j$ moves to location k at time t , conditional on the realized productivity vector θ_t and signal $s_{it} = s$. Given the agent's optimal information and mobility strategies f_t and σ_t , this probability is

$$p_{jkt}(s, \theta_t, \omega_{it}) = f_t(s | \theta_t, \omega_{it}) \mathbf{1}\{\sigma_t(s, \omega_{it}) = k\} \mathbf{1}\{l_{it-1} = j\}. \quad (11)$$

The post-migration population in location k is given by $L_{kt} = \sum_j L_{jkt}$, where

$$L_{jkt} = L_{j,t-1} \bar{p}_{jkt}(\theta_t, L_{t-1}), \quad (12)$$

and $\bar{p}_{jkt}(\theta_t, L_{t-1}) = \mathbb{E}[p_{jkt}(s, \theta_t, \omega_{it})]$ denotes the expected probability of moving from j to k at time t . The expectation is taken over agents' beliefs $\bar{\pi}_{l_{it-1}t}$, signals s , and idiosyncratic preference shocks ε_{it} .

To forecast wages, agents form rational expectations about the behavior of others. The equilibrium wage w_{kt} in each location depends on the realized population L_{kt} , which itself results from all agents' migration decisions. Given their understanding of the economic environment—including the distributions of productivity and preference shocks—agents can correctly predict equilibrium migration flows $\bar{p}_{jkt}(\theta_t, L_{t-1})$ conditional on the current population L_{t-1} and a conjectured productivity realization θ_t .²⁰ Given (12), agents use their predicted transition probabilities to infer the expected population vector L_t .

2.6 Local Information Sharing and Belief Dynamics

Workers may benefit from local interactions that provide information relevant to future migration decisions. For example, individuals who remain in their original location can learn about payoffs elsewhere by talking

²⁰Just as observing wages w_{kt} at the end of period t could in principle allow workers to infer the realized productivity shocks by inverting the wage-setting equation (2), observing the mobility flows L_t could also allow them to back out θ_t by inverting the law of motion (12). In Appendix A.3, I explore this case, where agents fully recover the realization of θ_t after observing L_t . Even then, because agents have heterogeneous preference shocks and location-specific information costs, they continue to adopt distinct information strategies, and migration patterns still reflect information frictions.

with newly arrived migrants, who possess relatively accurate knowledge of their origin markets. Such second-hand information can, in turn, influence future mobility choices. To capture this process of local information diffusion, I assume that, before deciding how much information to acquire individually, agents first gather information about migration opportunities through communication with others in the same location.

Unlike the rational inattention channel described above, local information sharing is assumed to be *costless* and *non-strategic*: it does not result from deliberate search or investment decisions, but rather from serendipitous social interactions within local networks. This modeling choice aims to capture the passive exchange of relevant information that naturally arises among co-located individuals.

In standard models of rational learning, individuals would update their beliefs by fully applying Bayes' rule to all new information. However, in the context of large social networks with many interacting decision-makers, such a requirement places unrealistic cognitive and informational demands on agents.²¹ To maintain tractability, I instead assume that beliefs are aggregated locally through a simple heuristic rule. After information sharing, agents in the same location hold a set of independent belief distributions about the productivity in each destination, whose means and variances are simple transformations of the corresponding moments of their residents' prior beliefs.

The belief distribution that prevails in location k after local information sharing, denoted $\bar{\pi}_{kt}$, is specified in the following assumption.

Assumption 3. *After residents of location k exchange information in period t , and before they acquire individual signals, their shared belief about the productivity vector θ_t consists of J independent distributions,*

$$\bar{\pi}_{kt} = (\bar{\pi}_{klt})_{l=1,\dots,J},$$

where $\bar{\pi}_{klt}$ represents the belief about productivity in location l . Each belief distribution $\bar{\pi}_{klt}(\cdot|L_{t-1})$ follows $F(\bar{\mu}_{klt}, \bar{\sigma}_{kt}^2)$, a distribution with mean $\bar{\mu}_{klt}$ and variance $\bar{\sigma}_{kt}^2$, defined by

$$\bar{\sigma}_{kt}^2 = \left(\frac{1}{L_{kt-1}} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2} \right)^{-1}, \quad \bar{\mu}_{klt} = \frac{\bar{\sigma}_{kt}^2}{L_{kt-1}} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2} \bar{\mu}_{jkt}. \quad (13)$$

Here, $\bar{\sigma}_{jkt}^2$ is the variance of beliefs held by a representative agent who moved from location j to k between periods $t-1$ and t with idiosyncratic preference shocks equal to their mean value (i.e., $\varepsilon_{ikt} = 0$ for all $k \in \mathcal{J}$), given their migration strategy $\mathbb{E}_\theta[p_{jkt-1}(\theta_{t-1}, L_{t-1}, 0)]$, and $\bar{\mu}_{jkt}$ is the mean of their marginal beliefs about θ_{lt} .

The aggregation rule in Assumption 3 has an intuitive interpretation. Beliefs that are shared by a larger fraction of the local population receive greater weight in the aggregate belief, and individuals with more precise beliefs (i.e., smaller variances) also exert greater influence.²²

²¹Starting with Degroot (1974), a rich literature has proposed tractable functional forms for belief updating rules that approximate social learning in networks while maintaining analytical and computational feasibility.

²²This contrasts with the heuristic of Degroot (1974), under which only the expectation of an individual's belief determines their influence. For example, under the log-linear learning rule, an individual holding a uniform (uninformative) belief would

The mean and variance of the shared beliefs $\bar{\pi}_{klt}$ are obtained from the means and variances of the beliefs of migrants who moved under the average strategy. Relying on strategies averaged over past realizations of productivity θ_{t-1} and idiosyncratic preferences ε_{it-1} simplifies the characterization of information sharing in two ways. First, it is as if one representative agent from each origin—rather than many individuals with heterogeneous beliefs—contributed to the aggregation of local beliefs. Second, the beliefs of this representative agent do not depend on the actual realization of productivity θ_{t-1} .²³ As a result, shared beliefs depend only on the previous population distribution L_{t-1} .

Assumption 3 follows much of the social learning literature by positing a heuristic yet tractable learning rule that captures key aspects of information diffusion. It is inspired by Molavi et al. (2018), who derive axioms under which the aggregation rule in (13) is in fact optimal. In Appendix A.2.5, I show that when agents' pre-sharing beliefs are normally distributed and information is exchanged via a log-linear learning rule, the resulting shared belief distribution F is also normal, with mean and variance given by (13).²⁴

Together, local information sharing and individual information acquisition determine how agents' beliefs evolve within a given period t . Across periods, beliefs are connected through Bayes' rule. At the beginning of period $t + 1$, each agent's prior about the new productivity vector θ_{t+1} is derived from their posterior at the end of t :

$$\pi_{it+1}(\theta_{t+1}) = \int_{\theta} \pi_{it|s}(\theta) \gamma(\theta_{t+1}|\theta) d\theta, \quad (14)$$

where s is the signal received by agent i in period t , and $\gamma(\cdot|\theta)$ is the pdf of a normal distribution with mean $\rho\theta$ and variance σ_{ξ}^2 .

2.7 Equilibrium

2.7.1 Dynamic Rational Inattention Equilibrium

We are now ready to define a competitive equilibrium. In this economy, a competitive equilibrium is a set of information-acquisition and mobility strategies such that all agents maximize their expected lifetime utility, taking as given the laws of motion for population and beliefs.

Definition 2. *Given an initial population distribution L_0 and initial shared beliefs $\{\bar{\pi}_{j0}\}_{j \in \mathcal{J}}$, an equilibrium consists of:*

- **Information strategies:** a system of signal distributions $f_t : \Theta \times \Omega \rightarrow \Delta S$,
- **Mobility strategies:** a system of mappings $\sigma_t : S \times \Omega \rightarrow \mathcal{J}$,

such that:

have no effect on the aggregate belief, whereas in Degroot (1974) such an individual would still be influential as long as their expectation differs from others'.

²³That posterior beliefs may not depend on the realization of the payoff-relevant variable, even though actions do, is a feature of rationally inattentive behavior under certain conditions (Caplin and Dean, 2015). I discuss this property further in Appendix A.2.3.

²⁴Following Molavi et al. (2018), I show that in a complete network, the log-linear learning rule is the unique aggregation rule when agents exhibit *imperfect recall*—that is, they treat their neighbors' current beliefs as sufficient statistics for all available information, ignoring how those beliefs were originally formed. This formalizes the idea that real-world individuals cannot fully reconstruct the entire history of beliefs or actions within a social network.

1. **Utility maximization:** Agents solve the problem in (10).
2. **Population dynamics:** The population evolves according to (12).
3. **Belief updating:** Posterior beliefs are derived from priors by Bayes' rule (7), shared locally via (13), and propagated across periods according to (14).

I next establish a lemma—analogous to Lemma 1 in Steiner et al. (2017)—that greatly simplifies the characterization of agents' location choices by restricting attention to a special class of information strategies in which each signal directly corresponds to an action. The intuition is straightforward: it is never optimal to design two distinct signals that induce the same mobility decision. Because information carries a convex entropy-based cost, combining such redundant signals into a single one leaves the distribution of actions unchanged while weakly reducing the cost. Hence, the optimal information structure can be represented as a distribution over *recommendations*, where each signal realization directly prescribes a location choice.²⁵

For each location k , only one signal realization s induces agent i to choose k . Hence, the migration probability $p_{jkt}(s, \theta_t, \omega_{it})$ defined in (11) can be written as $p_{jkt}(\theta_t, \omega_{it})$ —the probability that agent i receives the signal associated with location k . I refer to this probability as a *mobility rule*, denoted p , which specifies for each possible (θ_t, ω_{it}) a distribution over destinations $\mathcal{J}$. The next lemma shows that, instead of solving for both the optimal information and mobility strategies in (10), it is sufficient to solve directly for the associated mobility rule p .

Lemma 1. *Any equilibrium strategy (f, σ) that solves the dynamic rational inattention problem in (10) generates a choice rule p , consisting of a sequence of mappings $p_t : \Theta \times \Omega \rightarrow \Delta\mathcal{J}$, that solves*

$$\max_p \mathbb{E} \left[\sum_{t=1}^{\infty} \delta^t \left(u_{l_{it-1}l_{it}}(\theta_t, \omega_{it}) - I(\omega_{it}) \right) \right], \quad (15)$$

where the flow of utility $u_{l_{it-1}l_{it}}(\theta_t, \omega_{it})$ is defined in (6) (after substituting L_t from the law of motion (12)), and the information cost is

$$I(\omega_{it}) = \lambda_{l_{it-1}} \left(H(\bar{\pi}_{l_{it-1}t}) - \sum_k q_{l_{it-1}kt}(\omega_{it}) H(\pi_{it|k}) \right), \quad (16)$$

with $\pi_{it|k} \equiv \pi_{it|s_t=k}$ denoting the posterior belief conditional on choosing location k . For $j = l_{it-1}$, the ex-ante probability of receiving the recommendation to move to k is $q_{jkt}(\omega_{it}) = \int_{\theta} p_{jkt}(\theta, \omega_{it}) \bar{\pi}_{jt}(\theta) d\theta$. Population and beliefs evolve according to the laws of motion (12)–(14).

Proof. See Appendix A.2.1. □

²⁵In dynamic settings, one must also ensure that agents have no incentive to construct a richer signal structure today for use in the future. Steiner et al. (2017) show that this is ruled out by the additivity of entropy: since information costs are linear in mutual information and agents discount future payoffs, delaying information acquisition cannot reduce total cost, regardless of other information obtained later.

In what follows, I dispense with explicit references to signal realizations s_t and work directly with actions l_t . Any rule p that solves (15) together with the population and belief dynamics constitutes a solution to the dynamic rational inattention equilibrium.

Proposition 1. *There exists a solution to the dynamic rational inattention equilibrium.*

Proof. See Appendix A.2.2. □

Proposition 1 extends the existence result in Steiner et al. (2017) to an environment with a continuum of agents, a non-finite state space, and endogenous payoffs. The proof relies on showing that the strategy space is compact and that agents' objective functions are continuous in their strategies, which follows from the boundedness of payoffs and the presence of discounting.

2.7.2 Stochastic Steady-State Equilibrium

I now introduce a set of assumptions that yield a tractable characterization of the dynamic rational inattention equilibrium in a stochastic steady state. A stochastic steady state consists of a mobility rule $p(\cdot)$ and shared beliefs $\bar{\pi}(\cdot)$ that are time-invariant: they map the same states to the same actions and probability distributions in every period. Although mobility flows and earnings vary each period with the realization of new productivity shocks, population and beliefs in each location evolve within an ergodic distribution. Hence, the time index t can be dropped from p and $\bar{\pi}$.

Definition 3. *A stochastic steady-state equilibrium is a mobility rule p , consisting of a sequence of mappings $p_t : \Theta \times \Omega \rightarrow \Delta\mathcal{J}$, and shared prior beliefs $\bar{\pi}$, consisting of mappings $\bar{\pi}_t : \Omega \rightarrow \Delta\Theta$, such that p solves (15), population and beliefs evolve according to (12)–(14), and for all $j \in \mathcal{J}$, $\omega \in \Omega$, and $t > 0$, $p_{jkt}(\theta, \omega) = p_{jkt+1}(\theta, \omega)$, $\bar{\pi}_{jt}(\theta|\omega) = \bar{\pi}_{jt+1}(\theta|\omega)$.*

Migration decisions depend on the state variables that include agents' prior beliefs at the beginning of the period. Even if two periods share identical realizations of productivity, population distribution, and preferences, differences in prior beliefs can lead agents to acquire different amounts of information and make distinct mobility decisions.

The definition of a stochastic steady-state equilibrium does not impose any restriction on how the mobility rule p_t and the shared priors $\bar{\pi}_t$ depend on the observable state ω_{it} . Proposition 2 below shows that, under Assumption 3 and the rational inattention structure described above, any stochastic steady-state equilibrium admits a representation in which the mobility rule and continuation values are Markov in $(\theta_t, L_{t-1}, \varepsilon_{it})$, and shared beliefs depend on L_{t-1} only.

Proposition 2 characterizes the solution to the stochastic steady state. As shown by Matějka and McKay (2015) in a static framework and extended by Steiner et al. (2017) to dynamic settings, the information cost in (9), based on Shannon entropy, leads to a modified logit form of the decision rule. In the stochastic steady state, the mobility rule inherits a similar logit structure.

Proposition 2. *In any stochastic steady-state equilibrium, there exists a representation in which, for each agent located in j at time $t-1$, the optimal mobility rule can be written as a function of $(\theta_t, L_{t-1}, \varepsilon_{it})$ only,*

and $p_{jk}(\theta_t, L_{t-1}, \varepsilon_{it})$ satisfies

$$p_{jk}(\theta_t, L_{t-1}, \varepsilon_{it}) = \frac{q_{jk}(L_{t-1}, \varepsilon_{it}) \exp(u_{jk}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_k(\theta_t, L_{t-1}))^{1/\lambda_j}}{\sum_{l=1}^J q_{jl}(L_{t-1}, \varepsilon_{it}) \exp(u_{jl}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_l(\theta_t, L_{t-1}))^{1/\lambda_j}}, \quad (17)$$

where $q_{jk}(L_{t-1}, \varepsilon_{it}) = \int_{\theta} p_{jk}(\theta, L_{t-1}, \varepsilon_{it}) \bar{\pi}_j(\theta | L_{t-1}) d\theta$ is the unconditional mobility rule, and $\bar{V}_k(\theta_t, L_{t-1}) = \mathbb{E}[V_k(\theta_{t+1}, L_t, \varepsilon_{it+1}) | \theta_t, L_{t-1}]$ is the expected continuation value. Continuation values solve

$$V_j(\theta_t, L_{t-1}, \varepsilon_{it}) = \lambda_j \log \left(\sum_{l=1}^J q_{jl}(L_{t-1}, \varepsilon_{it}) \exp(u_{jl}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_l(\theta_t, L_{t-1}))^{1/\lambda_j} \right), \quad (18)$$

population and beliefs evolve according to (12)–(14), and shared beliefs satisfy $\bar{\pi}_j(\cdot | \omega_{it}) = \bar{\pi}_j(\cdot | L_{t-1})$ for all $j \in \mathcal{J}$ and $t \geq 1$.

Proof. See Appendix A.2.3. □

Conditional on $(\theta_t, L_{t-1}, \varepsilon_{it})$, the agent's decision remains stochastic because it depends on the realization of their signal. Under full information, an agent in j would observe all payoffs $u_{jk} + \delta \bar{V}_k$ and select the destination k^* offering the highest value, so the mobility rule p_{jk} would reduce to a deterministic indicator equal to one for k^* and zero otherwise. This corresponds to the limit of (17) as $\lambda_j \rightarrow 0$.

When information is costly ($\lambda_j > 0$), (17) implies that destinations offering higher expected payoffs $u_{jk} + \delta \bar{V}_k$ are more likely to be chosen after information acquisition. The rule also contains an endogenous bilateral term, the ex-ante mobility probability q_{jk} , which captures a form of path dependence: if the prior beliefs or past flows make k a likely destination from j , it remains more likely to be chosen even for similar realized payoffs. This endogenous predisposition quantifies how information frictions reinforce established migration links and amplify persistence in mobility patterns.

The expected value function (18) resembles a dynamic logit formulation. The value of residing in j rises with the payoffs expected from all potential destinations, weighted by their ex-ante probabilities q_{jl} . Under complete information, this collapses to the payoff in the optimal destination k^* . With incomplete information, each location's payoff contributes in proportion to the likelihood that it is chosen, while the parameters λ_j and q_{jk} summarize the influence of information frictions on perceived opportunities and the dispersion of migration choices.

At this stage, mobility decisions vary across preference draws. To express aggregate bilateral migration flows for each combination of productivity and population (θ_t, L_{t-1}) , I integrate these individual choices over the distribution of preference shocks ε_{it} , after imposing two distributional assumptions.

First, I assume that preference shocks follow the conjugate of a Type-1 Extreme Value (EV1) distribution. Second, I assume that the distribution of shared beliefs F , introduced in Assumption 3, is also conjugate EV1.²⁶ The key property of the conjugate EV1 distribution is that if a random variable X is drawn from an EV1 distribution and another variable Y from a conjugate EV1, then $Y + X$ is also

²⁶Brown and Jeon (2024) impose a similar structure in their static model of optimal health-insurance choice, while Dasgupta and Mondria (2018) apply it to productivity shocks in a model of international trade.

EV1-distributed (Cardell, 1997).²⁷

Assumption 4. For any $k, l = 1, \dots, J$, the marginal distribution of shared beliefs $\bar{\pi}_{klt}$ is conjugate EV1, $C(\mu_{kl}(L_{t-1}), \sigma_k^2(L_{t-1}))$, with mean $\mu_{kl}(L_{t-1})$ and variance $\sigma_k^2(L_{t-1})$.

Assumption 5. Preference shocks follow a conjugate EV1 distribution, $C(0, \nu^2)$, with mean zero and variance ν^2 .

Before deriving the migration probability $\bar{p}_{jk}(\theta_t, L_{t-1}) = \mathbb{E}[p_{jk}(\theta_t, L_{t-1}, \varepsilon_{it})]$, I define several auxiliary objects. Let $\bar{u}_{jk}$ denote the average instantaneous utility flow, $\bar{u}_{jk} = \mathbb{E}[u_{ijk}]$, integrated over ε_{it} . Define the expected value of moving from j to k given (θ_t, L_{t-1}) as $\bar{v}_{jk}(\theta_t, L_{t-1}) = \bar{u}_{jk}(\theta_t, L_{t-1}) + \delta \bar{V}_k(\theta_t, L_{t-1})$. Finally, define the expectational error as $\chi_{jk}(\theta_t, L_{t-1}) = \bar{\mu}_{jk}(L_{t-1}) - \theta_{kt}$, which measures the difference between the expected and realized productivity in location k from the perspective of agents residing in j .

Proposition 3. Under Assumptions 4 and 5, the average mobility rule in the presence of information frictions and preference heterogeneity is

$$\bar{p}_{jk}(\theta_t, L_{t-1}) = \frac{\exp(\eta_j \chi_{jk}(\theta_t, L_{t-1}) + \bar{v}_{jk}(\theta_t, L_{t-1}))^{\frac{1}{\phi_j}}}{\sum_l \exp(\eta_j \chi_{jl}(\theta_t, L_{t-1}) + \bar{v}_{jl}(\theta_t, L_{t-1}))^{\frac{1}{\phi_j}}}, \quad (19)$$

where $\chi_{jk}(\theta_t, L_{t-1}) = \bar{\mu}_{jk}(L_{t-1}) - \theta_{kt}$ is the expectational error made by agents in j about θ_{kt} , and continuation payoffs satisfy

$$V_j(\theta_t, L_{t-1}) = \phi_j \log \left(\sum_l \exp(\eta_j \chi_{jl}(\theta_t, L_{t-1}) + \bar{v}_{jl}(\theta_t, L_{t-1}))^{\frac{1}{\phi_j}} \right), \quad (20)$$

with $\phi_j = \nu \left(1 + \lambda_j^2 (1 - \eta_j)^2 \right)^{1/2}$ and $\eta_j = \left(1 + \frac{6\bar{\sigma}_j^2}{\pi^2 \lambda_j^2 \nu^2} \right)^{-1/2} \in (0, 1)$, and $\pi = 3.1415 \dots$

When information costs vanish, $\bar{\mu}_{jk}(L_{t-1}) \rightarrow \theta_{kt}$ and $\phi_j \rightarrow \nu$, so the model reduces to a standard preference-based migration model:

$$\bar{p}_{jk}(\theta_t, L_{t-1}) \xrightarrow{\lambda_j \rightarrow 0} \frac{\exp(\bar{v}_{jk}(\theta_t, L_{t-1}))^{1/\nu}}{\sum_l \exp(\bar{v}_{jl}(\theta_t, L_{t-1}))^{1/\nu}}. \quad (21)$$

When the dispersion of preferences ν tends to zero, the solution becomes

$$\bar{p}_{jk}(\theta_t, L_{t-1}) \xrightarrow{\nu \rightarrow 0} \frac{\exp(\rho_j \chi_{jk}(\theta_t, L_{t-1}) + \bar{v}_{jk}(\theta_t, L_{t-1}))^{\frac{1}{\psi_j}}}{\sum_l \exp(\rho_j \chi_{jl}(\theta_t, L_{t-1}) + \bar{v}_{jl}(\theta_t, L_{t-1}))^{\frac{1}{\psi_j}}}, \quad (22)$$

where $\psi_j = \lambda_j (1 - \rho_j)$ and $\rho_j = \left(1 + \frac{6\bar{\sigma}_j^2}{\pi^2 \lambda_j^2} \right)^{-1/2} \in (0, 1)$.

Proof. See Appendix A.2.4. □

²⁷ Appendix E illustrates the relation between the EV1 and conjugate EV1 distributions, which are qualitatively similar.

The role of information frictions λ_j in shaping migration responses is clear in (19). The elasticity of migration with respect to observed payoffs $\bar{u}_{jk}$ equals $1/\phi_j$, whereas the elasticity with respect to the unobserved component θ_t is smaller, equal to $(1 - \eta_j)/\phi_j$, because agents only imperfectly observe productivity. Since $\eta_j \in (0, 1)$, migration flows react less to variations in θ_t than to changes in directly observed payoffs. As information costs rise, $\eta_j \rightarrow 1$ and migration becomes completely unresponsive to θ_t . When information costs vanish, prior beliefs become perfectly accurate, $\bar{\mu}_{jk}(L_{t-1}) = \theta_{kt}$, and (21) shows that the elasticity with respect to all payoff components equals $1/\nu$. Hence, the effect of information frictions can be identified from the differential elasticity of migration with respect to unobserved (θ_t) and observed (e.g., L_{t-1}) determinants of payoffs. Section 3.2.2 develops a strategy to estimate these elasticities ($1/\phi_{jt}, (1 - \eta_{jt})/\phi_{jt}$) and, by extension, the underlying parameters (λ_j, ν) .

Information frictions also influence levels of migration flows. As seen in (19), the mean prior beliefs $\bar{\mu}_{jk}(L_{t-1})$ shift the baseline magnitude of bilateral migration rates. From (13), beliefs held by residents in k about their own productivity θ_{kt} are likely upward-biased: migrants to k received favorable signals about k , raising the local consensus on its productivity. This optimism increases the probability of staying in k next period. More generally, optimistic beliefs about a destination raise its expected inflows, reinforcing the endogenous predispositions discussed in relation to (17).

Failing to account for these information frictions leads to overestimation of true bilateral migration costs κ_{jk} (see Section 3.4). Moreover, the interaction between beliefs and realized productivity drives systematic differences in migration patterns. As developed in Section 4, the covariance between mean beliefs and realized productivity varies across pairs of locations, implying that distant destinations—where migration costs are higher—elicit weaker informational responses. Agents both interact less with well-informed migrants from faraway places and acquire less precise information about them individually, generating a declining migration elasticity with distance consistent with empirical evidence.

3 Quantification

3.1 Data Sources

In this section, I present the main sources of data and provide descriptive statistics on migration patterns in Brazil. I provide more information about how I constructed the sample in Appendix B.

The main source of data is the *Relação Anual de Informações Sociais* (RAIS), which is collected annually by the Brazilian Ministry of Labor and contains matched employer-employee information for every formally employed worker in Brazil.²⁸ It includes demographic, occupational and income characteristics for employees, with individual identifiers so that workers can be followed from year to year. I use 15 consecutive years of data, corresponding to the period between 2000 and 2014. RAIS also includes the geographic location of each employment contract at the municipality level. For every formal job and

²⁸In Brazil, informal and self-employed workers constitute a large fraction of the labor force, peaking at 45% in 2000 (Bosch et al., 2012). Consequently, my analysis leaves out a significant fraction of the labor force. Workers may consider transitions into and out of the formal sector as an alternative to geographic mobility. To limit the prevalence of these transitions, I restrict my analysis to workers with a high attachment to the formal sector by focusing on workers spending at least seven consecutive years in the formal sector over the sample period.

year, I exploit information on the duration of the job spell, the average monthly wage, the number of hours stipulated in the contract, and certain characteristics of the plant where the worker is employed.²⁹ Specifically, I use information on the micro and mesoregion in which the plant is located.³⁰

Workers in the sample often appear to be performing multiple different jobs in the same year. If a worker has more than one job in a year, I assign them to the job they held for longest during that year. However, to determine the total labor income of a worker in a year, I add labor income from all jobs that worker held.

The data contains no information on the residence of a worker before their first job in the formal sector. Consequently, in the analysis, I focus on the migration decisions of workers and do not model the decision to enter the labor force or acquire college education. For this reason, I limit the data to workers that are over 25 years of age, since, for the majority of the population, education decisions are taken before this age. Similarly, I do not model the retirement decision of workers and, consequently, I limit the data workers below 65 years of age.

Besides the information on workers' labor market histories contained in the RAIS database, I also use information on the population of each municipality in each year between 2000 and 2014 from the population census collected by *Instituto Brasileiro de Geografia e Estatística* (IBGE). I compute population by mesoregion by aggregating the population of all municipalities included in the corresponding mesoregion.

Finally, I collect information on the degree of internet penetration by microregion. Specifically, from the *Agência Nacional de Telecomunicações*, ANATEL, a government agency in charge of regulating and supervising telecommunications in Brazil, I obtain information on the number of broadband connections by municipality and year between 2007 and 2014. I use the population data obtained from the population census to construct a municipality-specific measure of the number of broadband connections per capita, and again use the municipality-level population data to construct an equivalent variable at the mesoregion level. As I do not have access to information on the number of broadband internet connections for the years 2000 to 2006 and the available data indicates that the overall number of broadband connections per inhabitant is less than 1% in 2007, I assume that the number of broadband connections per capita equals zero in every mesoregion before 2007.

The resulting dataset includes 372,454,979 worker-year pairs that correspond to 45,958,805 workers. According to the RAIS data, the 137 mesoregions had on average 250,000 legal workers in 2014 (the median microregion had close to 62,000 legal workers). The mesoregions with the highest labor force are located in the South and along the coast. The average yearly rate of migration across mesoregions is 3.4%.

²⁹In the analysis of migration patterns at the individual level described in section 4.1, I also exploit the 2-digit occupation (according to the *Classificação Brasileira de Ocupações*, CBO), the 2-digit industry of production of the establishment for the main job spell (according to the *Classificação Nacional de Atividades Econômicas*, CNAE), as well as information on the workers' gender, age, and level of education.

³⁰Brazilian microregions are groups of municipalities that span the entirety of the Brazilian territory and are the closest equivalent to commuting zones. During our sample period, there were 558 microregions which are grouped into 137 mesoregions.

3.2 Estimation Procedure

In this section, I estimate the parameters of the model, guided by the mobility rule derived in the previous section. I estimate the model separately over two time periods: 2000-2007 and 2008-2014. I will treat each of these periods as steady states, and compare the parameters, in particular information costs, during each period. I follow a two-step procedure for each period: first, I estimate the production parameters that can be inferred directly from the data without simulating the model. Second, I use the gravity equation of migration predicted by the model to estimate the remaining parameters consisting of the amenities, migration costs, preference heterogeneity, and information costs.

The estimation uses data from RAIS described in Section 3.1. I construct yearly bilateral migration flows between each of the 137 mesoregions in Brazil, and average wages in each region and year, from individual records of earnings and location each year.

3.2.1 Step 1: Production parameters

I begin by estimating $J+3$ parameters that can be obtained directly from the data. This step only imposes assumptions on the production side of my model. These are the parameters governing the decreasing returns in labor, $1 - \alpha$, the persistence and volatility of the productivity process, (ρ, σ_ξ^2) , and the regional baseline productivities A_k .

I first set the Cobb-Douglas share of labor, $1 - \alpha$, used in production equal to the average labor share computed from national statistics over each period.³¹ The labor share, $1 - \alpha$, in Brazil is stable between 2000 and 2014 at a value of 0.60.

I then estimate the parameters of the productivity process. The Brazilian economy is growing over the sample period. The total employment recorded in RAIS is also growing each year, both from demographic change and increased transitions to the formal labor market. To be able to interpret the data as closely as possible to a steady state with constant total population, I project wages on year fixed effects and normalize all population stocks so that the total population in the economy is constant at its 2000 level over the sample. I then estimate the AR(1) process associated with the observed wages after substituting year fixed effects and adjusting for population, $\tilde{w}_{kt} = \log w_{kt} + \alpha \log L_{kt} - \log(1 - \alpha)$. From the first order condition determining wages (2) and the evolution of productivity (3), the adjusted wages can be expressed as

$$\tilde{w}_{kt} \equiv \log w_{kt} + \alpha \log L_{kt} - \log(1 - \alpha) = A_k + \theta_{kt}. \quad (23)$$

The baseline productivities are therefore recovered as the average of $\tilde{w}_{kt}$ over the period, since $\mathbb{E}[\theta_{kt}] = 0$. The persistence of the productivity process ρ is equal to the covariance of $\tilde{w}_{kt}$ and $\tilde{w}_{kt-1}$ divided by the variance of $\tilde{w}_{kt}$. The volatility of the productivity process is then computed as $(1 - \rho^2)\text{Var}(\tilde{w}_{kt})$. Finally, I recover the productivity shocks θ_{kt} implied from the observed $\tilde{w}_{kt}$ and estimated productivities A_k as the residuals from (23).³² As reported in Table 1, the estimated productivities are on average equal to 4.49

³¹See Restrepo-Echavarria and Reinbold (2018). The data available in RAIS only contains payments made to formally employed workers and does not report information on value added. The labor share is computed from the Penn World Tables.

³²Since the production specification does not account for the use of capital or other inputs, any variation in observed wages that is caused by a shock to these variables is interpreted as a productivity shock.

with a standard deviation of 0.87 across regions between 2000 and 2007. For 2008–2014, the mean and standard deviations are 4.72 and 0.83 respectively. The persistence of the productivity process ρ is 0.76 in the first period and 0.69 in the second period. The volatility of the productivity process σ_ξ^2 is 0.33 in the first period and 0.38 in the second period.

3.2.2 Step 2: Simulated Method of Moments

In the second stage, I use the method of simulated moments to estimate the remaining parameters $\vartheta = (\nu, \lambda_j, B_j, \kappa_{jk})$, which consist of the dispersion of idiosyncratic preferences, information costs, amenities, and migration costs. The total number of parameters to be estimated in this step is large: there are as many as $2J + (J - 1)^2 = 18770$ of them.³³ A grid search over the parameter space is therefore not practical. To circumvent this issue, I develop an iterative algorithm that updates the parameter guesses in a simple intuitive way and delivers fast convergence. First, I use the mobility rule in (19) to obtain a regression equation predicted by the model. This regression equation offers moments that I target to identify the parameters of interest. I then simulate the model given parameter estimates from the first step and guesses for the parameters to be estimated. I run the predicted regression in the model and update the parameter guesses using the estimated coefficients of the regression.

To obtain the regression equation at the core of my estimation, I exploit the gravity structure of the model. Denote by $\bar{p}_{jkt}$ the migration share between j and k conditional on the productivity and population at time t . Taking the log of the migration share $\bar{p}_{jkt}$ divided by the share of stayers in j , $\bar{p}_{jjt}$, and using the expression of $\bar{u}_{jk}$ in (6) yields the following expression:

$$\log \frac{\bar{p}_{jkt}}{\bar{p}_{jjt}} = \frac{\eta_j}{\phi_j} (\chi_{jkt} - \chi_{jjt}) + \frac{1}{\phi_j} \left(\log \frac{w_{kt}}{w_{jt}} + D_{jk} + \delta (V_{kt+1} - V_{jt+1}) \right) + e_{jkt}, \quad (24)$$

where $D_{jk} = B_k - B_j - \kappa_{jk}$ is a composite bilateral “resistance” term combining the contributions of amenity differences between regions and migration costs. The error term $e_{jkt} = (\zeta_{jjt} - \zeta_{jkt}) / \phi_j$ is composed of expectational errors about future values $\zeta_{jkt} = \delta (V_{kt+1} - E_{j,t} V_{kt+1})$. Equation (24) has a “gravity” structure in that the magnitude of bilateral flows is increasing in the wage (and future value) differentials between any two regions, and is decreasing in the “distance” between regions captured by D_{jk} . The new contributor to this gravity structure is the “optimism differential” that agents express towards the destination relative to their origin, and is represented by the difference between the expectational errors made by agents in j about θ_{kt} and θ_{jt} , $(\chi_{jkt} - \chi_{jjt})$.

Since the error term e_{jkt} only arises because of the irreducible uncertainty about future productivity θ_{t+1} due to the innovation of the AR(1) process, it is orthogonal to all other regressors. As a result, the coefficients $\hat{\beta}$ obtained from estimating (24) by ordinary least squares (OLS) are the method of moments estimator for the moment condition associated to the orthogonality of the regressors and the error term.

There is a unique mapping between the coefficients of the regression $\beta = (\eta_j / \phi_j, 1 / \phi_j, \{D_{jk} / \phi_j\})$, and the parameters of interest $\vartheta = (\nu, \lambda_j, B_j, \kappa_{jk})$.

³³There are J information costs, $J - 1$ amenities since one can be normalized to 0, $(J - 1)^2$ migration costs since I normalize $\kappa_{jj} = 0$ for all j , and the preference heterogeneity ν .

Intuitively, η_j and ϕ_j together identify ν and λ_j , while projecting D_{jk} on origin and destination fixed effects allows me to recover amenities, B_j , and fixed costs, κ_{jk} , after normalizing $\kappa_{jj} = 0$ for all j .³⁴

The estimation algorithm I employ relies on a standard fixed point iteration on the OLS equation (24). A satisfactory set of parameters ϑ should deliver beliefs and values such that the estimates $\hat{\beta}$ from (24) map exactly to $\hat{\vartheta}(\hat{\beta}) = \vartheta$.

First, I guess initial values $\vartheta^0 = (\nu^{(0)}, \lambda_j^{(0)}, B_j^{(0)}, \kappa_{jk}^{(0)})$ for the parameters to be estimated.³⁵ Second, I simulate the model using the production parameters $(\alpha, \rho, \sigma_\xi^2)$ estimated in the first block, and the current parameters $\vartheta^{(0)}$. This delivers mean belief functions $\mu^{(0)}(\cdot)$ and values $V^{(0)}(\cdot)$. Third, I evaluate beliefs and values at observed populations L_{t-1}^{obs} and recovered θ_t^{obs} , and estimate (24) by OLS.³⁶ Fourth, from the estimated η_j and ϕ_j and fixed effects, I update the parameters to $\vartheta^{(1)}$ and return to the second step using $\vartheta^{(1)}$. I keep iterating until my estimator converges to a fixed point ϑ^* . By construction, this procedure minimizes the method of moment objective associated to the moment conditions of the OLS regression (24). I describe the iterative algorithm in more detail in Appendix C.1. Despite the absence of theoretical results on the convergence of this algorithm, I find that in practice, it converges quickly and always to the same solution ϑ^* for different initial values $\vartheta^{(0)}$.

I estimate the model over the two different periods, 2000–2007 and 2008–2014, and report the results in Table 1. I describe the information costs in Section 3.3. The estimated dispersion of preferences is equal to 2.31 in the first period and 2.62 in the second period. The dispersion in preferences appears to have increased over time as information costs decreased. This reflects the fact that migration flows have become less responsive to wages overall, but have become relatively more responsive to unobserved productivity shocks. Finally, I separate the amenities from the migration costs by projecting the estimated D_{jk} on origin and destination fixed effects. I normalize the lowest estimates of amenity across regions to zero, and obtain that the standard deviation is 0.83 in the first period and 0.91 in the second period. The average migration costs are 0.56 in the first period and 0.47 in the second period. The decline in migration costs between the two periods is consistent with the increase in migration flows between the two periods.³⁷ Overall, parameter values are fairly similar over the two periods.

3.3 Information Costs and Internet Access

Information costs are on average equal to 3.11 in the first period and 2.23 in the second period. The standard deviation in the information costs is 1.43 in the first period and 1.32 in the second period. Figure 1a and Figure 1b display the estimated information costs λ_j for each of the 137 mesoregions for the first and second periods. The information costs appear strongly correlated with the economic development of regions, with the lowest costs of information in the most densely populated and richest regions of the South, including the metropolitan areas of São Paulo, Rio de Janeiro, and Brasilia. In contrast, the

³⁴The mapping between (η_j, ϕ_j) and (ν, λ_j) is not direct as it depends on the variance of beliefs $\bar{\sigma}_j$, itself a function of all parameters. See Appendix C.1 for more details.

³⁵To obtain reasonable initial guesses, I first run (24) omitting the mean beliefs and future values.

³⁶In the data, out of the $(J-1)^2 = 136^2 = 18,496$ migration trips that could be undertaken between any two mesoregions, only 10,382 of them have positive flows recorded more than once over the 15 years of data. This implies that no fixed bilateral resistance term can be estimated for these pairs, and I assign them a prohibitive migration cost.

³⁷See Appendix B.2 for a description of the yearly migration rates.

Table 1: Estimated Parameters

Parameter	Related Moment	Statistic	Time Period	
			2000-2007	2008-2014
α	Inverse Labor Share	Value	0.41	0.40
ρ	Persistence of Wages	Value	0.76	0.69
σ_ξ^2	Volatility of Wages	Value	0.33	0.38
ν	Migration Elast. wrt wages	Value	2.31	2.62
λ_j	Migration Elast. wrt Productivity	Mean	3.11	2.23
A_j	Average Wages	Std Dev	0.87	0.83
B_j	Average Population	Std Dev	0.83	0.91
κ_{jk}	Average Migration Flows	Mean	0.56	0.47

Estimated values of the parameters of the production process, regional baseline productivities, amenities, and information costs, as well as bilateral migration costs.

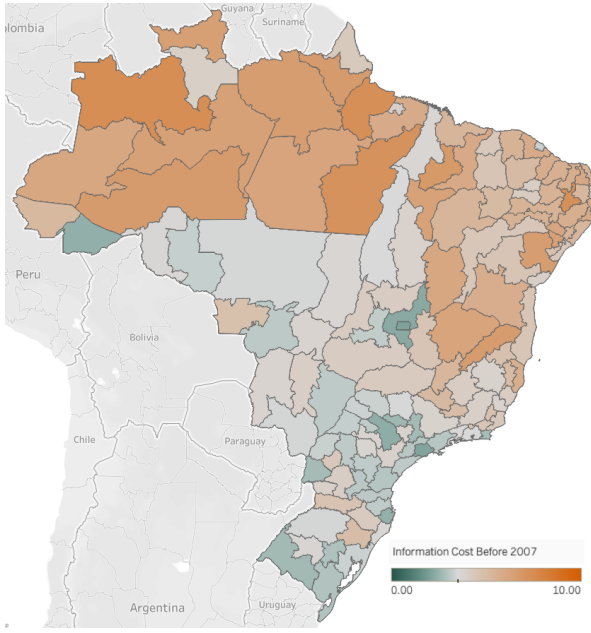
less developed North-Eastern regions and remote regions in the Amazon appear to have a much higher cost of information. It is worth emphasizing that these information costs were recovered as fixed-slope coefficients with no parametric assumption regarding their correlation with any observable variable. The lower information costs obtained for Southern regions result from the fact that workers in these regions appear to be relatively more responsive to the unobserved productivity shocks than workers in other regions.

Since these information costs are novel parameters that have not yet been estimated in the literature, I now investigate their variation along observable regional characteristics. From inspecting Figure 1a and Figure 1b, these information costs appear to be correlated with population density and average income. One intuitive shifter for the technological cost of information acquisition is the availability of internet in the region. There may also be persistent determinants of the information costs at the region level, such as geographic accessibility. Motivated by these remarks, I propose a simple parameterization of information costs:

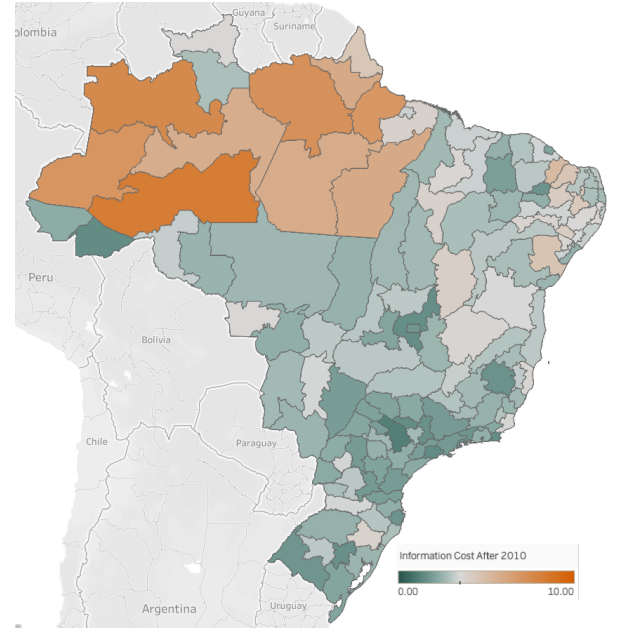
$$\lambda_{jt} = \ell_1 \text{int}_{jt} + \ell_2 \log w_{jt} + \ell_3 \log \text{popdens}_{jt} + \varsigma_j + u_{jt}, \quad (25)$$

where $t \in \{1, 2\}$ corresponds to each of the two periods 2000–2007 and 2008–2014, int_{jt} is the average fraction of residents of j with an active internet connection over the years corresponding to the period $t = 2$, and zero for the period $t = 1$, $\log w_{jt}$ is the log average wage in region t over the years corresponding to the period t , $\log \text{popdens}_{jt}$ is the average log of the population density in region j during each period, ς_j captures unobserved determinants of the information costs that are constant across the two periods, and u_{jt} captures the unobserved time varying determinants of the information costs.

I estimate (25) in first differences by OLS and report the results in the first column of Table 2. The coefficient on internet access is large and significant at the 1% level, and indicates that conditional on income and population density, increasing the fraction of residents with an internet connections from zero to one is associated with a decline in the information cost by 0.97 units. The coefficient on log income, also significant at the 1% level, implies that a 1% increase in local income is associated with a decrease in the information cost by 0.00824 units. The coefficient on population density is only significant at the



(a) Information Costs 2000-2007



(b) Information Costs 2008-2014

Table 2: Determinants of Information Costs

	OLS	IV
Internet Connections / Inhabitants	-0.971 ^a (0.321)	-0.831 ^b (0.461)
Log Income	-0.824 ^a (0.241)	-0.793 ^a (0.262)
Log Pop. Density	-0.310 ^c (0.190)	-0.323 ^c (0.202)
Observations	137	137
R^2	0.219	0.192

a denotes 1% significance, *b* denotes 5% significance, *c* denotes 10% significance. In parenthesis, I report standard errors.

10% level and reveals that a 1% increase in population density is associated with a 0.0031 decrease in the information cost.

In the analysis of counterfactual exercises described in Section 5, I am interested in reproducing the plausible decrease in information costs brought about by the expansion of internet access at the local level. From inspecting (25), one may suspect that there could be unobserved time varying factors in the error u_{jt} that are correlated with the fraction of households with an internet connection. For example, changes in local public spending on transportation or communication infrastructure may have directly reduced the cost of information, and facilitated internet expansion. These omitted variables may lead to an upward bias in the estimation of the causal effect of internet penetration on the information cost.

To circumvent this issue, I instrument the fraction of residents with an internet connection with a dummy variable equal to 1 if the region is located less than 250 km away from a backbone cable.³⁸ Only agents residing in the vicinity of a backbone cable can expect to have access to high speed internet. Importantly, the geographic coverage of these backbones, including the ones deployed over the period 2008–2014, follows other infrastructure that existed prior to 2008. This provides a plausible source of variation for the extent of internet penetration that is unlikely to be affected by later changes in local economic conditions. To construct the instrument, I follow Tian (2021); see Appendix C.2. I report the results from the instrumental variable regression in column 2 of Table 2. The magnitude of the coefficient of internet connections per resident is slightly lower, but remains large and significant at the 5% level.

3.4 Migration Costs With and Without Information Frictions

I now illustrate the implications of my model for the magnitude of migration costs and preference heterogeneity. Since information frictions are a source of both endogenous migration costs and limited migration elasticity, smaller exogenous migration costs κ_{jk} and preference heterogeneity ν will be necessary to explain the observed migration patterns. I confirm this prediction by estimating the model under the assumption that there are no information frictions, so $\lambda_j = 0$ for every region j . This corresponds to the mobility rule (21).

Since there are no unobserved expectational errors to control for, it is possible to devise a direct estimation strategy without simulating the model. This method relies on renewal actions and compares migration paths that visit the same locations at t and $t+2$ but differ at $t+1$ (Artuç et al., 2010; Traiberman, 2019; Caliendo et al., 2021). From $t+2$ onwards, these paths offer the same continuation value, so the difference in the payoffs they offer can be expressed as a function of the wages in the visited regions at time $t+1$. If I select the origin destination at t to be j , the location at $t+1$ to be either j or k , and the destination at $t+2$ to be k , I can write the estimation equation

$$\Lambda_{jkt} = \frac{1}{\nu} (\Delta \log w_{jkt} + D_{jk} + e_{jkt}), \quad (26)$$

³⁸“Backbones” are national trunk infrastructure that brings traffic from international submarine cables in coastal regions to inland parts of the country. Backbones consist of high-capacity fiber optic cables. The quality of broadband connection decreases exponentially with distance to the backbone, and I follow Tian (2021) in setting the threshold of 250 km.

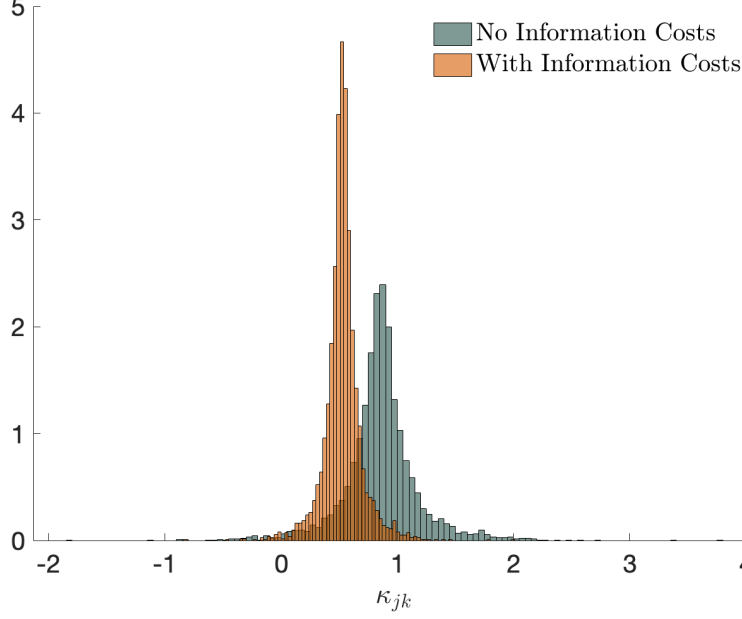


Figure 2: Estimated Migration Costs with and without Information Cost

where $D_{jk} = B_k - B_j - \kappa_{jk}$ and $\Lambda_{jkt} = \log \left(\frac{p_{jkt} p_{kkt+1}^\delta}{p_{jjt} p_{jkt+1}^\delta} \right)$ is the relative discounted probability of the paths $j \rightarrow k \rightarrow k$ and $j \rightarrow j \rightarrow k$. The residuals χ_{jkt} are a collection of expectational errors that are orthogonal to the regressors:

$$\chi_{jkt} = \zeta_{jkt} - \zeta_{jkt} + \delta(\zeta_{jkt+1} - \zeta_{kkt+1}), \quad \zeta_{jkt} = \delta(V_{kt+1} - E_{j,t}V_{kt+1}).$$

The iterative estimation algorithm described in Section 3.2.2 can also be applied to this particular case. I obtain almost exactly the same estimates as in the direct estimation, which validates my iterative estimation method.

The distribution of bilateral migration costs resulting from the estimation of (26)—after projecting on origin and destination fixed effects to net out amenities—is displayed in Figure 2 for the period 2000–2007. Out of the $(J - 1)^2 = 136^2 = 18,496$ migration trips that could be undertaken between any two mesoregions, only 10,382 have positive flows recorded more than once during the first period. This implies that my procedure does not estimate a fixed bilateral resistance term for the pairs with no consistent flows. In figure 2, the average migration cost among the 10,382 estimated bilateral costs is 0.94. For comparison, I report the distribution of the estimated migration costs for the same 10,382 origin-destination pairs in the model with information frictions. With information frictions, the average migration cost in the first period is 0.56, so the average migration cost is 40% smaller once we allow for information frictions. This difference illustrates the quantitative relevance of the endogenous predispositions emanating from the information frictions.

To think of these costs as a fraction of income, consider what an agent earning some initial income w would have to receive as additional income Δw to be perfectly compensated for paying the average

Table 3: Determinants of Migration Costs

	With Info Frictions		Without Info Frictions	
	2000-2007	2008-2014	2000-2007	2008-2014
Log Travel Time	0.384 ^a (0.053)	0.370 ^a (0.051)	0.619 ^a (0.089)	0.568 ^a (0.077)
Log Distance	0.263 ^a (0.032)	0.219 ^a (0.029)	0.440 ^a (0.052)	0.401 ^a (0.045)
Dummy Contiguous	-0.211 ^a (0.012)	-0.182 ^a (0.013)	-0.331 ^a (0.021)	-0.301 ^a (0.016)
Observations	68,152	68,152	68,152	68,152
R^2	0.334	0.291	0.352	0.301

^a indicates significant at the 1% level. In parenthesis, I report standard errors.

migration cost of 0.56. The additional income would require that $\log(w + \Delta w) = \log w - 0.56$, which corresponds to an increase $\Delta w/w$ of 75%, or an annualized flow of 3% of earnings, using a discount factor $\delta = 0.96$.

The estimated inverse elasticity from (26) in the first period is equal to $\nu = 2.92$, compared with 2.31 in the model with information frictions. The preference heterogeneity is then 21% smaller after accounting for information frictions. For the second period, I estimate $\nu = 3.12$, compared with 2.62, which corresponds to a 16% difference. All of these values are well in the range of the few existing estimates of the migration elasticity in the literature. For example, using a model similar to (21), Caliendo et al. (2021) estimate $\nu = 2.43$ in the context of migration between European countries during a similar period.

3.5 Migration Costs and Distance

I have estimated migration costs as origin-specific fixed effects. I now investigate how these migration costs vary with common measures of distance between regions. In particular, I express the migration costs κ_{jk} as a function of the euclidean distance between the population centroids of each region, as well as measures of bilateral travel times on the road network and a dummy for whether the two regions are contiguous. I rely on geo-referenced maps of the Brazilian road network from the Brazilian Ministry of Transportation for the year 2010 and compute travel time measures between each pair of regions using the Open Source Routing Machine. I estimate the following regression by OLS:

$$\kappa_{jk} = \beta_1 \log dist_{jk} + \beta_2 \log travelttime_{jk} + \beta_3 contiguous_{jk} + e_{jk}. \quad (27)$$

Table 3 reports the results from estimating (27) in each period. Migration costs are increasing with travel time and distance, and are smaller between contiguous regions. My estimates are slightly smaller in the second period, indicating migration costs have become slightly less dependent on distance and travel time. For comparison, the last two columns show the coefficients obtained after regressing the migration

costs estimated from the model with no information frictions. The role of distance and travel time appear significantly more pronounced.

4 Testing Predictions of the Model

In this section, I present two exercises that illustrate the success of my model at describing migration patterns. In the first exercise, I uncover a set of new facts on migration patterns in Brazil. I show that the migration elasticity with respect to wages is decreasing with the distance between regions, increasing with the intensity of past migration flows connecting them, and increasing with the internet penetration at the origin. I replicate the same empirical exercise in my model and show that the same patterns arise, with similar magnitudes, in the model with information frictions. In the model with complete information, the elasticities are constant. Second, the migration response to observed local positive shocks is more delayed for more distant origins and origins with lower internet penetration. Taken together, these two exercises support both the qualitative and quantitative relevance of the new mechanisms introduced in my model.

Before describing the two exercises, it is worth pointing out that logit models of migration with complete information such as (21) are typically considered quite successful at matching observed migration patterns. This is in large part thanks to their flexibility with respect to the bilateral migration costs κ_{jk} , which allow them to exactly match the average levels of bilateral migration flows between any two regions. Hence, in order to further test the ability of such models to accurately describe migration decisions, I focus on the cross-sectional variation in migration elasticities rather than levels, and on dynamic responses rather than time averages.

4.1 Heterogeneous Migration Elasticities

I start by presenting a simple empirical approach that provides sharply distinct predictions depending on whether information frictions are present or not. Note that in the model with complete information (21), there is a straightforward strategy for recovering the migration elasticity $1/\nu$, described in (26). I can however allow for some heterogeneity in the migration elasticity along some variable Z_{jkt} :

$$\Lambda_{jkt} = \beta_1 \Delta \log w_{jkt} + \beta_2 Z_{jkt} \times \Delta \log w_{jkt} + \beta_3 Z_{jkt} + D_{jk} + e_{jkt}. \quad (28)$$

For any variable Z_{jkt} , the model with complete information predicts that $\beta_2 = \beta_3 = 0$. However in the presence of information frictions, from equation (24), it is clear that the responsiveness of migration flows between two regions j and k with respect to productivity shocks θ_{kt} depends on the responsiveness of mean beliefs to θ_{kt} . If the beliefs held by agents in j about θ_{kt} increase when θ_{kt} increases, their migration elasticity is higher. These posterior beliefs are in turn determined by individual information acquisition and local information sharing.

In the presence of information frictions, the migration elasticity should decrease with distance. If, say, region k is close to region j , so that the migration cost κ_{jk} is small, then agents in j will pay quite a lot of attention to payoffs in k . Hence, upon receiving a recommendation to go to k , agents in j will update their beliefs significantly upward since they know this recommendation is likely to reflect the true

productivity in k . This makes beliefs to close regions more responsive to nearby regions. In addition, when the productivity in this nearby region k is high at time $t - 1$, people in k are more likely to stay in k and less likely to move to j . At the end of $t - 1$ in j , there are fewer people from k , who tend to be pessimistic about k —people who leave their region tend to think their region is less attractive—and the shared beliefs in j about k become more optimistic. Since the productivity in k is persistent, it is likely to be high at time t as well, and agents in j are now likely to think productivity is high too. For these two reasons, beliefs about nearby regions are more responsive to productivity shocks, making the migration elasticity larger for nearby origin-destination pairs.

Second, the migration elasticity should also be increasing with the size of the past migration flows connecting an origin to another destination. In practice, if a large number of people in region j were in region k in the past, they are likely to be informed about this region and could pass along relevant information about the payoffs in k that will make migration more responsive to opportunities in k . In my model, the effect of past flows from k to j on the responsiveness of contemporaneous migration from j to k is happening through local information sharing. If j tends to welcome a large number of people from k over time—maybe because the two regions are geographically close—beliefs in j about k tend to be accurate since they are largely influenced by the beliefs held by people coming from k . When a positive shock happens in region k , the flow from k to j will decrease as more people decide to stay. This translates into an increase in the shared mean beliefs about k in j due to the adjustment of population on the extensive margin.

Third, the migration elasticity should be increasing with local internet penetration. In practice, people with access to internet should have access to cheaper information and decide to gather more accurate information about migration opportunities. This should induce these people to take advantage of positive shocks and make them less likely to move when the conditions in the destination are not favorable. The idea that expanded access to information technology could explain changes in aggregate migration patterns was advanced by Kaplan and Schulhofer-Wohl (2017). In my model, regions with better access to internet also appear to have lower information costs λ_j . They are therefore able to form beliefs that are more accurate on average, so that in particular their beliefs about a region’s payoffs are more likely to be high when the payoffs are high.

In Table 4, I report the results from estimating equation (28) over the period 2008–2014 for three different variables Z_{jkt} : the log distance between the origin and destination, the average fraction of residents with an active internet connection over the period, and the log number of individuals who moved from k to j in the past year. In columns 1, 4, and 7, I report the results from estimating equation (28) with observed migration flows and wages over the period 2008–2014. Columns 2, 5, and 8 contain the results from estimating equation (28) with migration flows and wages predicted by the model using the estimated parameters in the period 2008–2014. Once I solve for the beliefs functions, value functions and mobility rule, I evaluate them at the observed L_{t-1} and recovered θ_t from the data. The interaction coefficients therefore reflect the differential correlation between the wage gap and the omitted expectational errors along the variable Z_{jkt} of interest. Columns 3, 6, and 9 report the results from estimating (28) with migration flows and wages predicted by the model with no information frictions. As expected, the

Table 4: Heterogeneity of Migration Elasticities in the Data and in the Model

	(1) Data	(2) Model Info	(3) Model No Info	(4) Data	(5) Model Info	(6) Model No Info	(7) Data	(8) Model Info	(9) Model No Info
Income gap	0.391 ^a (0.012)	0.342 ^a (0.007)	0.362 ^a (0.008)	0.398 ^a (0.016)	0.306 ^a (0.008)	0.319 ^a (0.009)	0.292 ^a (0.018)	0.241 ^a (0.008)	0.311 ^a (0.008)
Log dist × Inc. gap	-0.012 ^a (0.004)	-0.009 ^a (0.002)	0.002 (0.002)						
Internet × Inc. gap				0.132 ^a (0.015)	0.101 ^a (0.004)	0.003 (0.004)			
Log past flows × Inc. gap							0.003 ^a (0.001)	0.006 ^a (0.001)	0.000 (0.001)
Observations	68,152	68,152	68,152	68,152	68,152	68,152	68,152	68,152	68,152
R^2	0.179	0.515	0.613	0.157	0.513	0.614	0.168	0.582	0.634

^a indicates significant at the 1% level. In parentheses, I report two-way cluster robust standard errors that allow for correlation in regression residuals at the origin-year level and at the destination-year level. I include origin-destination and year fixed effects in each regression. The income gap is the difference in the log average wages between the destination and the origin. The effect of the level of Z_{jkt} is absorbed by the origin-destination fixed effect for both distance and average internet access, and is not reported for log past flows.

coefficients on the interactions from these regressions are all precisely estimated zeros.

In column 1, the estimated coefficient of 0.391 for the income gap corresponds to the migration elasticity between adjacent pairs, while the negative coefficient for the interaction of the wage gap with log distance indicates that a 1 percentage point increase in distance is associated with a -0.012 decrease in the migration elasticity. For a pair of regions separated by the average distance of 500km, the estimated elasticity is therefore 0.3164. Note that this estimate is in line with the average inverse elasticity of $\nu = 3.12$ estimated in Section 3.4. There is, therefore, significant variation of the migration elasticity with distance.³⁹ In column 2, the regression in the model provides strikingly similar results. Distance has a similar negative effect on the migration elasticity, slightly smaller but of the same order of magnitude. I cannot reject that the coefficients are the same at the 10% level. This similarity strongly emphasizes the quantitative relevance of information frictions, as it indicates that most of the decline in the responsiveness of migration flows with distance can be accounted for by the less effective acquisition and transmission of information between remote regions.

In column 4, the coefficient of 0.281 represents the migration elasticity for individuals in regions with no internet penetration at all. The large coefficient of 0.132 indicates that the estimated elasticity is 0.413 for a region with complete internet penetration. In column 5, I report the results from running the same regression in the model, also interacting the income gap with the measure of internet penetration at the origin. Here too, the model predicts a significant effect of having better internet access on the elasticity of

³⁹In their analysis of international migration flows, Bertoli et al. (2020) estimate a static gravity equation and find, similarly to my results, that the migration elasticity is decreasing with distance.

migration. The positive coefficient reflects the strong negative correlation between internet access and λ_j documented in Section 3.3. The fact that the model can generate this positive interaction coefficient with internet access is another success, especially since the local information costs λ_j were estimated as fixed effects without imposing any relationship with respect to internet access. The model is able to account for 76% of the variation of the migration elasticity with local internet access observed in the data.

In column 6, the coefficient of 0.362 corresponds to the estimated elasticity of migration if the log of the past migration flow was zero, namely if virtually all of the individuals in origin j were in k at the previous period. In practice, the fraction of current residents in a region who were in any other particular region k in the previous period is of the order of 10^{-6} , leading to an average migration elasticity of 0.319. In column 6, I report the results from the same regression in the model. As expected, the migration elasticity with respect to wages in a destination is significantly larger when this destination is connected to the origin via large recent migration flows. The effects are about twice as large in the model as in the data, which is not surprising since the past migration flow is exactly the metric governing the adjustment of beliefs through local information sharing in the model. Taken together, these results show that introducing information frictions in the model delivers predictions that match important characteristics of migration flows.

4.2 Delayed Response to Local Shocks

I now confront the predictions of the model regarding the dynamic response of migration flows to local shocks. I show that the model is able to replicate a differential delay observed in the data in the migration response from some regions depending on their distance to the shock and depending on their access to internet. In this section, I rely on an empirical approach developed by Porcher et al. (2024) to study the migration response to local shocks in Brazil.

In the model with perfect information (21), the delay with which individuals react to a positive local shock in another location is determined by the past migration flows between their origin and the location of the shock. If a large number of workers moved from the origin to the location of the shock in the period before the shock, this indicates that a significant number of agents would move to this location were the payoffs to increase. In this case, the migration response would be faster, with a large influx of migrants in the early periods after the shock is realized. In the model with information frictions, the delay in the migration response from a particular origin can vary even between regions that sent similar amounts of migrants to the location of the shock in the previous period. This is because people in two regions sending a similar fraction of migrants to a given destination in a period may have different information about the destination. For example, the past migration flow to the destination from some distant origin may be the same as from another nearby origin if, say, the distant origin experienced a negative shock so that many people decided to leave. In this case, agents from the distant region may not be as well informed about the destination as migrants from closer regions.

In order to describe the speed of the migration response of an origin region j to the destination k following a local shock in k at time t , I define the rate of migration from j to k at $t + s$ with respect to a horizon $(\underline{t}, \bar{t})$ as the fraction of total migrants who move from j to k at $t + s$, relative to the total number of migrants who will move from j to k between $t + \underline{t}$ and $t + \bar{t}$. Denoting by $L_{jkt+s} = p_{jkt+s} L_{jt+s-1}$ the

gross migration flow between j and k at $t + s$, the rate of migration $p_{jkt+s}^{(t,\bar{t})}$ from j to k at $t + s$ with respect to a horizon $(\underline{t}, \bar{t})$ is

$$p_{jkt+s}^{(t,\bar{t})} = \frac{L_{jkt+s}}{\sum_{s'=\underline{t}}^{\bar{t}} L_{jkt+s'}}.$$

I also define the expected time of migration between j and k over a horizon $(\underline{t}, \bar{t})$ as $\sum_{s=0}^{\bar{t}} sp_{jkt+s}^{(t,\bar{t})}$. I then say that region j has a faster migration response to k than j' at the horizon $(\underline{t}, \bar{t})$ if the expected time of migration from j is smaller than from j' . According to this definition, a region j has a faster migration response to k than another region j' if the individuals who migrate from j to k do so earlier than migrants from j' .

For the labor demand shocks listed in Appendix D.3, I consider migration flows to the shocked region in the two years prior or in the four years subsequent to the labor demand shock, so that $\underline{t} = -2$ and $\bar{t} = 4$, and estimate the following regression model:

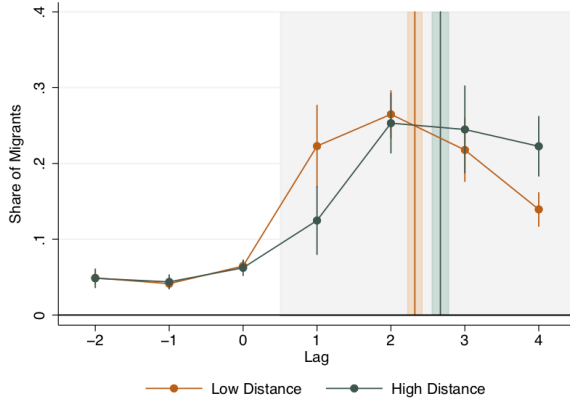
$$p_{jkt+s}^{(t,\bar{t})} = \sum_{s'=-2}^4 \mathbb{1}\{s = s'\} \left(\gamma_{1s'} \text{int}_{jt} + \gamma_{2s'} \log \text{dist}_{jk} + \gamma_{3s'} \log p_{jk}^{\text{past}} \right) + u_{jkt+s}, \quad (29)$$

where t indicates the year in which the labor demand shock in the region of interest took place, int_{jt} is the internet penetration in region j at the year of the shock, dist_{jk} is the distance between j and k , p_{jk}^{past} is the average migration probability from j to k in the three years preceding $t - 2$, u_{jkt+s} captures unobserved determinants of the rate of migration at $t + s$, and $\{\gamma_{nt+s}; n = 1, 2, 3\}$ is the parameter vector of interest.

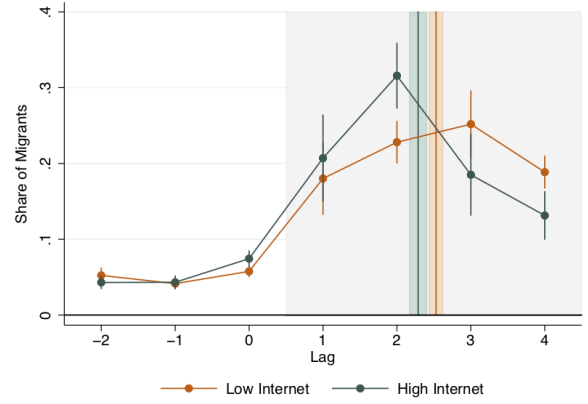
Figure 3a and Figure 3b display the estimates of $\{\gamma_{1t+s}, \gamma_{2t+s}\}$ that indicates how internet access, distance and past migration flows affect the timing of migration. I focus here on the largest labor demand shock in the sample, which took place in Ipojuca in the region of Recife in 2009 following the construction of a large refinery. The figures illustrate, for each year between $t - 2$ and $t + 4$, the predicted migration probability when the corresponding covariate, distance or internet access, is set to its 25th percentile, labeled as “Low”, or to its 75th percentile, labeled as “High”, while other covariates are set to their mean values. The whiskers attached to each dot represent the 95% confidence interval for each predicted migration probability. The vertical lines indicate the expected time of migration between j and k once the shock is realized, and the shaded regions illustrate the corresponding 95% confidence intervals. Everything else equal, workers living in geographically close regions or in regions with higher broadband internet penetration react faster to the positive labor demand shocks happening in Ipojuca.

I then run regression (29) in the model, starting with the model with information frictions. Selecting the destination of interest k to be the mesoregion of Recife, I evaluate the migration flows predicted by the model under the observed population and recovered productivity vectors. The recovered productivity features a persistent increase starting in 2009 (see Appendix D.4), leading to an inflow of migrants similar to what is observed in the data. Although the model predicts that most of the migration should happen in the first period after the shock, Figure 3c illustrates that the estimated coefficients on distance lead to a significant delay for regions that are farther from Recife. Figure 3d shows that the model also predicts a significant delay from regions with lower internet access.

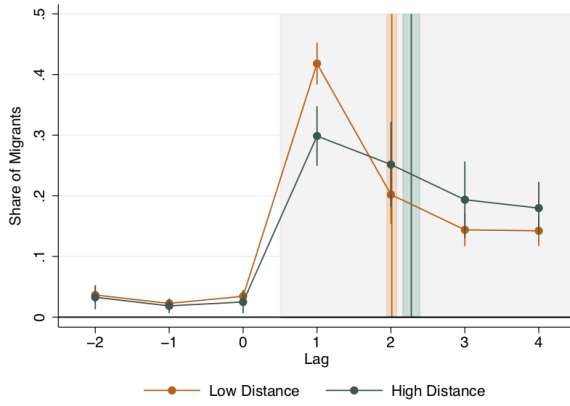
Figure 3: Delay in the Migration Response to the Local Shock in Ipojuca



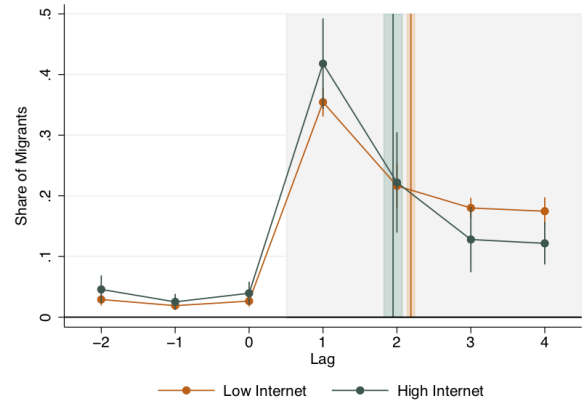
(a) Distance: Data



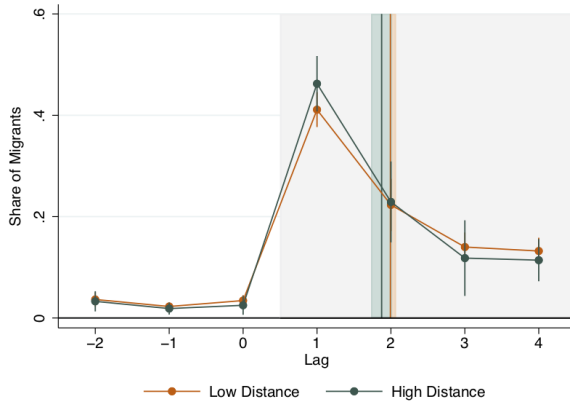
(b) Internet: Data



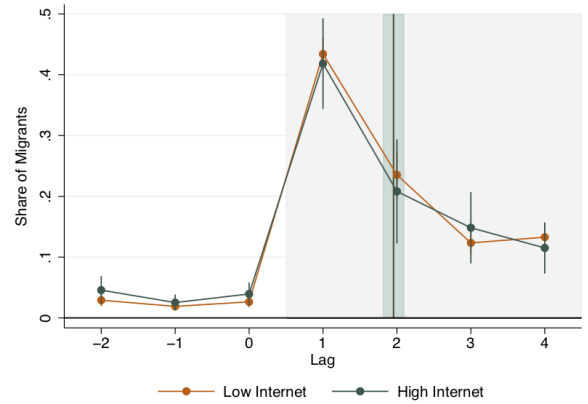
(c) Distance: Model with information frictions



(d) Internet: Model with information frictions



(e) Distance: Model without information frictions



(f) Internet: Model without information frictions

Note: I compute standard errors of the implied expected probability and the expected time of migration with the Delta method, and I cluster standard errors for estimates of the parameter $\{\gamma_{1t+s}, \gamma_{2t+s}\}$ by year and mesoregion of origin.

In the model with no information frictions, Figures 3e and 3f reveal that the migration patterns resulting from the exact same simulated shock do not result in any significant delay along distance or internet access once we control for the past migration shares. This result confirms that the model with information frictions can generate migration patterns similar to those observed in the data, here the differential dynamic response to local shocks. In Appendix D.4, I show that this differential delay holds for other local shocks occurring in Brazil between 2000 and 2014.

5 Counterfactual Exercises

I have discussed several implications of information frictions on migration patterns. The main results of interest are that individuals feature endogenous predispositions towards moving to some regions, leading to lower estimated migration costs, and that their responsiveness to variations in payoffs is limited, leading to heterogeneous migration elasticities and delayed migration responses along geographic distance or internet access. I have verified these predictions in the data, and shown that the magnitude of the effects in the estimated model are close to the observed ones. To further illustrate the quantitative implications of my model for the spatial allocation of workers and welfare, I conduct a number of counterfactual exercises. First, I compare the outcomes in the economy estimated in the first period to a hypothetical one without information frictions. In a second exercise, I reduce information frictions by the amount corresponding to the estimated effect of increased internet access in the second period. For each of the exercises, I decompose the welfare gains into several contributions highlighting the role of the adjustment of the information structure for the overall effects.

5.1 Removing Information Costs

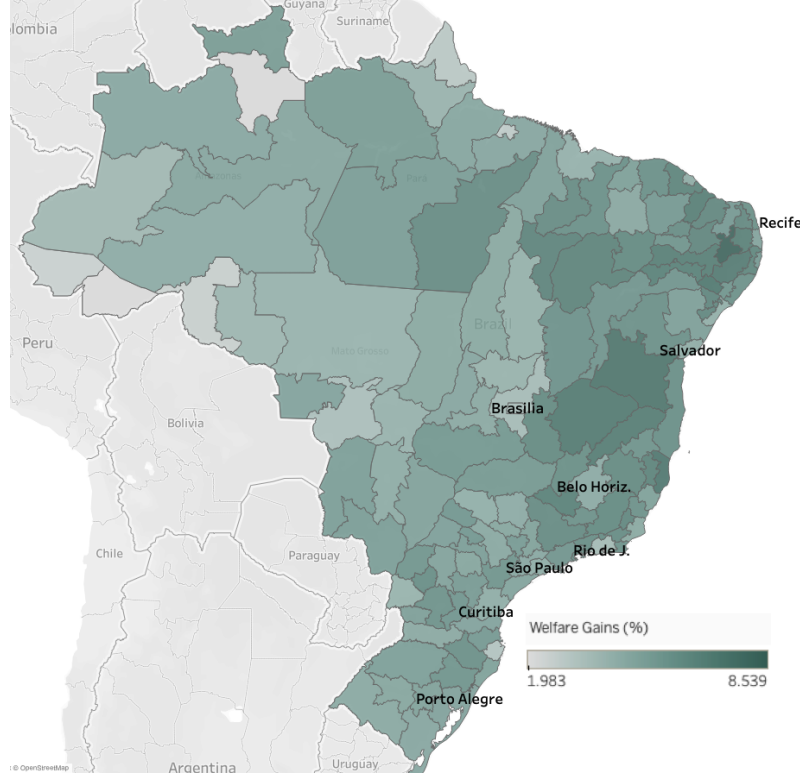
In this first counterfactual exercise, I evaluate the potential gains from completely removing information frictions. This exercise speaks to the initial question formulated in the introduction: what is the scope for improving the spatial allocation of workers by expanding their access to information, while taking into account that information must be acquired at a cost and that it can be shared locally?

To evaluate the effects of removing all information frictions, I first compute the expected value of being in a given region in the steady state of the first period, 2000–2007. In the stochastic steady state, the value of residing in a given region varies over time as different productivity shocks and population vectors are realized. Therefore, I report the expected value over possible realizations of (θ_t, L_{t-1}) in the ergodic distribution. Denoting by $\mathcal{V}_j$ the expected value of a region,

$$\mathcal{V}_j = \mathbb{E}_{\theta_t, L_{t-1}} \left[\sum_k \bar{p}_{jk}(\theta_t, L_{t-1}) (\bar{u}_{jk}(\theta_t, L_{t-1}) + \delta \bar{V}_k(\theta_t, L_{t-1})) - I_j(\pi_j(L_{t-1})) \right]. \quad (30)$$

The expected value $\mathcal{V}_j$ can be computed by simulating the model over a large number of periods, so that the visited states $\{(\theta_t, L_{t-1})\}_t$ are representative of the ergodic distribution of possible states, and taking the simple average of $V_j(\theta_t, L_{t-1})$ over time. I then simulate a new steady state with all the parameters

Figure 4: Welfare Gains by Mesoregion after Setting All Information Costs to Zero



set to their value estimated in the first period, except for the information costs, which I set to zero. In this new stochastic steady state, I compute the expected values $\mathcal{V}_j^*$ in each region.

Figure 4 displays the percentage change in the expected value of each region $\Delta\mathcal{V}_j/\mathcal{V}_j$ between the two equilibria, where I now define $\Delta X = X' - X$. The average welfare gain across regions is 5.55%. There is important heterogeneity across regions, with gains ranging from only 2% in the most remote regions to 8.5% in a region close to Recife in the North-East. By looking at the initial distribution of information costs illustrated in Figure 1a, one might have expected the largest gains to accrue to regions starting with initially high information costs, such as the remote regions in the Amazon and regions of the North-East, and the lowest gains for regions with initially quite low information costs, such as Brasília, São Paulo, and Rio de Janeiro. Figure 4 illustrates that this is mostly verified, although some regions in the Amazon do not seem to benefit as much as expected.

In the steady state with complete information, the standard deviation of earnings across space decreases by 15%. This is a clear illustration of the improved arbitrage of local shocks: a positive local shock attracts more immigrants in this economy, driving the local wage down. Similarly, when a negative productivity shock hits a region, more people leave to other regions offering better payoffs, thereby alleviating the negative wage effects in their origin region.

Perhaps surprisingly, the overall migration flows, computed as the fraction of the population moving to another region each period, decrease by 4% in the model with complete information. This net decline in migration flows masks two countervailing forces, both stemming from the fact that agents make fewer

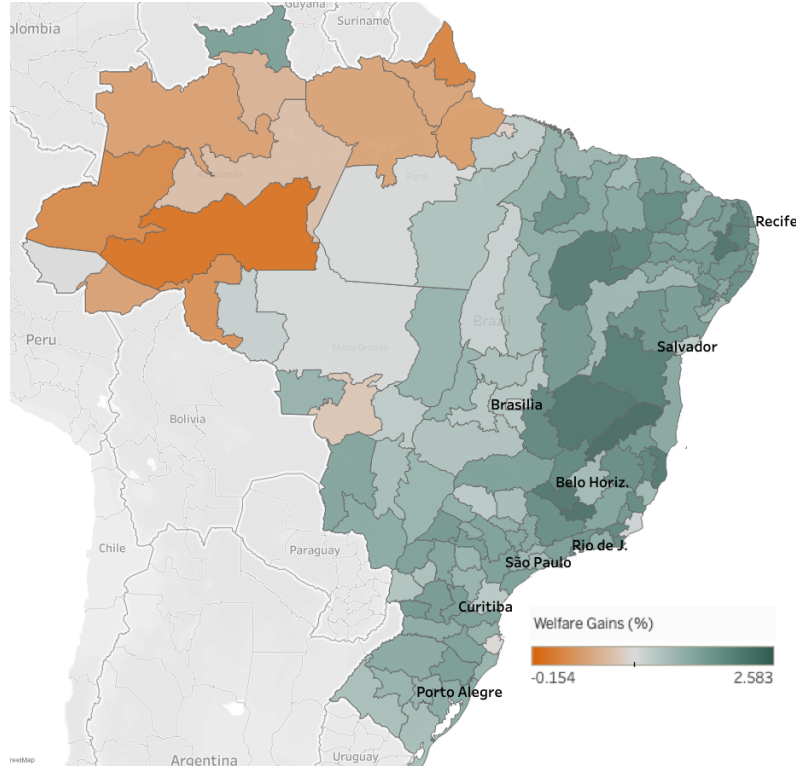
mistakes. First, and counteracting the decline in migration, agents with complete information no longer feature any predisposition towards staying in their current location. In the presence of positive costs of acquiring information about other regions, this predisposition was a reason for agents to stay in their current region. Second, agents now only move to regions that offer high payoffs, and no longer visit a region by mistake, so they move less often. The persistence in the productivity shocks allows workers to benefit from a mobility decision for several periods, reducing their propensity to move. This second effect appears to be dominating and leads to the 4% decline in overall geographic mobility. Note that this decline in migration flows corresponds to a comparison of the two steady states, once each region's population has reached its ergodic distribution. One may expect that the transition from the first steady state to the second may lead to a temporary increase in mobility flows as workers relocate to the regions offering higher average payoffs.

To describe the forces at play even further, I decompose the welfare gains $\Delta\mathcal{V}_j$ into the contribution of three intuitive channels. Denote by $\bar{U}_j = (\bar{u}_{jk} + \delta\bar{V}_k)_{k=1,\dots,J}$ the vector composed of the sum of flow payoffs and future values for all possible destinations and states, (θ_t, L_{t-1}) . Similarly, denote by $\bar{p}_j = (\bar{p}_{jk})_{k=1,\dots,J}$ the vector composed of the mobility probabilities to all destinations for all states. I can express the difference in expected value in a region j from (30) as

$$\Delta\mathcal{V}_j = \underbrace{\mathbb{E}_{\theta,L} [\Delta\bar{p}_j \cdot \bar{U}'_j]}_{\text{better sorting}} + \underbrace{\mathbb{E}_{\theta,L} [\bar{p}_j \cdot \Delta\bar{U}_j]}_{\text{better outcomes}} - \underbrace{\mathbb{E}_L [\Delta I_j]}_{\text{lower info cost}} . \quad (31)$$

The first term in the decomposition represents the welfare gains arising from better sorting of agents in the second equilibrium, relative to the initial equilibrium. It represents the expected gains in utility coming from the different mobility choices made by agents in the new equilibrium, maintaining the payoffs at their new value $\bar{U}'_j$. This better sorting can come about because agents have better information, leading them to choose the locations offering the highest payoffs. The second term reflects the gains due to the change in the payoffs themselves. The reallocation of population across regions can change the average wages offered in a region. Moreover, since the future value of residing in a region incorporates the expected payoff resulting from future mobility decisions, it can increase if the quality of the information available in the region has improved. The third term corresponds to the gains from spending less effort to acquire information. This decline can arise for two reasons. First, the cost of acquiring a given amount of information (λ_j) is cheaper. The second reason is due to the change in the precision of agents' prior beliefs. If the residents of the region have better information, local information sharing will allow agents to start with better information, so that they need to acquire less on their own. I find that 21% of the 5.5% welfare gains can be attributed to the “better sorting” channel, 58% to “better outcomes”, and 21% to “lower information costs”. It is remarkable that most of the gains seem to come from the response of mobility and earnings, so that the mechanical effect of lowering the information cost λ_j to zero accounts at most for about 20% of the gains.

Figure 5: Welfare Gains by Mesoregion after Expansion of Internet Access



5.2 Effect of the Expansion of Internet Access

I now evaluate a counterfactual decrease in information costs in each region by a magnitude equal to the estimated contribution of increases in local internet access. I use the measure of internet penetration observed in every region in 2014, the last year of the sample. Starting from the equilibrium in the first period, I decrease the information cost by an amount equal to the local internet access in 2014 (assuming that internet penetration was zero in the years 2000–2007), multiplied by the estimated coefficient ℓ_1 obtained after projecting the information costs on internet access. The average reduction in the information cost is about 0.73 units, starting from an initial average of 3.11.

Before describing the results, it is worth emphasizing that this exercise assumes that the only effect of increased local internet penetration is to allow workers to gather information more easily. It is likely that the expansion of internet access for households has been accompanied by a parallel expansion in access for firms, and that local firms' performance may have been altered by internet services. There is an extensive literature studying the effects of the development of information and communication technologies (ICT) on how production is organized. Hence, the results reported below should be interpreted as the effects of a hypothetical policy that would reduce information costs by a magnitude similar to the result of expanded internet access, but without affecting the local productivity process.

Figure 5 depicts the geographic distribution of welfare gains from the counterfactual exercise. The average welfare gains amount to 1.63%. The standard deviation of earnings across space decreases by 4.01%, illustrating the better arbitrage of local shocks in the economy with lower information costs. The

average decomposition of the welfare gains into the three channels described in Section 5.1 is almost exactly the same as in the previous exercise: better sorting, better outcomes and lower information costs account for 22%, 57% and 21% of the average welfare gains respectively.

In contrast to the previous counterfactual exercise, there are some regions that experience a mild decline in expected value in the equilibrium with lower information costs. This is the case for a few sparsely populated regions in the North-West of the country. Interestingly, these negative effects arise even though these regions have benefitted from increased internet access—although to a lesser extent than most other regions. For instance, the region of Manaus experiences a decline in welfare of 0.15%, despite experiencing an increase in internet penetration of 23 percentage points.

To understand why some regions do not benefit from the episode of internet expansion, it is useful to decompose their welfare gains into the three channels described above. Focusing again on the example of Manaus, the contribution of information costs is actually negative, equal to -0.33 . This means that agents in Manaus have to spend more to gather information than they did before. This happens because the information they can obtain from their local network has deteriorated. Indeed, in the new equilibrium, the population of Manaus has decreased by 7%, reflecting the fact that fewer workers from other regions now decide to locate there. Workers are now better informed and rarely find it optimal to move to the remote region of Manaus. In the initial equilibrium, Manaus would welcome workers with relatively good information who may have made a “mistake” by moving there. With fewer well informed visitors, local information sharing has become less effective in Manaus, and workers located there need to acquire more information on their own. Agents end up holding less precise information even after individual acquisition, and their migration decisions lead them to regions offering lower payoffs on average. This is reflected by a negative contribution of -0.36 of the “sorting” channel. Only the “outcomes” channel is working in favor of workers in Manaus, with a contribution of 0.54, as the lower population in the region tends to increase the average wage.

One additional reason why outcomes do not improve enough in Manaus to compensate for the decline due to worse information is that the information cost has decreased more in most other regions. As a result, when a positive local shock is realized, the people from better informed regions move faster to the location of the shock—as discussed in Section 4.2—and reap the benefits in the form of high wages before more people arrive and put downward pressure on wages.

6 Conclusion

In this paper, I propose a theory of migration under incomplete information. In my model, information about opportunities in other regions, earnings, and mobility patterns are all determined in equilibrium. I model, in a tractable way, both the incentives that agents have for acquiring information about some regions, and the possibility for workers to benefit from the information circulating in their local networks. When agents face costs of acquiring information, they collect limited information about regions offering a priori lower expected payoffs, making them less likely to move to these regions. Accounting for these endogenous default rules that hinder mobility appears to be quantitatively important for the estimation

of bilateral migration costs.

I show that my model with information frictions can rationalize the observed heterogeneity in migration elasticities, as well as the differential delay in migration responses to local shocks from origins that are more distant or benefit from lower internet access. More generally, the model can help explain why the net inflow of migrants in response to positive local shocks often appears limited. Agents in regions where information about these shocks is more difficult to obtain, either because internet access is limited or because local networks can provide little relevant information, will be less likely to respond. I discuss how policies that could reduce information frictions can generate important welfare gains. Information acquired individually can be shared with other agents, inducing wide-ranging benefits from this positive externality. Yet, the distributional consequences from such policy interventions are not trivial. Some regions risk becoming “information traps” where agents struggle to gather accurate information.

The role of information frictions in migration decisions is likely to be more complex than the model I put forward in this paper. For example, local information sharing is likely to be more prominent between individuals of the same demographic characteristics and working in similar occupations. Incorporating a richer description of local interactions could lead to interesting insights on migration decisions. Finally, studying an empirical setting featuring a clear distinction between payoffs observed and unobserved by workers could help further identify the contribution of information frictions.

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Online Appendix for “Migration with Costly Information”

Charly Porcher

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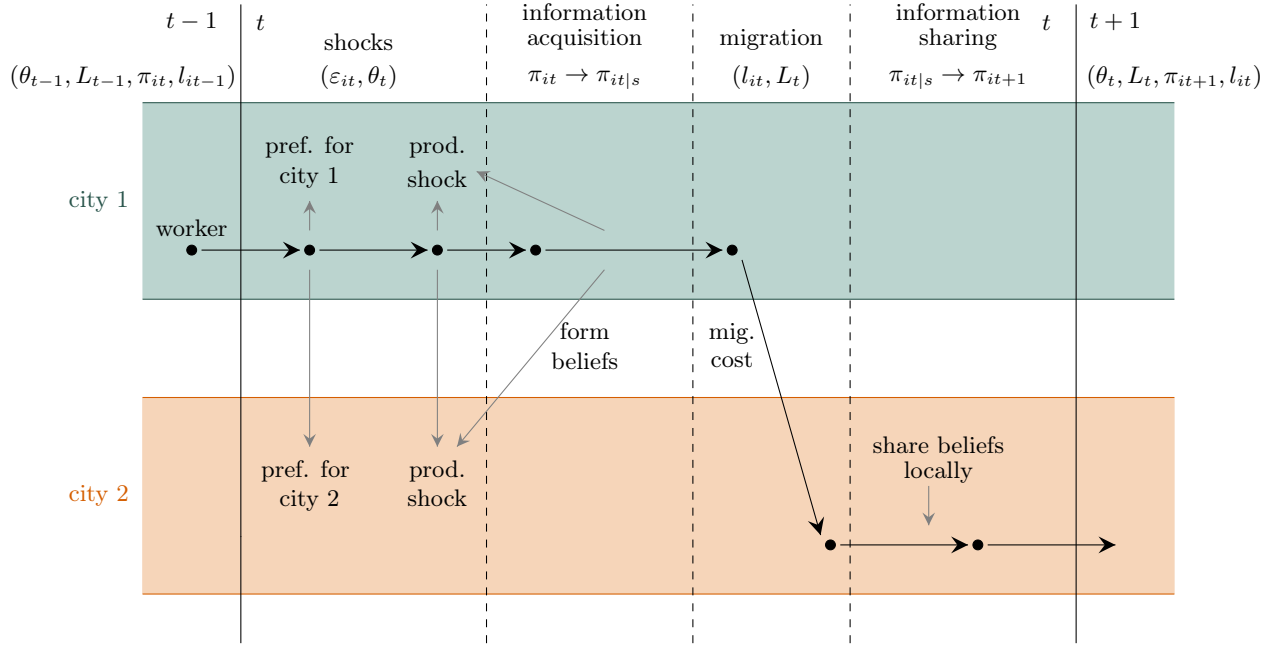


Figure A.1: Sequence of actions during any period t . Example of a worker moving from city 1 to city 2.

A Appendix to Section 2

A.1 Timeline of Actions

As represented in Figure A.1, each period has four steps.

A.2 Proofs

A.2.1 Proof of Lemma 1

Proof. Adapted from Lemma 1 in Steiner et al. (2017). The argument exploits the concavity of the Shannon entropy to show that the dynamic problem over signal-action strategies (f, σ) is equivalent to the reduced problem over the induced choice rule p .

Let (f, σ) be any feasible strategy, and let p denote the corresponding choice rule:

$$p_{l_{it-1}l_{it}}(\theta_t, L_{t-1}, \bar{\pi}_{l_{it-1}t}, \varepsilon_{it}) = \Pr(\sigma_{it}(s_{it}, L_{t-1}, \bar{\pi}_{l_{it-1}t}, l_{it-1}, \varepsilon_{it}) = l_{it} \mid \theta_t, L_{t-1}, \bar{\pi}_{l_{it-1}t}, l_{it-1}, \varepsilon_{it}).$$

Because (f, σ) and p generate the same distribution of actions conditional on states, they yield identical streams of expected gross utilities. Hence, to compare the two problems it suffices to compare their associated information costs.

Consider an agent i located in j at the start of period t , with common prior $\bar{\pi}_{jt}$. From (9), the period- t information cost under (f, σ) is

$$I(f_t | \omega_{it}) = \lambda_j [H(\bar{\pi}_{jt}) - \mathbb{E}_s [H(\pi_{it|s})]],$$

while the cost associated with the induced rule p is, from (16),

$$I(p_t | \bar{\pi}_{jt}) = \lambda_j \left[H(\bar{\pi}_{jt}) - \sum_k q_{jkt}(\omega_{it}) H(\pi_{it|k}) \right],$$

where $q_{jkt}(\omega_{it}) = \int_{\theta} p_{jkt}(\theta, \omega_{it}) \bar{\pi}_{jt}(\theta) d\theta$ is the ex-ante probability that the signal realization corresponds to recommendation k .

Since the realized location l_{it} is measurable with respect to the signal s_{it} , the posterior $\pi_{it|l_{it}=k}$ can be written as a convex combination of the posteriors $\pi_{it|s}$ over all signal realizations s :

$$\pi_{it|k}(\cdot) \equiv \Pr(\theta_t \in \cdot \mid l_{it} = k, \omega_{it}) = \int_{S_k} \pi_{it|s}(\cdot) \Pr(s \mid l_{it} = k, \omega_{it}) ds,$$

By concavity of Shannon entropy $H(\cdot)$,

$$H(\pi_{it|k}) \geq \int_{S_k} H(\pi_{it|s}) \Pr(s \mid l_{it} = k, \omega_{it}) ds.$$

Multiplying by $q_{jkt}(\omega_{it})$ and summing over k yields

$$\sum_k q_{jkt}(\omega_{it}) H(\pi_{it|k}) \geq \sum_k q_{jkt}(\omega_{it}) \int_{S_k} H(\pi_{it|s}) \Pr(s \mid l_{it} = k, \omega_{it}) ds.$$

Since $q_{jkt}(\omega_{it}) = \Pr(l_{it} = k \mid \omega_{it})$,

$$q_{jkt}(\omega_{it}) \Pr(s \mid l_{it} = k, \omega_{it}) = \Pr(s, l_{it} = k \mid \omega_{it}).$$

Moreover, $l_{it} = k$ if and only if $s \in S_k$, so $\sum_k \Pr(s, l_{it} = k \mid \omega_{it}) = \Pr(s \mid \omega_{it})$. Therefore,

$$\sum_k q_{jkt}(\omega_{it}) \int_{S_k} H(\pi_{it|s}) \Pr(s \mid l_{it} = k, \omega_{it}) ds = \int_S H(\pi_{it|s}) \Pr(s \mid \omega_{it}) ds = \mathbb{E}_s[H(\pi_{it|s})].$$

We conclude that

$$\mathbb{E}_s[H(\pi_{it|s})] \leq \sum_k q_{jkt}(\omega_{it}) H(\pi_{it|k}),$$

as claimed. It follows that

$$I(p_t | \bar{\pi}_{jt}) \leq I(f_t | \omega_{it}),$$

and therefore

$$\mathbb{E} \left[\sum_{t=1}^{\infty} \delta^t I(p_t | \bar{\pi}_{jt}) \right] \leq \mathbb{E} \left[\sum_{t=1}^{\infty} \delta^t I(f_t | \omega_{it}) \right].$$

Hence the discounted expected payoff under the reduced-form rule p is at least as large as that under the original strategy (f, σ) .

Conversely, given any choice rule p , one can construct signal and action mappings (f, σ) that reproduce the same joint distribution over (θ_t, l_{it}) and the same information cost (16). Thus the discounted expected payoff associated with (f, σ) equals the value of the objective in (15) given p . Combining the two directions establishes the equivalence claimed in the lemma. $\square$

A.2.2 Proof of Proposition 1

Proof. The existence proof in Steiner et al. (2017) relies on Weierstrass's theorem on a *compact* strategy space, obtained as a product of finite sets; compactness of the product is ensured by Tychonoff's theorem. In our setting, the state includes continuously distributed, *unbounded* components (θ, ε) , and the one-period utility is *unbounded*, so the compactness and boundedness preconditions that support the Weierstrass argument are unavailable. In addition, the population allocation L and the shared beliefs $\bar{\pi}$ are *endogenous aggregates* that evolve with the population's choices. We therefore prove existence via a two-step construction.

Step A (auxiliary control problem). Fix a candidate law of motion Ψ for the endogenous aggregates $(L, \bar{\pi})$ (generated by the population's policy profile), and analyze the *individual* discounted control problem with $(L, \bar{\pi})$ treated as state variables evolving according to Ψ . Because each individual is atomistic, a single agent takes Ψ as given. We show that this auxiliary problem satisfies the hypotheses of Hernández-Lerma and Lasserre (1996, Thm. 8.3.6): compact actions, continuity in actions, and a per-period cost bounded by some weight function w , where w itself is bounded in terms of drift. The theorem then yields that the Bellman operator is a contraction on

the weighted space $\mathbb{L}_w$, the value function V^* is *unique*, and a measurable *optimal decision rule* exists.

Step B (consistency of aggregates). Using the solution of the auxiliary problem, we construct a correspondence that maps any conjectured aggregate law Ψ into a *best-response* population policy, and—via the population and belief laws of motion—back into a *new* aggregate law. We verify that this correspondence is nonempty, convex-valued, and upper hemicontinuous on a compact domain; hence, by Kakutani–Fan–Glicksberg, there exists a fixed point Ψ^* consistent with the induced policy. At such a fixed point, the individual optimal policy and the induced evolution of $(L, \bar{\pi})$ jointly constitute a dynamic rational inattention equilibrium.

Step A: Auxiliary Control Problem.

We start by setting up the worker’s auxiliary problem as a Markov control model, introducing the notation used in (Hernández-Lerma and Lasserre, 1996, henceforth HLL). After Lemma 1, each agent’s information design is represented by a *mixed action*—a probability distribution over destinations—and an associated entropy cost. Let the state be $x = (\theta, L, \bar{\pi}, j, \varepsilon) \in X$, and let actions $a \in A(x)$ belong to the simplex $A(x) = \Delta(\mathcal{J})$. Each $a = (a_k)_{k \in \mathcal{J}}$ represents the probabilities of moving to each destination k . The one-stage cost (the expected negative utility plus information cost) is then

$$c(x, a) = - \sum_{k \in \mathcal{J}} a_k [A_k + \theta_k - \alpha \log L_k + B_k - \kappa_{jk} + \nu \varepsilon_k] + \lambda_j \left(H(\bar{\pi}_j) - \sum_{k \in \mathcal{J}} a_k H(\pi_{|k}(x, a)) \right). \quad (\text{A.1})$$

In the auxiliary control problem, we treat the evolution of the endogenous aggregates $(L, \bar{\pi})$ as given by an exogenous *aggregate transition rule*. Specifically, let

$$\tilde{p} : (\theta, L, \bar{\pi}) \mapsto L', \quad \tilde{q} : (\theta, L, \bar{\pi}) \mapsto \bar{\pi}',$$

denote measurable mappings that specify how the current aggregate state and productivity shock determine next-period population and belief aggregates. The pair $\Psi = (\tilde{p}, \tilde{q})$ therefore represents a conjectured law of motion for the endogenous states. Intuitively, $\tilde{p}$ summarizes how migration flows translate the current population distribution into the next one, while $\tilde{q}$ captures how shared beliefs are updated across periods.

These mappings are taken as given when solving the individual decision problem in Step A; they close the agent’s state transition but are not yet required to reflect equilibrium behavior. In Step B below, we impose *consistency* by requiring that the aggregate law of motion generated by the optimal individual policies coincide with the conjectured pair $(\tilde{p}, \tilde{q})$. The fixed-point argument in Step B establishes that such a self-consistent aggregate transition rule exists.

Given such a candidate $\Psi = (\tilde{p}, \tilde{q})$, the transition kernel of the individual state $x' = (\theta', L', \bar{\pi}', j', \varepsilon')$ is

$$Q_\Psi(\cdot \mid x, a) = \Gamma(\cdot \mid \theta) \otimes \text{Cat}(a) \otimes \Xi \otimes \delta_{\tilde{p}(\theta, L, \bar{\pi})} \otimes \delta_{\tilde{q}(\theta, L, \bar{\pi})}, \quad (\text{A.2})$$

where:

- $\Gamma(\cdot \mid \theta)$ is the AR(1) law of motion for the productivity vector θ ;
- $\text{Cat}(a)$ is the categorical distribution governing the next location j' implied by the mixed action a ;
- Ξ is the law of the i.i.d. shocks ε' ; and
- the two Dirac measures $\delta_{\tilde{p}(\theta, L, \bar{\pi})}$ and $\delta_{\tilde{q}(\theta, L, \bar{\pi})}$ fix the next-period aggregates $(L', \bar{\pi}')$ according to the conjectured transition rule Ψ .

Given the one-stage cost $c(x, a)$ and the discount factor $\alpha \in (0, 1)$, the *discounted total cost* under a (possibly randomized) policy p is

$$V^p(x) \equiv \mathbb{E}_x^p \left[\sum_{t=0}^{\infty} \alpha^t c(x_t, a_t) \right], \quad V^*(x) \equiv \inf_p V^p(x),$$

where $\mathbb{E}_x^p[\cdot]$ denotes expectation under the law of motion induced by the policy p and the transition kernel Q_Ψ . Define the *dynamic programming operator* T by

$$(Tv)(x) \equiv \inf_{a \in A(x)} \left\{ c(x, a) + \alpha \int v(y) Q_\Psi(dy \mid x, a) \right\}.$$

The corresponding α -discounted cost optimality equation (DCOE) is

$$V^*(x) = \inf_{a \in A(x)} \left\{ c(x, a) + \alpha \int V^*(y) Q_\Psi(dy | x, a) \right\}, \quad (\text{A.3})$$

where V^* is the optimal value function.

To prove that the auxiliary control problem admits a solution, we rely on the general existence result of Hernández-Lerma and Lasserre (1996, Section 8.3, pp. 45–48). In this section we restate the relevant assumptions and theorem in a form adapted to our notation, and we discuss the economic meaning of each requirement. The goal is to verify in the next subsection that our auxiliary problem satisfies them.

Assumption 8.3.1 (Regularity of state, action, and kernel). *For every state $x \in X$:*

- (a) *the admissible action set $A(x)$ is compact;*
- (b) *the one-stage cost function $c(x, a)$ is lower semicontinuous in $a \in A(x)$;*
- (c) *for every bounded measurable function u on X , the map $a \mapsto \int u(y) Q_\Psi(dy | x, a)$ is continuous on $A(x)$.*

These conditions ensure that, for each state, the set of feasible actions is closed and bounded, the instantaneous cost does not jump upward with small changes in the control, and the transition kernel behaves continuously with respect to the control variable. Together, they guarantee that minimization over $a \in A(x)$ is well posed and that the Bellman operator is measurable and continuous in standard topologies.

Assumption 8.3.2 (Cost and drift bounds). *There exist constants $\bar{c} \geq 0$ and β with $1 \leq \beta < 1/\alpha$, and a measurable weight function $w : X \rightarrow [1, \infty)$ such that for every $x \in X$,*

$$\sup_{a \in A(x)} |c(x, a)| \leq \bar{c} w(x), \quad (\text{Growth Bound}) \quad (\text{A.4})$$

$$\sup_{a \in A(x)} \int w(y) Q_\Psi(dy | x, a) \leq \beta w(x). \quad (\text{Drift Bound}) \quad (\text{A.5})$$

The first inequality bounds the instantaneous cost by a multiple of the weight $w(x)$; the second bounds the expected growth of w across transitions, acting as a *drift condition* that prevents the process from exploding in expectation. When w is constant, this reduces to the standard assumption that costs are uniformly bounded; here, the weighted version allows unbounded costs and states while still keeping the discounted problem well behaved. The requirement $\alpha\beta < 1$ ensures that the expected discounted future weight shrinks geometrically, which is the key to establishing contraction of the Bellman operator.

Assumption 8.3.3 (Continuity of the expected weight). *For every $x \in X$, the function $a \mapsto \int w(y) Q_\Psi(dy | x, a)$ is continuous on $A(x)$.*

This assumption guarantees that small perturbations in the control do not cause discontinuous jumps in the expected future weight, which is needed for measurable selection and for continuity of the Bellman operator under the weighted norm.

Proposition 8.3.9 (a) and Theorem 8.3.6 (b) (HLL, p. 48). *Suppose that Assumptions 8.3.1, 8.3.2 and 8.3.3 hold. Let β be the constant in (A.5), and define $\gamma \equiv \alpha\beta$. Then:*

- (a) *T is a contraction operator on $\mathbb{L}_w(X)$ with modulus γ ; that is, T maps $\mathbb{L}_w(X)$ into itself and*

$$\|Tu - Tu'\|_w \leq \gamma \|u - u'\|_w, \quad \forall u, u' \in \mathbb{L}_w(X), \quad \|u\|_w \equiv \sup_{x \in X} \frac{|u(x)|}{1 + w(x)},$$

- (b) *There exists a measurable rule $p^* : X \rightarrow A(x)$ such that $p^*(x)$ attains the minimum in (A.3) for every x , i.e.,*

$$V^*(x) = c(x, p^*(x)) + \alpha \int V^*(y) Q_\Psi(dy | x, p^*(x)) \quad \forall x \in X,$$

The measurable rule p^ is therefore an optimal (time-invariant) policy for the discounted problem.*

Proposition 8.3.9 (a) shows that under the growth and drift bounds (A.5) the Bellman operator contracts distances between value functions by the factor $\gamma = \alpha\beta < 1$. Hence T has a unique fixed point V^* in the weighted space $\mathbb{L}_w(X)$, and successive applications of T converge geometrically to this limit. Theorem 8.3.6 (b) then ensures that there exists a measurable optimal decision rule f^* attaining the minimum in the DCOE. Economically, this guarantees that even with unbounded productivity shocks and idiosyncratic tastes, the combination of discounting and mean-reverting state dynamics keeps the dynamic problem well defined and yields a unique optimal mobility rule.

Verification of Assumptions 8.3.1–8.3.3.

8.3.1(a) Compactness of the action set. For every state $x \in X$, the admissible action set is the probability simplex $A(x) = \Delta(\mathcal{J}) = \left\{ a = (a_k)_{k \in \mathcal{J}} \in \mathbb{R}^J : a_k \geq 0, \sum_{k \in \mathcal{J}} a_k = 1 \right\}$. This set is a closed and bounded subset of the finite-dimensional Euclidean space $\mathbb{R}^J$, hence compact. Moreover, $A(x)$ does not depend on x , so compactness holds uniformly for all states.

8.3.1(b) Lower semicontinuity of $c(x, a)$ in a . Fix $x = (\theta, L, \bar{\pi}, j, \varepsilon)$ and examine each term of $c(x, a)$ in (A.1). The first summation is affine in a , hence continuous and lower semicontinuous on $A(x)$. The entropy component involves a through $-\lambda_j \sum_k a_k H(\pi_{|k}(x, a))$. For fixed x , each posterior $\pi_{|k}(\cdot | x, a)$ is obtained from Bayes' rule applied to the prior $\bar{\pi}_j(\theta)$ and to the conditional signal distribution $f_k(s | \theta, a)$ associated with choice k . Under the Shannon information structure, the densities $f_k(s | \theta, a)$ depend smoothly on the choice probabilities a_k and are uniformly bounded. Because both $\bar{\pi}_j$ and f_k are continuous and bounded, the posteriors $\pi_{|k}(\cdot | x, a)$ vary continuously in $L^1(\Theta)$ with a . Since the integrand $-\pi \log \pi$ is continuous and dominated by an integrable function, the Shannon entropy $H(\pi_{|k}(x, a))$ is continuous in a by the dominated convergence theorem. Consequently, $a \mapsto a_k H(\pi_{|k}(x, a))$ is continuous for each k , and so is the finite sum $\sum_k a_k H(\pi_{|k}(x, a))$. Combining the payoff and entropy components, $c(x, a)$ is continuous (and therefore lower semicontinuous) in a for every fixed x .

8.3.1(c) Continuity of the kernel in a . Let $u : X \rightarrow \mathbb{R}$ be any bounded measurable function and define

$$u'(x, a) \equiv \int u(y) Q_\Psi(dy | x, a).$$

From the definition of the kernel $Q_\Psi(\cdot | x, a)$ in (A.2), the only factor depending on a is the categorical distribution $\text{Cat}(a)$ that governs the next location j' . Therefore,

$$u'(x, a) = \sum_{k \in \mathcal{J}} a_k \int u(\theta', \tilde{p}(\theta, L, \bar{\pi}), \tilde{q}(\theta, L, \bar{\pi}), k, \varepsilon') (\Gamma \otimes \Xi)(d\theta', d\varepsilon').$$

This expression is an affine function of a because the integrand does not depend on a . Hence $u'(x, a)$ is linear, and in particular continuous, in a for every bounded u and every state x . This verifies Assumption 8.3.1(c).

8.3.2 Growth bound on the cost function. Assumption 8.3.2 requires the existence of a measurable *weight function* $w : X \rightarrow [1, \infty)$ and constants $\bar{c} \geq 0$ and $\beta \in [1, 1/\alpha)$ such that, for every $x \in X$, (A.4) and (A.5) are satisfied.

The first condition bounds the instantaneous cost by the weight w , while the second ensures that the expected value of w under the transition kernel does not grow faster than a factor β . Together, these inequalities guarantee that the expected discounted cost remains finite and that the Bellman operator maps $\mathbb{L}_w(X)$ into itself.

We define the following weight function,

$$w(x) \equiv 1 + \|\theta\|^2 + \|\varepsilon\|^2 + \|\log L\|^2, \tag{A.6}$$

where for any vector $z = (z_1, \dots, z_J)$, $\|z\|^2 \equiv \sum_{k=1}^J z_k^2$ denotes the squared Euclidean norm. The function w is measurable, satisfies $w(x) \geq 1$, and grows quadratically in the unbounded components of the state x .

We first show that the absolute value of the one-stage cost is bounded by a multiple of $w(x)$ uniformly over actions a . From the definition of $c(x, a)$ in (A.1),

$$|c(x, a)| \leq \sum_k a_k |A_k + \theta_k - \alpha \log L_k + B_k - \kappa_{jk} + \nu \varepsilon_k| + \lambda_j \left| H(\bar{\pi}_j) - \sum_k a_k H(\pi_{|k}(x, a)) \right|.$$

Focusing first on the payoff component, note that by the triangle inequality and $\sum_k a_k = 1$,

$$\sum_{k \in \mathcal{J}} a_k \left| A_k + \theta_k - \alpha \log L_k + B_k - \kappa_{jk} + \nu \varepsilon_k \right| \leq \sum_{k \in \mathcal{J}} a_k \left(|A_k + B_k - \kappa_{jk}| + |\theta_k| + \alpha |\log L_k| + \nu |\varepsilon_k| \right).$$

Define $C_0(j) \equiv \max_{k \in \mathcal{J}} |A_k + B_k - \kappa_{jk}|$ and $C_0 \equiv \max_{j \in \mathcal{J}} C_0(j)$, so that $\sum_k a_k |A_k + B_k - \kappa_{jk}| \leq C_0(j) \leq C_0$. For the remaining terms $z \in (\theta, \varepsilon, \log L)$, use $|z| \leq \frac{1}{2}(1 + z^2)$ and the facts that $\sum_k a_k = 1$ and $a_k \geq 0$ to obtain $\sum_k a_k |z_k| \leq \frac{1}{2}(1 + \|z\|^2)$. Combining these bounds gives

$$\begin{aligned} \sum_k a_k \left| A_k + \theta_k - \alpha \log L_k + B_k - \kappa_{jk} + \nu \varepsilon_k \right| &\leq \left(C_0 + \frac{1+\alpha+\nu}{2} \right) + \frac{1}{2} \|\theta\|^2 + \frac{\alpha}{2} \|\log L\|^2 + \frac{\nu}{2} \|\varepsilon\|^2 \\ &\leq \bar{c}_1 w(x), \end{aligned}$$

with the choice $\bar{c}_1 \equiv C_0 + \frac{1+\alpha+\nu}{2} + \max\{\frac{1}{2}, \frac{\alpha}{2}, \frac{\nu}{2}\}$. This bound is uniform in $a \in \Delta(\mathcal{J})$ and $j \in \mathcal{J}$, and depends only on the primitives $(A, B, \kappa, \alpha, \nu)$.

Let us now turn to the Entropy component of $|c(x, a)|$. We work with differential (Shannon) entropy on $\Theta \equiv \mathbb{R}^J$,

$$H(\pi) = - \int_{\Theta} \pi(\theta) \log \pi(\theta) d\theta.$$

Let $\lambda_{\max} \equiv \max_{j \in \mathcal{J}} \lambda_j < \infty$ denote the largest information-cost weight, and let $\ell_{\max}(\Sigma)$ denote the largest eigenvalue of a covariance matrix Σ . Under Assumption 2, for each k we have $\theta_{kt} = \rho \theta_{k,t-1} + \xi_{kt}$ with $\xi_{kt} \sim \mathcal{N}(0, \sigma_{\xi}^2)$ and $|\rho| < 1$, so the stationary variance $\bar{\sigma}^2 = \sigma_{\xi}^2 / (1 - \rho^2)$ is finite. Since conditioning cannot increase variance, for all priors $\bar{\pi}_j$ and posteriors $\pi_{|k}$,

$$\ell_{\max}(\text{Cov}[\theta_t | \bar{\pi}_j]) \leq \bar{\sigma}^2, \quad \ell_{\max}(\text{Cov}[\theta_t | \pi_{|k}]) \leq \bar{\sigma}^2.$$

By the maximum-entropy property of the Gaussian,

$$H(\pi) \leq \frac{1}{2} \log((2\pi e)^J \det \Sigma_{\pi}) \leq \frac{J}{2} \log(2\pi e \ell_{\max}(\Sigma_{\pi})).$$

Using the uniform eigenvalue bound gives the global constant $\bar{h} \equiv \frac{J}{2} \log(2\pi e \bar{\sigma}^2)$ such that $H(\bar{\pi}_j) \leq \bar{h}$ and $H(\pi_{|k}(x, a)) \leq \bar{h}$ for all j, k .

Since conditioning reduces entropy, $\sum_k a_k H(\pi_{|k}(x, a)) \leq H(\bar{\pi}_j)$, and hence

$$0 \leq H(\bar{\pi}_j) - \sum_k a_k H(\pi_{|k}(x, a)) \leq H(\bar{\pi}_j) \leq \bar{h}.$$

Multiplying by λ_j and using $\lambda_j \leq \lambda_{\max}$ yields

$$0 \leq \lambda_j \left(H(\bar{\pi}_j) - \sum_k a_k H(\pi_{|k}(x, a)) \right) \leq \lambda_{\max} \bar{h} =: \bar{c}_2.$$

Combining payoff and entropy bounds, set $\bar{c} \equiv \max\{\bar{c}_1, \bar{c}_2\}$ to obtain

$$|c(x, a)| \leq \bar{c} w(x), \quad \forall (x, a) \in X \times A(x),$$

which verifies the growth bound (A.4).

8.3.2 Drift bound on the weighth function. We now show that the expected next-period weight grows at most linearly in $w_K(x)$, where we define the offset weight

$$w_K(x) \equiv K + w(x), \quad K \geq 0.$$

Under the kernel $Q_{\Psi}(\cdot | x, a)$, the next state $x' = (\theta', L', \pi', j', \varepsilon')$ satisfies:

- $\theta' \sim \Gamma(\cdot | \theta)$ follows the AR(1) process $\theta' = \rho\theta + \xi$, with $\xi \sim \mathcal{N}(0, \sigma_{\xi}^2 I_J)$;

- $\varepsilon' \sim \Xi$ is i.i.d. across periods with finite second moment;
- $L' = \tilde{p}(\theta, L, \bar{\pi})$ and $\bar{\pi}' = \tilde{q}(\theta, L, \bar{\pi})$ are deterministic functions of the current aggregates.

Hence,

$$\mathbb{E}[w(x') \mid x, a] = 1 + \mathbb{E}[\|\theta'\|^2 \mid \theta] + \mathbb{E}[\|\varepsilon'\|^2] + \sum_k (\log L'_k)^2.$$

Because in equilibrium each location receives a positive mass of agents, we can restrict attention to aggregate laws of motion that are *uniformly positive*: there exists $\eta \in (0, 1]$ such that for all $(\theta, L, \bar{\pi})$, $L'_k = \tilde{p}_k(\theta, L, \bar{\pi}) \geq \eta \bar{L}$ for every k . Then $|\log L'_k| \leq C_L(\eta) \equiv \max\{|\log(\eta \bar{L})|, |\log \bar{L}|\}$ and $\sum_k (\log L'_k)^2 \leq \bar{C}_L(\eta) \equiv J C_L(\eta)^2$. With $\bar{c}_\varepsilon \equiv \mathbb{E}\|\varepsilon'\|^2 < \infty$,

$$\mathbb{E}[w(x') \mid x, a] \leq 1 + J\sigma_\xi^2 + \bar{c}_\varepsilon + \bar{C}_L(\eta) + \rho^2 \|\theta\|^2 = \bar{c}_6(\eta) + \rho^2 \|\theta\|^2,$$

where $\bar{c}_6(\eta) \equiv 1 + J\sigma_\xi^2 + \bar{c}_\varepsilon + \bar{C}_L(\eta) < \infty$. Since $w(x) \geq \|\theta\|^2$, it follows that

$$\sup_{a \in A(x)} \int w(y) Q_\Psi(dy \mid x, a) \leq \bar{c}_6(\eta) + \rho^2 w(x).$$

To obtain a multiplicative drift bound with coefficient strictly below $1/\alpha$, consider the offset weight $w_K(x)$. Then

$$\sup_a \int w_K(y) Q_\Psi(dy \mid x, a) \leq (\bar{c}_6(\eta) + (1 - \rho^2)K) + \rho^2 w_K(x).$$

For any $\beta \geq \rho^2$, this implies

$$\sup_a \int w_K(y) Q_\Psi(dy \mid x, a) \leq \beta w_K(x)$$

whenever $\bar{c}_6(\eta) + (1 - \rho^2)K \leq (\beta - \rho^2)w_{K,\min}$, where $w_{K,\min} \equiv \min_x w_K(x) = K + 1$. Equivalently,

$$\beta \geq \rho^2 + \frac{\bar{c}_6(\eta) + (1 - \rho^2)K}{K + 1}.$$

Hence, for any $\beta_\star \in [1, 1/\alpha)$, choosing

$$K \geq K_\star(\eta) \equiv \frac{\bar{c}_6(\eta) - (\beta_\star - \rho^2)}{\beta_\star - 1}$$

guarantees that the drift inequality (A.5) holds for w_K , with $\beta = \beta_\star < 1/\alpha$. Because $w(x) \leq w_K(x)$, the growth bound (A.4) also holds for w_K . Therefore, Assumption 8.3.2 is satisfied.

8.3.3 Continuity of the expected weight. Fix $x = (\theta, L, \bar{\pi}, j, \varepsilon) \in X$ and define

$$g(a) \equiv \int w(y) Q_\Psi(dy \mid x, a), \quad a \in A(x) = \Delta(\mathcal{J}).$$

From the definition of the transition kernel $Q_\Psi(\cdot \mid x, a)$ in (A.2), all components are independent of a except for the next-location coordinate j' , which follows the categorical distribution $\text{Cat}(a)$. Because the weight $w(y)$ in (A.6) depends on $(\theta', L', \bar{\pi}', \varepsilon')$ but not on j' , integration over $\text{Cat}(a)$ eliminates any dependence on a :

$$g(a) = \int w(\theta', L', \bar{\pi}', j', \varepsilon') \Gamma(d\theta' \mid \theta) \text{Cat}(a)(dj') \Xi(d\varepsilon') = \int w(\theta', \tilde{p}(\theta, L, \bar{\pi}), \tilde{q}(\theta, L, \bar{\pi}), \varepsilon') \Gamma(d\theta' \mid \theta) \Xi(d\varepsilon').$$

The right-hand side is independent of a ; hence $g(a)$ is constant on $A(x)$ and therefore continuous. This verifies Assumption 8.3.3.

Conclusion of Step A. Assumptions 8.3.1–8.3.3 are satisfied for the auxiliary control problem. By Theorem 8.3.6 of Hernández-Lerma and Lasserre (1996), the Bellman operator T is a contraction on the weighted space $\mathbb{L}_w(X)$; the value function V^* is the unique fixed point of T and the unique solution of the discounted cost optimality equation (A.3); and there exists a measurable optimal policy attaining the infimum. This completes Step A.

Step B: Consistency of Aggregates.

We begin by defining the spaces of policies and aggregate laws over which the fixed point is established.

Spaces of policies and aggregate laws. Let $\mathcal{P}$ denote the compact, convex set of measurable mixed mobility rules $p : X \rightarrow \Delta(\mathcal{J})$ endowed with the topology of convergence in measure. Let

$$\mathcal{S} := \left\{ \Psi = (\tilde{p}, \tilde{q}) : \tilde{p} : (\theta, L, \bar{\pi}) \mapsto L' \in \mathcal{L}, \tilde{q} : (\theta, L, \bar{\pi}) \mapsto \bar{\pi}' \in (\Delta\Theta)^J, \text{ continuous and bounded} \right\},$$

the set of continuous candidate laws of motion for the aggregate state $(L, \bar{\pi})$, where $\mathcal{L} = \{L \in \mathbb{R}_+^{J \times J} : \sum_{j,k} L_{jk} = \bar{L}\}$. Under the bounded-moment restrictions and compact support adopted in Step A, $\mathcal{S}$ is a compact, convex subset of the Banach space of continuous functions endowed with the uniform norm.

From mobility rules to aggregate laws. By Lemma 1, any optimal rational-inattention strategy (f, σ) can be represented without loss by a *mobility rule* $p : X \rightarrow \Delta(\mathcal{J})$, for which the canonical one-signal-per-action implementation induces posteriors

$$\pi_{j|k}(\theta) = \frac{p_{jk}(\theta, L, \bar{\pi}) \bar{\pi}_j(\theta)}{\int p_{jk}(\tilde{\theta}, L, \bar{\pi}) \bar{\pi}_j(\tilde{\theta}) d\tilde{\theta}}.$$

Hence we can work directly with these implementable mobility rules p , knowing that they are supported by a rational-inattention structure.

Given $(\theta, L, \bar{\pi})$ and a candidate rule p , define:

(i) *Population update*:

$$L'_{jk} = L_j \bar{p}_{jk}(\theta, L, \bar{\pi}), \quad \bar{p}_{jk}(\theta, L, \bar{\pi}) := \mathbb{E}_\varepsilon [p_{jk}(\theta, L, \bar{\pi}, \varepsilon)],$$

and write $L' = \tilde{p}(\theta, L, \bar{\pi}, p)$.

(ii) *Belief update*:

$$\begin{aligned} \pi_{j|k}^+(\theta') &= \int \gamma(\theta' | \theta) \pi_{j|k}(\theta) d\theta, \\ \bar{\pi}'_k &= \text{Agg} \left(\{(\pi_{j|k}^+, L'_{jk}/L'_k)\}_{j=1}^J \right), \end{aligned}$$

and write $\bar{\pi}' = \tilde{q}(\theta, L, \bar{\pi}, p)$, where **Agg** is the local information-sharing rule from (13).

Define the aggregate-law map

$$\mathcal{G}(p) := (\tilde{p}(\cdot, \cdot, \cdot, p), \tilde{q}(\cdot, \cdot, \cdot, p)).$$

Each component of $(\theta, L, \bar{\pi}, p) \mapsto (L', \bar{\pi}')$ is continuous: (i) $p \mapsto \bar{p}_{jk}$ by dominated convergence; (ii) $(p, \bar{\pi}_j) \mapsto \pi_{j|k}$ by dominated convergence with denominators bounded away from zero on the compact support used in Step A/B; (iii) $\pi_{j|k} \mapsto \pi_{j|k}^+$ by continuity of Gaussian convolution; and (iv) **Agg** by Assumption 3. Hence $\mathcal{G}$ is single-valued and continuous, mapping the compact convex policy space $\mathcal{P}$ into the compact convex space of aggregate laws $\mathcal{S}$.

Fixed point over aggregate laws. For any $\Psi \in \mathcal{S}$, Step A ensures a nonempty, convex, compact set $\mathcal{B}(\Psi)$ of optimal mobility rules for the auxiliary control problem with kernel Q_Ψ . Define the correspondence

$$\Phi : \mathcal{S} \rightrightarrows \mathcal{S}, \quad \Phi(\Psi) := \{ \mathcal{G}(p) : p \in \mathcal{B}(\Psi) \}.$$

Nonemptiness and convexity follow from the existence of measurable minimizers and the convexity of $A(x) = \Delta(\mathcal{J})$. Upper hemicontinuity of Φ follows from Berge's Maximum Theorem in the weighted setting and the continuity of $\mathcal{G}$. By the Kakutani–Fan–Glicksberg fixed-point theorem, there exists $\Psi^* \in \mathcal{S}$ and $p^* \in \mathcal{B}(\Psi^*)$ such that $\Psi^* = \mathcal{G}(p^*)$. By Lemma 1, p^* is supported by a rational-inattention structure, and the posteriors used in $\tilde{q}$ are exactly those induced by the canonical one-signal-per-action implementation of p^* . Hence (Ψ^*, p^*) jointly constitute a dynamic rational-inattention equilibrium, completing the proof of Proposition 1. $\square$

A.2.3 Proof of Proposition 2

We prove Proposition 2 with the help of three additional lemmas.

Following Steiner et al. (2017), the dynamic rational inattention problem (15) can be rewritten as a control problem with observable states in which the agent pays an explicit cost for deviating from a default choice rule. Lemma 2 establishes this equivalence and shows that it applies to any stochastic equilibrium, including the stochastic steady-state equilibrium characterized in Proposition 2.

Lemma 3 then shows that, conditional on any fixed default rule q' , the optimal choice probabilities that solve (A.7) take the dynamic logit form (17), and that the associated value function satisfies the corresponding log-sum representation. For the remainder of the section, we omit the individual index i for notational simplicity.

Finally, Lemma 4 shows that in a stochastic steady state, the mobility rule and the continuation value can be written as functions of $(\theta_t, L_{t-1}, \varepsilon_{it})$ only. This conclusion relies on the restricted time dependence of the shared beliefs implied by Assumption 3, together with the Locally Invariant Posteriors property of Shannon-entropy rational inattention problems derived in Caplin et al. (2022).

Lemma 2. *A stochastic mobility rule $p = (p_\tau)_\tau$, together with some default rule $q = (q_\tau)_\tau$, solves the dynamic RI problem if and only if it solves:*

$$\max_{q,p} \mathbb{E} \left[\sum_{t=1}^{\infty} \delta^t (u_{l_{it-1}l_{it}}(\theta_t, \omega_{it}) + \lambda_{l_{it-1}} (\log q_{l_{it-1}l_{it}t}(\omega_{it}) - \log p_{l_{it-1}l_{it}t}(\theta_t, \omega_{it}))) \right]. \quad (\text{A.7})$$

where the expectation is with respect to the joint distribution generated by $\bar{\pi}$, p , and $\gamma(\cdot|\theta)$.

Proof of Lemma 2. Recall that, according to Lemma 1, the dynamic RI problem is equivalent to finding an optimal mobility rule:

$$\max_p \mathbb{E} \left[\sum_{t=1}^{\infty} \delta^t (u_{l_{it-1}l_{it}}(\theta_t, \omega_{it}) - I(\omega_{it})) \right], \quad (\text{A.8})$$

where the information cost is expressed as a function of the prior and posterior beliefs $\pi_{it|k} \equiv \pi_{it|s_t=k}$,

$$I(\omega_{it}) = \lambda_{l_{it-1}} \left(H(\bar{\pi}_{l_{it-1}t}) - \sum_k q_{l_{it-1}kt}(\omega_{it}) H(\pi_{it|k}) \right), \quad (\text{A.9})$$

and for $j = l_{it-1}$, $q_{jkt}(\omega_{it}) = \int_{\theta} p_{jkt}(\theta, \omega_{it}) \bar{\pi}_{jt}(\theta) d\theta$ is the ex-ante probability of receiving the recommendation to move to k .

As noted by Steiner et al. (2017), by symmetry of the entropy, instead of expressing the information cost as a function of the uncertainty about θ_t before and after receiving the signal, we can rewrite it as a function of the uncertainty about the distribution of mobility choices l_t before and after observing θ_t . Formally, for any random variables X, Y, Z and the entropy H defined in (8), the conditional mutual information to which the information cost is proportional, is defined as

$$\mathcal{I}(X; Y|Z) \equiv H(X|Z) - \mathbb{E}_Y[H(X|Y, Z)|Z],$$

and satisfies $\mathcal{I}(X; Y|Z) = \mathcal{I}(Y; X|Z)$. Applying this property to $X = \theta_t$, $Y = s$ and $Z = \omega_{it}$, we note that $(X|Z) = (\theta_t|\omega_{it}) \sim \bar{\pi}_{l_{it-1}t}$, $(Y|Z) = (s|\omega_{it}) \sim q_{l_{it-1}t}(\omega_{it})$, $(X|Y, Z) = (\theta_t|s, \omega_{it}) \sim \pi_{it|s}$, $(Y|X, Z) = (s|\theta_t, \omega_{it}) \sim p_{l_{it-1}t}(\theta_t, \omega_{it})$, we get from (A.9) that

$$\begin{aligned} I(\omega_{it}) &= \lambda_{l_{it-1}} \mathcal{I}(\theta_t; s|\omega_{it}) \\ &= \lambda_{l_{it-1}} \mathcal{I}(s; \theta_t|\omega_{it}) \\ &= \lambda_{l_{it-1}} (H(q_{l_{it-1}t}(\omega_{it})) - \mathbb{E}_{\theta} [H(p_{l_{it-1}t}(\theta, \omega_{it}))|\omega_{it}]). \end{aligned} \quad (\text{A.10})$$

Recall that the expectation in (A.8) is taken over all possible future paths, so that the location choice l_{it} in period t is given inside the expectation. Given Lemma 1, the distribution of signals s conditional on l_{it} is degenerate, with $s_{it} = l_{it}$. Hence, the entropy inside the expectation in (A.10), after conditioning on the realization of l_{it} , is equal to $-\log p_{l_{it-1}l_{it}t}(\theta_t, \omega_{it})$.

We express the first entropy term in (A.10) as the result of a maximization problem by using the property of properness of the entropy. Properness implies that for any random variable X with finite support $\mathcal{S}$ and distribution

$g(x) \in \Delta(\mathcal{S})$,

$$H(g) = - \max_{h \in \Delta(\mathcal{S})} \mathbb{E}_g [\log h(x)].$$

To interpret the properness property, we can consider an agent who believes that X is distributed according to g and is offered to report a distribution $h \in \Delta(\mathcal{S})$ before observing the realization of X , with the promise of a reward of $\log h(x)$ if the realized value is x . Properness implies that the truthful report $h = g$ maximizes the expected reward. The use of properness relies on the information cost being proportional to the reduction in entropy.

We express the first entropy term in (A.10) using the properness of Shannon entropy. For any distribution g on a finite set $\mathcal{S}$,

$$H(g) = - \max_{h \in \Delta(\mathcal{S})} \mathbb{E}_g [\log h(X)].$$

Applying this to the distribution $q_{l_{it-1} \cdot t}(\omega_{it})$ of s_t conditional on ω_{it} , we obtain

$$\begin{aligned} H(q_{l_{it-1} \cdot t}(\omega_{it})) &= - \max_{q' \in \Delta(\mathcal{J})} \mathbb{E}_{s \sim q_{l_{it-1} \cdot t}(\omega_{it})} [\log q'_{l_{it-1} st}(\omega_{it})] \\ &= - \max_{q' \in \Delta(\mathcal{J})} \mathbb{E} [\log q'_{l_{it-1} l_{it} t}(\omega_{it}) \mid \omega_{it}], \end{aligned}$$

where in the second line we used that, by Lemma 1, the signal is degenerate at $s_{it} = l_{it}$ conditional on ω_{it} . Conditional on the realized location l_{it} , the information cost can therefore be written as

$$\begin{aligned} I(\omega_{it} \mid l_{it}) &= \lambda_{l_{it-1}} \left(H(q_{l_{it-1} \cdot t}(\omega_{it})) - \mathbb{E}_\theta [-\log p_{l_{it-1} l_{it} t}(\theta_t, \omega_{it}) \mid \omega_{it}] \right) \\ &= \max_{q' \in \Delta(\mathcal{J})} \mathbb{E}_\theta \left[\lambda_{l_{it-1}} \left(-\log q'_{l_{it-1} l_{it} t}(\omega_{it}) + \log p_{l_{it-1} l_{it} t}(\theta_t, \omega_{it}) \right) \mid \omega_{it} \right]. \end{aligned}$$

Substituting this expression for the information cost into (A.8) gives the result in (A.7). $\square$

Lemma 3. *Given any default rule q' , the control problem with fixed q' :*

$$\max_p \mathbb{E} \left[\sum_{t=1}^{\infty} \delta^t \left(u_{l_{t-1} l_t t}(\theta_t, \omega_t) + \lambda_{l_{t-1}} \left(\log q'_{l_{t-1} l_t t}(\omega_t) - \log p_{l_{t-1} l_t t}(\theta_t, \omega_t) \right) \right) \right], \quad (\text{A.11})$$

has the following optimal mobility rule:

$$p_{jkt}(\theta_t, \omega_t) = \frac{q'_{jkt}(\omega_t) \exp(u_{jkt}(\theta_t, \omega_t) + \delta \bar{V}_{kt+1}(\theta_t, \omega_t))^{1/\lambda_j}}{\sum_l q'_{jlt}(\omega_t) \exp(u_{jlt}(\theta_t, \omega_t) + \delta \bar{V}_{lt+1}(\theta_t, \omega_t))^{1/\lambda_j}} \quad (\text{A.12})$$

and the value function satisfies:

$$V_{jt}(\theta_t, \omega_t) = \lambda_j \log \left(\sum_l q'_{jlt}(\omega_t) \exp(u_{jlt}(\theta_t, \omega_t) + \delta \bar{V}_{lt+1}(\theta_t, \omega_t))^{1/\lambda_j} \right).$$

Proof of Lemma 3. For a given default rule $q'_{jkt}(\omega_t)$, let the continuation value in the control problem for q' be given by:

$$V_{l_{t-1} t}(\theta_t, \omega_t) = \max_{p_\tau} \sum_{\tau=t}^{\infty} \delta^{\tau-t} \mathbb{E}_{\theta_\tau} \left[\sum_k p_{l_{\tau-1} k \tau}(\theta_\tau, \omega_\tau) \left(u_{l_{\tau-1} k \tau}(\theta_\tau, \omega_\tau) + \lambda_{l_{\tau-1}} \left(\log q'_{l_{\tau-1} k \tau}(\omega_\tau) - \log p_{l_{\tau-1} k \tau}(\theta_\tau, \omega_\tau) \right) \right) \right],$$

where the expectation is conditional on (θ_t, ω_t) . The value V_{jt} satisfies the recursion:

$$V_{jt}(\theta_t, \omega_t) = \max_{p_t} \sum_k p_{jkt}(\theta_t, \omega_t) \left(u_{jkt}(\theta_t, \omega_t) + \lambda_j \left(\log q'_{jkt}(\omega_t) - \log p_{jkt}(\theta_t, \omega_t) \right) + \delta \bar{V}_{kt+1}(\theta_t, \omega_t) \right), \quad (\text{A.13})$$

where $\bar{V}_{kt+1}(\theta_t, \omega_t) = \mathbb{E}[V_{kt+1}(\theta_{t+1}, \omega_{t+1}) | \theta_t, \omega_t]$.

To solve the maximization problem in (A.13), we can write the first-order condition with respect to $p_{jkt}(\theta_t, \omega_t)$:

$$u_{jkt}(\theta_t, \omega_t) + \lambda_j (\log q'_{jkt}(\omega_t) - (\log p_{jkt}(\theta_t, \omega_t) + 1)) + \delta \bar{V}_{kt+1}(\theta_t, \omega_t) = \mu_{jt}(\theta_t, \omega_t),$$

where $\mu_{jt}(\theta_t, \omega_t)$ is the Lagrange multiplier associated with the constraint $\sum_k p_{jkt}(\theta_t, \omega_t) = 1$. Rearranging the first-order condition gives:

$$p_{jkt}(\theta_t, \omega_t) = \exp(\log q'_{jkt}(\omega_t) - 1 + 1/\lambda_j (u_{jkt}(\theta_t, \omega_t) + \delta \bar{V}_{kt+1}(\theta_t, \omega_t) - \mu_{jt}(\theta_t, \omega_t))).$$

Since $\sum_k p_{jkt}(\theta_t, \omega_t) = 1$, we obtain:

$$\begin{aligned} p_{jkt}(\theta_t, \omega_t) &= \frac{\exp(\log q'_{jkt}(\omega_t) - 1 + 1/\lambda_j (u_{jkt}(\theta_t, \omega_t) + \delta \bar{V}_{kt+1}(\theta_t, \omega_t) - \mu_{jt}(\theta_t, \omega_t)))}{\sum_l \exp(\log q'_{jlt}(\omega_t) - 1 + 1/\lambda_j (u_{jlt}(\theta_t, \omega_t) + \delta \bar{V}_{lt+1}(\theta_t, \omega_t) - \mu_{jt}(\theta_t, \omega_t)))} \\ &= \frac{q'_{jkt}(\omega_t) \exp(u_{jkt}(\theta_t, \omega_t) + \delta \bar{V}_{kt+1}(\theta_t, \omega_t))^{1/\lambda_j}}{\sum_l q'_{jlt}(\omega_t) \exp(u_{jlt}(\theta_t, \omega_t) + \delta \bar{V}_{lt+1}(\theta_t, \omega_t))^{1/\lambda_j}}. \end{aligned}$$

Substituting into (A.13) gives the recursion:

$$\begin{aligned} V_{jt}(\theta_t, \omega_t) &= \sum_k p_{jkt}(\theta_t, \omega_t) \lambda_j \log \left(\sum_l q'_{jlt}(\omega_t) \exp(u_{jlt}(\theta_t, \omega_t) + \delta \bar{V}_{lt+1}(\theta_t, \omega_t))^{1/\lambda_j} \right) \\ &= \lambda_j \log \left(\sum_l q'_{jlt}(\omega_t) \exp(u_{jlt}(\theta_t, \omega_t) + \delta \bar{V}_{lt+1}(\theta_t, \omega_t))^{1/\lambda_j} \right) \sum_k p_{jkt}(\theta_t, \omega_t) \\ &= \lambda_j \log \left(\sum_l q'_{jlt}(\omega_t) \exp(u_{jlt}(\theta_t, \omega_t) + \delta \bar{V}_{lt+1}(\theta_t, \omega_t))^{1/\lambda_j} \right). \end{aligned}$$

□

Lemma 4. *In the stochastic steady state, in each period $t \geq 0$, the shared beliefs $\{\bar{\pi}_k\}_{k \in \mathcal{J}}$ about θ_t , defined in Assumption 3, only depend on the previous period population composition, L_{t-1} .*

Proof of Lemma 4. Fix a stochastic steady-state equilibrium and a period $t \geq 1$. Throughout the proof, we focus on beliefs about the payoff-relevant state θ_t held by a *representative mover* at each origin j . Following Assumption 3, the shared beliefs $\bar{\pi}_{kt}$ in location k after local information sharing at time t are constructed from the beliefs of such representative movers who arrived in k between $t-1$ and t . We take the representative mover to be the agent with idiosyncratic preference shocks equal to zero, $\varepsilon_{it} = 0$; this is the type whose posteriors enter the aggregation rule in Assumption 3.

We first describe the representative mover's dynamic problem in period $t-1$ and show that the payoff environment is fully summarized by L_{t-1} . Fix an origin location $j \in \mathcal{J}$. Consider the representative agent at j at the beginning of period $t-1$, with prior belief $\bar{\pi}_{j,t-1}$ about θ_{t-1} and population composition L_{t-2} . Given the equilibrium mobility rule and value functions, the representative's continuation problem from $t-1$ onward can be written as

$$\max_{\{f_\tau, \sigma_\tau\}_{\tau \geq t-1}} \mathbb{E} \left[\sum_{\tau=t-1}^{\infty} \delta^{\tau-(t-1)} \left(u_{l_{i,\tau-1}l_{i\tau}}(\theta_\tau, L_{\tau-1}, 0) - I(f_\tau | \omega_{i\tau}) \right) \middle| \theta_{t-1}, L_{t-2}, \bar{\pi}_{j,t-1} \right],$$

subject to the law of motion for $(\theta_\tau, L_{\tau-1})$ and Bayes' rule. Conditional on the realized population composition L_{t-1} in period $t-1$, the one-period payoff index for choosing destination k from origin j in that period can be written as

$$x_{jk,t-1}(\theta_{t-1}; L_{t-1}) \equiv u_{jk}(\theta_{t-1}, L_{t-1}, 0) + \delta \bar{V}_k(\theta_{t-1}, L_{t-1}),$$

where $\bar{V}_k(\theta_{t-1}, L_{t-1})$ denotes the equilibrium expected continuation value from period t onward, conditional on the realization (θ_{t-1}, L_{t-1}) at the end of period $t-1$. This index is a deterministic function of θ_{t-1} and L_{t-1} , together with time-invariant parameters of preferences and technology. From the standpoint of the representative agent in period $t-1$, the *payoff environment* is thus fully summarized by L_{t-1} and the family of functions $\{x_{jk,t-1}(\cdot; L_{t-1})\}_{k \in \mathcal{J}}$.

We now cast the representative mover's problem in period $t-1$ as a static RI problem and invoke the property of Locally Invariant Posteriors.

In period $t-1$, conditional on L_{t-1} , the representative agent's choice of information and action can be viewed as a static rational inattention problem with prior $\bar{\pi}_{j,t-1}$ over θ_{t-1} , payoff indices $\{x_{jk,t-1}(\cdot; L_{t-1})\}_{k \in \mathcal{J}}$, and information cost given by a Shannon-entropy mutual information term with coefficient $\lambda_j > 0$:

$$I(\omega_{i,t-1}) = \lambda_j \left(H(\bar{\pi}_{j,t-1}) - \mathbb{E}_{\theta_{t-1}} [H(\pi_{j,t-1}(\cdot | s_{t-1}))] \right),$$

where $H(\cdot)$ denotes Shannon entropy and $\pi_{j,t-1}(\cdot | s_{t-1})$ is the posterior belief over θ_{t-1} induced by signal s_{t-1} .

Let $Q_{j,t-1}$ denote an optimal information strategy for this static RI problem at period $t-1$, i.e., a distribution over posterior beliefs $\pi \in \Delta(\Theta)$ that is consistent with Bayes' rule under the prior $\bar{\pi}_{j,t-1}$. Let $\text{supp } Q_{j,t-1}$ denote the set of posterior beliefs used with positive probability under $Q_{j,t-1}$; this set is the *posterior support* or *posterior family* of the optimal strategy.

We now invoke a key result from Caplin et al. (2022). They show that the Shannon mutual information cost admits a *weakly uniformly posterior-separable* representation and satisfies the *Locally Invariant Posteriors* property. In particular, their Theorem 1 (stated here in adapted notation) implies the following:

Theorem (Caplin et al., 2022, Theorem 1) *Consider a static rational inattention problem with state θ , prior m , action set $\mathcal{K}$, and payoff indices $\{u_k(\theta)\}_{k \in \mathcal{K}}$. Suppose the information cost is weakly uniformly posterior-separable (a condition satisfied by the Shannon mutual-information cost). Let Q be an optimal information strategy for prior m , with posterior support $\mathcal{S} = \text{supp } Q$, and satisfying Bayesian plausibility $m = \int_{\mathcal{S}} \pi Q(d\pi)$. Then for any other prior m' in the convex hull of $\mathcal{S}$, there exists an optimal information strategy Q' for prior m' whose posterior support is also $\mathcal{S}$. In particular, the set $\mathcal{S}$ of posterior beliefs is determined solely by the payoff indices $\{u_k(\cdot)\}_{k \in \mathcal{K}}$ and is independent of the specific prior within the convex hull of $\mathcal{S}$.*

In our setting, the state is θ_{t-1} , the prior is $m = \bar{\pi}_{j,t-1}$, the actions are destinations $k \in \mathcal{J}$, the payoff indices are $x_{jk,t-1}(\cdot; L_{t-1})$, and the information cost is the Shannon mutual information with coefficient λ_j . The optimal strategy $Q_{j,t-1}$ at period $t-1$ satisfies Bayesian plausibility:

$$\bar{\pi}_{j,t-1} = \int_{\text{supp } Q_{j,t-1}} \pi Q_{j,t-1}(d\pi),$$

so $\bar{\pi}_{j,t-1}$ belongs to the convex hull of $\text{supp } Q_{j,t-1}$. Hence Theorem 1 applies: conditional on the payoff environment summarized by L_{t-1} and $\{x_{jk,t-1}(\cdot; L_{t-1})\}_{k \in \mathcal{J}}$, the *posterior support* $\text{supp } Q_{j,t-1}$ is uniquely determined by that payoff environment and is independent of the particular prior $\bar{\pi}_{j,t-1}$ within its convex hull. In particular, the family of posteriors associated with each action k at time $t-1$ is a function of L_{t-1} only.

In the canonical "one-signal-per-action" representation used in Lemma 1, we can associate with each destination k a single posterior $\pi_{j|k,t-1}$ over θ_{t-1} , interpreted as the belief held by the representative agent who moves from j to k in period $t-1$. By the argument above, the family $\{\pi_{j|k,t-1}\}_{k \in \mathcal{J}}$ is completely determined by the payoff environment in period $t-1$ and thus depends only on L_{t-1} .

The last step is to show that the dependence of posteriors on L_{t-1} at $t-1$ extends to shared priors at t .

Given a posterior $\pi_{j|k,t-1}$ over θ_{t-1} , the posterior over the next-period state θ_t induced by the exogenous transition kernel γ is

$$\tilde{\pi}_{j|k,t}(\theta_t) \equiv \int_{\Theta} \gamma(\theta_t | \theta_{t-1}) \pi_{j|k,t-1}(\theta_{t-1}) d\theta_{t-1}.$$

Since each $\pi_{j|k,t-1}$ depends only on L_{t-1} through the payoff environment, the same is true for the corresponding beliefs $\tilde{\pi}_{j|k,t}$ over θ_t .

Assumption 3 specifies that, after local information sharing in period t , the shared belief in location k ,

$$\bar{\pi}_{kt} = (\bar{\pi}_{klt})_{l=1,\dots,J},$$

is constructed from the means and variances of the beliefs of migrants who arrived in k between $t - 1$ and t under the representative strategy. Formally, for each component l , the distribution $\bar{\pi}_{klt}$ is given by $F(\bar{\mu}_{klt}, \bar{\sigma}_{kt}^2)$, with $(\bar{\mu}_{klt}, \bar{\sigma}_{kt}^2)$ defined in (13) as functions of the means and variances of the beliefs of migrants from each origin j to k and of the population weights $\{L_{jkt-1}\}_j$. By construction, these means and variances are computed from the family $\{\bar{\pi}_{j|k,t}\}_{j \in \mathcal{J}}$ of post-migration beliefs at time t , and all arguments entering (13) depend only on L_{t-1} .

Because the inputs to the aggregation rule in Assumption 3 (the migrants' beliefs and the weights L_{jkt-1}) depend only on L_{t-1} , the resulting means $\bar{\mu}_{klt}$ and variances $\bar{\sigma}_{kt}^2$ are functions of L_{t-1} only. Hence, for each destination k and each component l , the distribution $\bar{\pi}_{klt}(\cdot | L_{t-1})$ is a function of L_{t-1} alone. Collecting components, the same is true for $\bar{\pi}_{kt}$:

$$\bar{\pi}_{kt}(\cdot | \omega_{it}) = \bar{\pi}_k(\cdot | L_{t-1}) \quad \text{for all } k \in \mathcal{J}, t \geq 1.$$

In conclusion, we have shown that, in a stochastic steady-state equilibrium, conditional on the realized population composition L_{t-1} , the posterior beliefs of representative movers in period $t - 1$ depend only on L_{t-1} , and so do the shared beliefs $\bar{\pi}_{kt}$ after local information sharing in period t . Therefore, in each period $t \geq 1$, the shared beliefs $\{\bar{\pi}_{kt}\}_{k \in \mathcal{J}}$ are functions of L_{t-1} only, as claimed. $\square$

We are now ready to complete the proof of Proposition 2.

Proof of Proposition 2. To prove that the logit mobility rule in (A.12) is a solution to the dynamic RI equilibrium, it only remains to show that the optimal default rule q' from (A.7) is expressed as:

$$q'_{jkt}(\omega_t) = q_{jkt}(\omega_t) = \int_{\theta} p_{jkt}(\theta, \omega_t) \bar{\pi}_{jt}(\theta) d\theta.$$

By properness of the entropy, for any given mobility rule p , the term in (A.7) that depends on q' is maximized when $q'_{jct}(\omega_t)$ equals the true distribution of signals s_t induced by p , namely $q_{jct}(\omega_t)$. Hence, in the outer maximisation over q' , the optimal default rule satisfies $q' = q$.

Invoking Lemmas 1 to 4, we have proved Proposition 2, namely, the mobility rule is given by

$$p_{jk}(\theta_t, L_{t-1}, \varepsilon_{it}) = \frac{q_{jk}(L_{t-1}, \varepsilon_{it}) \exp(u_{jk}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_k(\theta_t, L_{t-1}))^{1/\lambda_j}}{\sum_l q_{jl}(L_{t-1}, \varepsilon_{it}) \exp(u_{jl}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_l(\theta_t, L_{t-1}))^{1/\lambda_j}},$$

where $q_{jk}(L_{t-1}, \varepsilon_{it}) = \int_{\theta} p_{jk}(\theta, L_{t-1}, \varepsilon_{it}) \bar{\pi}_j(\theta | L_{t-1}) d\theta$ and we define the expected future value as $\bar{V}_k(\theta_t, L_{t-1}) = \mathbb{E}[V_k(\theta_{t+1}, L_t, \varepsilon_{t+1}) | \theta_t, L_{t-1}]$.

The continuation values solve

$$V_j(\theta_t, L_{t-1}, \varepsilon_{it}) = \lambda_j \log \left(\sum_l q_{jl}(L_{t-1}, \varepsilon_{it}) \exp(u_{jl}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_l(\theta_t, L_{t-1}))^{1/\lambda_j} \right).$$

$\square$

A.2.4 Proof of Proposition 3

Proof. We begin from the control representation of the dynamic rational inattention problem in Lemma 2. Using the optimal mobility rule conditional on q from Lemma 3, which yields the continuation value in (18), the default choice probabilities $\{q_{jk}\}_{k \in \mathcal{J}}$ solve, for any vector of idiosyncratic preferences ε_{it} , origin location j , and previous population distribution L_{t-1} ,

$$\begin{aligned} \max_{\{q_{jk}\}} \left\{ \int_{\theta} \lambda_j \log \left(\sum_{k=1}^J q_{jk} \exp \left((u_{jk}(\theta, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_k(\theta, L_{t-1}))^{1/\lambda_j} \right) \right) \bar{\pi}_j(\theta | L_{t-1}) d\theta \right\} \\ \text{s.t.} \quad \sum_{k=1}^J q_{jk} = 1, \quad q_{jk} \geq 0. \end{aligned}$$

Recall that

$$u_{jk}(\theta, L_{t-1}, \varepsilon_{it}) = A_k + \theta_{kt} - \alpha \log L_{kt} + B_k - \kappa_{jk} + \nu \varepsilon_{ikt}.$$

We implement the following normalization: divide all payoffs by ν and define the normalized quantities

$$\begin{aligned} \tilde{u}_{jk}(\theta, L_{t-1}, \varepsilon_{it}) &\equiv \frac{u_{jk}(\theta, L_{t-1}, \varepsilon_{it})}{\nu} = \frac{A_k + \theta_{kt} - \alpha \log L_{kt} + B_k - \kappa_{jk}}{\nu} + \varepsilon_{ikt}, \\ \tilde{V}_k(\theta, L_{t-1}) &\equiv \frac{\delta \bar{V}_k(\theta, L_{t-1})}{\nu}. \end{aligned}$$

This rescaling does not affect the optimal q , since it multiplies all utilities by a positive constant.

With these definitions, the objective becomes

$$\int_{\theta} \lambda_j \log \left(\sum_{k=1}^J q_{jk} \exp \left(\left(\tilde{u}_{jk}(\theta, L_{t-1}, \varepsilon_{it}) + \tilde{V}_k(\theta, L_{t-1}) \right)^{1/\lambda_j} \right) \right) \bar{\pi}_j(\theta | L_{t-1}) d\theta.$$

For notational simplicity, we now drop tildes and use again u_{jk} and $\bar{V}_k$ to denote their normalized counterparts. Thus the recursive problem can be written as

$$\begin{aligned} \max_{\{q_{jk}\}_{k \in \mathcal{J}}} & \left\{ \int_{\theta} \lambda_j \log \left(\sum_{k=1}^J q_{jk} \exp \left(\left((\theta_k + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})) / \nu + \varepsilon_{ikt} \right)^{1/\lambda_j} \right) \right) \bar{\pi}_j(\theta | L_{t-1}) d\theta \right\} \\ \text{s.t.} \quad & \sum_{k=1}^J q_{jk} = 1, \quad q_{jk} \geq 0. \end{aligned} \tag{A.14}$$

Note that for any real vector $(v_k)_{k \in \mathcal{J}}$ and any scalar $\gamma > 0$,

$$\log \left(\sum_{k=1}^J e^{v_k/\gamma} \right) = \mathbb{E}_e \left[\max_k \{v_k + \gamma e_k\} \right] + C,$$

where $e_k \sim \text{EV1}$ and C is a constant that does not depend on (v_k) (Small and Rosen, 1981). Applying this identity to (A.14) yields

$$\begin{aligned} & \mathbb{E}_{\theta} \mathbb{E}_e \left[\max_{k \in \mathcal{J}} \left\{ \frac{A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\nu \lambda_j} + \log q_{jk} + \frac{\varepsilon_{ikt}}{\lambda_j} + \frac{\theta_k}{\lambda_j \nu} + e_k \right\} \right] + C \\ &= \mathbb{E}_{\theta} \mathbb{E}_e \left[\max_{k \in \mathcal{J}} \left\{ \frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\nu \lambda_j} + \log q_{jk} + \frac{\varepsilon_{ikt}}{\lambda_j} + \underbrace{\left(\frac{\tilde{\theta}_k}{\lambda_j \nu} + e_k \right)}_{\equiv \zeta_j \tilde{e}_k} \right\} \right] + C, \end{aligned}$$

where $\tilde{\theta}_k \equiv \theta_k - \bar{\mu}_{jk}(L_{t-1})$ has mean 0 and variance $\bar{\sigma}_j^2(L_{t-1})$ under $\bar{\pi}_j(\cdot | L_{t-1})$.

Under Assumption 4, $\tilde{\theta}_k$ follows a conjugate EV1 distribution. Hence the sum $(\tilde{\theta}_k/(\lambda_j \nu)) + e_k$ is again EV1-distributed up to a scale factor (see Nadarajah, 2008, Marques et al., 2015). Define $\zeta_j > 0$ so that

$$\zeta_j \tilde{e}_k \equiv \frac{\tilde{\theta}_k}{\lambda_j \nu} + e_k, \quad \text{Var}(\tilde{e}_k) = \frac{\pi^2}{6}.$$

This implies

$$\zeta_j = \left(1 + \frac{6\bar{\sigma}_j^2}{\pi^2 \lambda_j^2 \nu^2} \right)^{1/2}.$$

Using the log-sum-exp identity once more with shocks $\tilde{e}_k$ produces the closed form

$$\begin{aligned} & \mathbb{E}_{\tilde{e}} \left[\max_k \left\{ \frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\nu \lambda_j} + \log q_{jk} + \frac{\varepsilon_{ikt}}{\lambda_j} + \zeta_j \tilde{e}_k \right\} \right] \\ &= \log \left(\sum_{k=1}^J q_{jk}^{1/\zeta_j} \exp \left(\frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\nu \lambda_j \zeta_j} + \frac{\varepsilon_{ikt}}{\lambda_j \zeta_j} \right) \right) + C. \end{aligned}$$

We can therefore rewrite the agent's problem, conditional on the vector ε_{it} and on L_{t-1} , as

$$\begin{aligned} \max_{\{q_{jk}\}_{k \in \mathcal{J}}} & \left\{ \log \left(\sum_{k=1}^J q_{jk}^{1/\zeta_j} \exp \left(\frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\nu \lambda_j \zeta_j} + \frac{\varepsilon_{ikt}}{\lambda_j \zeta_j} \right) \right) \right\} \\ \text{s.t.} & \quad \sum_{k=1}^J q_{jk} = 1, \quad q_{jk} \geq 0. \end{aligned}$$

Denote

$$X_{jk}(\theta, L_{t-1}, \varepsilon_{it}) \equiv \frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\nu \lambda_j \zeta_j} + \frac{\varepsilon_{ikt}}{\lambda_j \zeta_j}.$$

Then the objective becomes

$$\log \left(\sum_{k=1}^J q_{jk}^{1/\zeta_j} e^{X_{jk}} \right).$$

Let φ be the Lagrange multiplier on the constraint $\sum_k q_{jk} = 1$. The first-order condition with respect to q_{jl} is

$$\frac{\partial}{\partial q_{jl}} \left[\log \left(\sum_{k=1}^J q_{jk}^{1/\zeta_j} e^{X_{jk}} \right) + \varphi \left(1 - \sum_{k=1}^J q_{jk} \right) \right] = 0,$$

that is,

$$\frac{(1/\zeta_j) q_{jl}^{1/\zeta_j - 1} e^{X_{jl}}}{\sum_{k=1}^J q_{jk}^{1/\zeta_j} e^{X_{jk}}} - \varphi = 0.$$

Rearranging,

$$q_{jl}^{1/\zeta_j - 1} e^{X_{jl}} = \varphi \zeta_j \sum_{k=1}^J q_{jk}^{1/\zeta_j} e^{X_{jk}} \equiv C,$$

where C does not depend on l . Substituting X_{jk} back in and normalizing over k gives the closed-form

$$q_{jk}(\omega_t) = \frac{\exp \left((\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})) / \nu + \varepsilon_{ikt} \right)^{\frac{1}{\lambda_j (\zeta_j - 1)}}}{\sum_{l=1}^J \exp \left((\bar{\mu}_{jl}(L_{t-1}) + A_l - \alpha \log L_{lt} + B_l - \kappa_{jl} + \delta \bar{V}_l(\theta, L_{t-1})) / \nu + \varepsilon_{ilt} \right)^{\frac{1}{\lambda_j (\zeta_j - 1)}}}.$$

With an expression for $q_{jk}(\omega_t)$ in hand, we can now derive an expression for the mobility probabilities after information acquisition. Recalling the optimal migration rule (17) from Proposition 2,

$$\begin{aligned} p_{jk}(\theta_t, \omega_t) &= \frac{q_{jk}(\omega_t) \exp \left((\theta_{kt} + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})) / \nu + \varepsilon_{ikt} \right)^{\frac{1}{\lambda_j}}}{\sum_{l=1}^J q_{jl}(\omega_t) \exp \left((\theta_{lt} + A_l - \alpha \log L_{lt} + B_l - \kappa_{jl} + \delta \bar{V}_l(\theta, L_{t-1})) / \nu + \varepsilon_{ilt} \right)^{\frac{1}{\lambda_j}}} \\ &= \frac{\exp \left(\frac{\theta_{kt}}{\nu} + (A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})) \frac{\zeta_j}{\nu(\zeta_j - 1)} + \frac{\bar{\mu}_{jk}(L_{t-1})}{\nu(\zeta_j - 1)} + \varepsilon_{ikt} \frac{\zeta_j}{\zeta_j - 1} \right)^{\frac{1}{\lambda_j}}}{\sum_{l=1}^J \exp \left(\frac{\theta_{lt}}{\nu} + (A_l - \alpha \log L_{lt} + B_l - \kappa_{jl} + \delta \bar{V}_l(\theta, L_{t-1})) \frac{\zeta_j}{\nu(\zeta_j - 1)} + \frac{\bar{\mu}_{jl}(L_{t-1})}{\nu(\zeta_j - 1)} + \varepsilon_{ilt} \frac{\zeta_j}{\zeta_j - 1} \right)^{\frac{1}{\lambda_j}}} \end{aligned}$$

Recognizing the expression of $p_{jk}(\theta_t, \omega_t)$ as the mobility rule that arises from a multinomial logit decision problem, we can rewrite the problem of agents as

$$\max_{k \in \mathcal{J}} \left\{ \frac{1}{\lambda_j} \left(\frac{\theta_{kt}}{\nu} + (A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})) \frac{\zeta_j}{\nu(\zeta_j - 1)} + \frac{\bar{\mu}_{jk}(L_{t-1})}{\nu(\zeta_j - 1)} + \varepsilon_{ikt} \frac{\zeta_j}{\zeta_j - 1} \right) + e'_{ikt} \right\},$$

where $e'_{ikt} \sim EV1$ and $\text{Var}(e'_{ikt}) = \pi^2/6$. We can then define $\varrho_j \tilde{e}'_{ikt} = \frac{\zeta_j}{\lambda_j(\zeta_j - 1)} \varepsilon_{ikt} + e'_{ikt}$. Since, under Assumption 5, ε_{ikt} is distributed according to the conjugate of a Type 1 extreme value distribution, this implies that $\tilde{e}'_{ikt} \sim EV1$, if we define ϱ_j so that $\text{Var}(\tilde{e}'_{ikt}) = \pi^2/6$,

$$\varrho_j = \left(1 + \frac{\zeta_j^2}{\lambda_j^2 (\zeta_j - 1)^2} \right)^{\frac{1}{2}}.$$

Therefore, we can rewrite the problem solved by agents as

$$\max_{k \in \mathcal{J}} \left\{ \frac{1}{\varrho_j \lambda_j} \left(\frac{\theta_{kt}}{\nu} + (A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})) \frac{\zeta_j}{\nu(\zeta_j - 1)} + \frac{\bar{\mu}_{jk}(L_{t-1})}{\nu(\zeta_j - 1)} \right) + \tilde{e}'_{ikt} \right\},$$

to which the solution is

$$\bar{p}_{jk}(\theta_t, L_{t-1}) = \frac{\exp \left(\theta_{kt} + \frac{\bar{\mu}_{jk}(L_{t-1})}{\zeta_j - 1} + (A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})) \frac{\zeta_j}{\zeta_j - 1} \right)^{\frac{1}{\lambda_j \nu \varrho_j}}}{\sum_l \exp \left(\theta_{lt} + \frac{\bar{\mu}_{jl}(L_{t-1})}{\zeta_j - 1} + (A_l - \alpha \log L_{lt} + B_l - \kappa_{jl} + \delta \bar{V}_l(\theta, L_{t-1})) \frac{\zeta_j}{\zeta_j - 1} \right)^{\frac{1}{\lambda_j \nu \varrho_j}}}.$$

To arrive at the expression in (19), define the following constants,

$$\eta_j = \frac{1}{\zeta_j} = \left(1 + \frac{6\bar{\sigma}_j^2}{\pi^2 \lambda_j^2 \nu^2} \right)^{-\frac{1}{2}}, \quad \phi_j = \frac{\lambda_j \nu \varrho_j (\zeta_j - 1)}{\zeta_j} = \nu \left(1 + \lambda_j^2 (1 - \eta_j)^2 \right)^{1/2}.$$

To obtain the limit in (21) when information costs tend to zero, i.e. $\lambda_j \rightarrow 0$, note first that

$$\lim_{\lambda_j \rightarrow 0} \phi_j = \nu, \quad \lim_{\lambda_j \rightarrow 0} \eta_j = 0.$$

Moreover, as $\lambda_j \rightarrow 0$ the cost of information vanishes, so the optimal signal structure is fully revealing: posterior beliefs collapse to point masses at the true shocks, and the shared means satisfy

$$\bar{\mu}_{jk}(L_{t-1}) \rightarrow \theta_{kt} \quad \Rightarrow \quad \chi_{jk}(\theta_t, L_{t-1}) = \bar{\mu}_{jk}(L_{t-1}) - \theta_{kt} \rightarrow 0.$$

Hence $\eta_j \chi_{jk}(\theta_t, L_{t-1}) \rightarrow 0$, and the expression in (19) reduces to the familiar multinomial logit mobility rule predicted by models of migration flows with EV1-distributed idiosyncratic preferences:

$$\bar{p}_{jk}(\theta_t, L_{t-1}) \xrightarrow{\lambda_j \rightarrow 0} \frac{\exp(\bar{v}_{jk}(\theta_t, L_{t-1}))^{1/\nu}}{\sum_l \exp(\bar{v}_{jl}(\theta_t, L_{t-1}))^{1/\nu}}.$$

To obtain the limit in (22) as the dispersion of idiosyncratic preferences vanishes, consider the case in which $\nu \rightarrow 0$. In this limit, the scale of the shocks ε_{ikt} tends to zero, so their contribution to utility is negligible relative to the other payoff components. Formally, the limiting problem corresponds to setting $\varepsilon_{ikt} = 0$ in (A.14). In this case, agents solve

$$\max_{\{q_{jk}\}_{k \in \mathcal{J}}} \left\{ \int_{\theta} \lambda_j \log \left(\sum_{k=1}^J q_{jk} \exp \left(\theta_k + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1}) \right)^{1/\lambda_j} \right) \bar{\pi}_j(\theta | L_{t-1}) d\theta \right\}$$

$$s.t. \quad \sum_k q_{jk} = 1, \quad q_{jk} \geq 0.$$

Using the same result as above to interpret the term with logarithm as the expected value of a maximization problem, we obtain,

$$\begin{aligned} & \mathbb{E}_\theta \mathbb{E}_e \left[\max_{k \in \mathcal{J}} \left\{ \frac{A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\lambda_j} + \log q_{jk} + \frac{\theta_k}{\lambda_j} + e_k \right\} \right] + C \\ = & \mathbb{E}_\theta \mathbb{E}_e \left[\max_{k \in \mathcal{J}} \left\{ \frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\lambda_j} + \log q_{jk} + \underbrace{\frac{\tilde{\theta}_k}{\lambda_j} + e_k}_{\equiv \zeta'_j \hat{e}_k} \right\} \right] + C \\ = & \mathbb{E}_{\hat{e}} \left[\max_{k \in \mathcal{J}} \left\{ \frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\lambda_j} + \log q_{jk} + \zeta'_j \hat{e}_k \right\} \right] + C \\ = & \log \left(\sum_{k=1}^J q_{jk}^{1/\zeta'_j} \exp \left(\frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\lambda_j \zeta'_j} \right) \right), \end{aligned}$$

where I defined $\zeta'_j \hat{e}_k = \frac{\tilde{\theta}_k}{\lambda_j} + e_k$, and $\tilde{\theta}_k = \theta_k - \bar{\mu}_{jk}(L_{t-1})$ has mean 0 and variance $\bar{\sigma}_j^2(L_{t-1})$ under the belief distribution $\bar{\pi}_j$. Note that since $e_k \sim EV1$ and, under Assumption 4, $\tilde{\theta}_k$ is distributed according to the conjugate of a Type 1 extreme value, it follows from Nadarajah (2008) and Marques et al. (2015) that $\zeta'_j \hat{e}_k \sim EV1$, and the variance of this error is:

$$\text{Var} \left(\frac{\tilde{\theta}_k}{\lambda_j} + e_k \right) = \frac{\bar{\sigma}_j^2}{\lambda_j^2} + \frac{\pi^2}{6},$$

where π^2 is here the square of the constant $\pi = 3.14159\dots$ I define the shifter ζ'_j to be such that $\text{Var}(\hat{e}_k) = \frac{\pi^2}{6}$, so that

$$\zeta'_j = \left(1 + \frac{6\bar{\sigma}_j^2}{\pi^2 \lambda_j^2} \right)^{\frac{1}{2}}.$$

We can therefore rewrite the problem of agents, for a given population distribution L_{t-1} , as:

$$\begin{aligned} & \max_{\{q_{jk}\}_{k \in \mathcal{J}}} \left\{ \log \left(\sum_{k=1}^J q_{jk}^{1/\zeta'_j} \exp \left(\frac{\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})}{\lambda_j \zeta'_j} \right) \right) \right\} \\ & s.t. \quad \sum_k q_{jk} = 1, \quad q_{jk} \geq 0. \end{aligned}$$

Solving for q_{jk} from the first-order condition associated with this problem, we obtain a closed-form expression for $q_{jk}(L_{t-1})$,

$$q_{jk}(L_{t-1}) = \frac{\exp(\bar{\mu}_{jk}(L_{t-1}) + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1}))^{\frac{1}{\lambda_j(\zeta'_j-1)}}}{\sum_{l=1}^J \exp(\bar{\mu}_{jl}(L_{t-1}) + A_l - \alpha \log L_{lt} + B_l - \kappa_{jl} + \delta \bar{V}_l(\theta, L_{t-1}))^{\frac{1}{\lambda_j(\zeta'_j-1)}}}.$$

With an expression for $q_{jk}(\omega_t)$ in hand, we can now derive an expression for the mobility probabilities after information acquisition. Recalling the optimal migration rule (17) from Proposition 2, this time with all preference shocks ε_{it} equal to zero,

$$\bar{p}_{jk}(\theta_t, L_{t-1}) = \frac{q_{jk}(L_{t-1}) \exp(\theta_{kt} + A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1}))^{\frac{1}{\lambda_j}}}{\sum_{l=1}^J q_{jl}(L_{t-1}) \exp(\theta_{lt} + A_l - \alpha \log L_{lt} + B_l - \kappa_{jl} + \delta \bar{V}_l(\theta, L_{t-1}))^{\frac{1}{\lambda_j}}}$$

$$\begin{aligned}
&= \frac{\exp\left(\theta_{kt} + (A_k - \alpha \log L_{kt} + B_k - \kappa_{jk} + \delta \bar{V}_k(\theta, L_{t-1})) \frac{\zeta'_j}{\zeta'_j - 1} + \frac{\bar{\mu}_{jk}(L_{t-1})}{\zeta'_j - 1}\right)^{\frac{1}{\lambda_j}}}{\sum_{l=1}^J \exp\left(\theta_{lt} + (A_l - \alpha \log L_{lt} + B_l - \kappa_{jl} + \delta \bar{V}_l(\theta, L_{t-1})) \frac{\zeta'_j}{\zeta'_j - 1} + \frac{\mu_{jl}(L_{t-1})}{\zeta'_j - 1}\right)^{\frac{1}{\lambda_j}}} \\
&= \frac{\exp(\rho_j \chi_{jk}(\theta_t, L_{t-1}) + \bar{v}_{jk}(\theta_t, L_{t-1}))^{\frac{1}{\psi_j}}}{\sum_l \exp(\rho_j \chi_{jl}(\theta_t, L_{t-1}) + \bar{v}_{jl}(\theta_t, L_{t-1}))^{\frac{1}{\psi_j}}},
\end{aligned}$$

where $\psi_j = \lambda_j (1 - \rho_j)$ and $\rho_j = \left(1 + \frac{6\bar{\sigma}_j^2}{\pi^2 \lambda_j^2}\right)^{-1/2} \in (0, 1)$. $\square$

A.2.5 Optimality of the Social Learning Rule for Beliefs with Imperfect Recall

In this section, I show that a log-linear learning rule such as the one stated in Assumption 3 is the unique social learning rule satisfying the four axioms introduced in Molavi et al. (2018): imperfect recall, label neutrality, monotonicity, and independence of irrelevant alternatives. These axioms impose natural restrictions on how individuals aggregate their neighbors' beliefs, and together they imply that any admissible social learning rule must take a log-linear form, with weights determined by the structure of the local information-sharing network. Moreover, when the beliefs being aggregated are Normally distributed, the resulting posterior mean and variance coincide exactly with those implied by the specific functional form in Assumption 3. Although I do not impose normality in the main analysis, this equivalence provides a strong conceptual motivation for the information-sharing rule used in the model.

The first assumption is that agents follow social learning rules with *imperfect recall*: they treat the current beliefs of their neighbors as sufficient statistics for all the information available to them, ignoring the history or origin of those beliefs.

Assumption 6 (Imperfect Recall). *For each location k , the shared belief $\bar{\pi}_{kt}$ depends only on the current belief profile $\{\pi_{jkt}\}_j$ and not on beliefs from earlier periods $\{\pi_{jk,\tau}\}_{\tau < t}$.*

Second, *label neutrality* requires that relabeling the underlying states has no effect on how agents process information—only the informational content matters.

Assumption 7 (Label Neutrality (LN)). *The social learning rule treats the labels of productivity states symmetrically. If the components of the productivity vector $\theta_t = (\theta_{1t}, \dots, \theta_{Jt})$ are relabeled by any permutation of indices, and all residents' beliefs are relabelled accordingly, then the shared belief $\bar{\pi}_{kt}$ is relabelled in the same way.*

Third, *monotonicity* requires the shared belief to move in the same direction as the individual beliefs. If all residents in k become more confident in state θ relative to state $\hat{\theta}$, then the shared belief must also place more weight on θ .

Assumption 8 (Monotonicity). *The shared belief $\bar{\pi}_{kt}$ is increasing in the residents' confidence in each state. If every resident in location k raises their belief in state θ relative to another state $\hat{\theta}$ —that is, if each individual's likelihood ratio $\pi_{jkt}(\theta)/\pi_{jkt}(\hat{\theta})$ weakly increases—then the shared belief must also assign (weakly) more relative weight to θ than to $\hat{\theta}$.*

Fourth, *independence of irrelevant alternatives* (IIA) requires that aggregation over any subset of states depends only on beliefs about that subset.

Assumption 9 (Independence of Irrelevant Alternatives (IIA)). *For any subset of states $\bar{\Theta} \subseteq \Theta$, the shared belief $\bar{\pi}_{kt}$ about the relative likelihood of states in $\bar{\Theta}$ depends only on the residents' beliefs about that same subset. Formally, if two profiles of beliefs yield identical likelihood ratios $\pi_{jkt}(\theta)/\pi_{jkt}(\hat{\theta})$ for all $\theta, \hat{\theta} \in \bar{\Theta}$ and all j , then the resulting shared beliefs must also coincide on $\bar{\Theta}$.*

Finally, we recall the representative-agent structure introduced in Assumption 3. For each origin j that supplies residents to location k (i.e., with $L_{jkt-1} > 0$), we define a representative agent who moved from j to k between $t-1$ and t with idiosyncratic preference shocks set to their mean value. These representative agents form a complete local network and each receives a weight equal to the mass of individuals they represent.

Assumption 10 (Complete Network of Representative Agents). *In each location k , the representative agents described above form a complete local network and each receives a weight equal to the mass of agents they represent, L_{jkt-1} .*

Molavi et al. (2018) show that, together with the four axioms above, social learning rules must combine neighbors' log-likelihood ratios linearly, with weights given by their centrality in the local network. Under Assumption 10, centralities are proportional to the masses L_{jkt-1} , yielding the following result.

Proposition 4. *Under Assumptions 6–10, local information sharing in period t leads to a log-linear learning rule. The shared beliefs in location k after information exchange in period t satisfy*

$$\log \bar{\pi}_{kt}(\theta_t) = C_{kt} + \sum_j L_{jkt-1} \log \pi_{jkt}(\theta_t), \quad (\text{A.15})$$

where π_{jkt} denotes the belief about θ_t held by the representative agent who moved from j to k in period $t-1$, L_{jkt-1} is the mass of such agents residing in k , and C_{kt} is a normalizing constant ensuring $\int_{\Theta} \bar{\pi}_{kt}(\theta) d\theta = 1$.

Corollary 1. *If, in addition to Assumptions 6–10, the beliefs π_{jkt} of the representative agents are Normally distributed with marginal means $\bar{\mu}_{jkl}$ and variances $\bar{\sigma}_{jkt}^2$, then the shared belief $\bar{\pi}_{kt}$ is also Normally distributed, with mean and variances given by*

$$\bar{\sigma}_{kt}^2 = \left(\frac{1}{L_{kt-1}} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2} \right)^{-1}, \quad \bar{\mu}_{klt} = \frac{\bar{\sigma}_{kt}^2}{L_{kt-1}} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2} \bar{\mu}_{jkl}.$$

Proof of Proposition 4. The proof is adapted from Theorem 1 in Molavi et al. (2018). For notational simplicity, we omit time indices. Fix two arbitrary states $\theta \neq \hat{\theta}$ and an arbitrary profile of beliefs $\pi = \{\pi_{jk}\}_{j,k=1,\dots,J} \in \Delta(\Theta)^{J^2}$, where π_{jk} denotes the belief of the representative agent who moved from j to k . Let $\bar{\Theta} = \{\theta, \hat{\theta}\}$, and denote by $\bar{\pi}(\pi) = \{\bar{\pi}_k(\pi)\}_{k=1,\dots,J}$ the shared beliefs implied by the social learning rule.

By definition of conditional probability, for every location k we have

$$\log \frac{\bar{\pi}_k(\pi)(\theta)}{\bar{\pi}_k(\pi)(\hat{\theta})} = \log \bar{\pi}_k(\text{cond}_{\bar{\Theta}}(\pi))(\theta) - \log \bar{\pi}_k(\text{cond}_{\bar{\Theta}}(\pi))(\hat{\theta}),$$

where $\text{cond}_{\bar{\Theta}}(\pi)$ denotes the profile of beliefs restricted to the two states in $\bar{\Theta}$.

By Independence of Irrelevant Alternatives (Assumption 9), the shared belief over $\bar{\Theta}$ depends only on residents' beliefs over $\bar{\Theta}$. Hence $\text{cond}_{\bar{\Theta}}(\pi)$ depends on π only through the collection of likelihood ratios $\{\pi_{jk}(\theta)/\pi_{jk}(\hat{\theta})\}_{j=1,\dots,J}$. Because the social learning rule is continuous in agents' beliefs, there exists a continuous function $g_k : \mathbb{R}^J \rightarrow \mathbb{R}$ such that

$$\log \frac{\bar{\pi}_k(\pi)(\theta)}{\bar{\pi}_k(\pi)(\hat{\theta})} = g_k \left(\log \frac{\pi_{1k}(\theta)}{\pi_{1k}(\hat{\theta})}, \dots, \log \frac{\pi_{Jk}(\theta)}{\pi_{Jk}(\hat{\theta})} \right), \quad (\text{A.16})$$

for all pairs of states $\theta \neq \hat{\theta}$ and all belief profiles π . Label neutrality (Assumption 7) guarantees that the function g_k does not depend on the particular choice of θ and $\hat{\theta}$.

Now consider three distinct states θ , $\hat{\theta}$, and $\tilde{\theta}$. Because (A.16) holds for any pair of states, we have

$$\begin{aligned} g_k \left(\log \frac{\pi_{1k}(\theta)}{\pi_{1k}(\hat{\theta})}, \dots, \log \frac{\pi_{Jk}(\theta)}{\pi_{Jk}(\hat{\theta})} \right) + g_k \left(\log \frac{\pi_{1k}(\hat{\theta})}{\pi_{1k}(\tilde{\theta})}, \dots, \log \frac{\pi_{Jk}(\hat{\theta})}{\pi_{Jk}(\tilde{\theta})} \right) &= \log \frac{\bar{\pi}_k(\theta)}{\bar{\pi}_k(\hat{\theta})} + \log \frac{\bar{\pi}_k(\hat{\theta})}{\bar{\pi}_k(\tilde{\theta})} \\ &= \log \frac{\bar{\pi}_k(\theta)}{\bar{\pi}_k(\tilde{\theta})} \\ &= g_k \left(\log \frac{\pi_{1k}(\theta)}{\pi_{1k}(\tilde{\theta})}, \dots, \log \frac{\pi_{Jk}(\theta)}{\pi_{Jk}(\tilde{\theta})} \right). \end{aligned}$$

Since the belief profile π was arbitrary, this implies that for any $x, y \in \mathbb{R}^J$,

$$g_k(x) + g_k(y) = g_k(x + y).$$

By continuity of the learning rule, this is Cauchy's functional equation, whose only solutions are linear. Hence there exist constants a_{jk} such that

$$g_k(x) = \sum_{j=1}^J a_{jk} x_j.$$

Monotonicity (Assumption 8) requires that increasing $\pi_{jk}(\theta)/\pi_{jk}(\hat{\theta})$ strictly increases the shared likelihood ratio, which implies $a_{jk} > 0$ for all j . Under the complete network of representative agents in location k (Assumption 10), each representative agent from origin j receives weight proportional to the mass of individuals they represent, so we set $a_{jk} = L_{jkt-1}$. Therefore,

$$\log \frac{\bar{\pi}_k(\theta)}{\bar{\pi}_k(\hat{\theta})} = \sum_{j=1}^J L_{jkt-1} \log \frac{\pi_{jk}(\theta)}{\pi_{jk}(\hat{\theta})}.$$

Fix any reference state $\hat{\theta}$ and rearrange to obtain

$$\log \bar{\pi}_k(\theta) = C_{kt} + \sum_{j=1}^J L_{jkt-1} \log \pi_{jk}(\theta),$$

where the constant

$$C_{kt} = \log \bar{\pi}_k(\hat{\theta}) - \sum_{j=1}^J L_{jkt-1} \log \pi_{jk}(\hat{\theta})$$

is chosen so that $\int_{\theta} \bar{\pi}_{kt}(\theta) d\theta = 1$. This proves the desired log-linear representation. $\square$

Proof of Corollary 1. Fix a location k in period t . By Proposition 4, the shared belief $\bar{\pi}_{kt}$ satisfies the log-linear representation

$$\log \bar{\pi}_{kt}(\theta) = C_{kt} + \sum_{j=1}^J L_{jkt-1} \log \pi_{jk}(\theta).$$

By assumption, each representative agent's belief π_{jk} is Normally distributed with mean vector $(\bar{\mu}_{jk1t}, \dots, \bar{\mu}_{jkJt})$ and variance $\bar{\sigma}_{jkt}^2$. Hence for each component θ_l we have the usual Normal likelihood expression:

$$\log \pi_{jkt}(\theta_l) = -\frac{1}{2\bar{\sigma}_{jkt}^2}(\theta_l - \bar{\mu}_{jklt})^2 + C_{jklt},$$

where C_{jklt} collects terms that do not depend on θ_l . Substituting this expression into (A.2.5) and collecting terms that depend on θ_l yields

$$\log \bar{\pi}_{kt}(\theta_l) = C'_{kt} - \frac{1}{2} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2} (\theta_l - \bar{\mu}_{jklt})^2. \quad (\text{A.17})$$

Expanding the quadratic terms in (A.17) shows that $\log \bar{\pi}_{kt}$ is a quadratic function of θ_l , and therefore $\bar{\pi}_{kt}$ is Normal. To identify the variance $\bar{\sigma}_{kt}^2$, expand the quadratic terms in (A.17). Using $(\theta_l - \bar{\mu}_{jklt})^2 = \theta_l^2 - 2\bar{\mu}_{jklt}\theta_l + \bar{\mu}_{jklt}^2$, the coefficient on θ_l^2 in (A.17) is

$$-\frac{1}{2} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2}. \quad (\text{A.18})$$

On the other hand, if $\bar{\pi}_{kt}(\theta_l)$ is Normal with variance $\bar{\sigma}_{kt}^2$ and total mass $L_{kt-1} = \sum_j L_{jkt-1}$, its log-density (up to a constant) is

$$\log \bar{\pi}_{kt}(\theta_l) = \tilde{C}_{kt} - \frac{L_{kt-1}}{2\bar{\sigma}_{kt}^2} (\theta_l - \bar{\mu}_{klt})^2,$$

whose coefficient on θ_l^2 is

$$-\frac{L_{kt-1}}{2\bar{\sigma}_{kt}^2}. \quad (\text{A.19})$$

Equating the coefficients in (A.18) and (A.19) yields the shared variance:

$$\frac{1}{\bar{\sigma}_{kt}^2} = \frac{1}{L_{kt-1}} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2}.$$

Next collect the linear terms in θ_l from (A.17). Matching coefficients with the Normal kernel

$$-\frac{1}{2\bar{\sigma}_{kt}^2}(\theta_l - \bar{\mu}_{klt})^2$$

implies

$$\frac{\bar{\mu}_{klt}}{\bar{\sigma}_{kt}^2} = \frac{1}{L_{kt-1}} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2} \bar{\mu}_{jkl},$$

and substituting the expression for $\bar{\sigma}_{kt}^2$ gives

$$\bar{\mu}_{klt} = \frac{\bar{\sigma}_{kt}^2}{L_{kt-1}} \sum_{j=1}^J \frac{L_{jkt-1}}{\bar{\sigma}_{jkt}^2} \bar{\mu}_{jkl},$$

which establishes the second expression. $\square$

A.3 Alternative Assumptions on Learning About Productivity and Population

This section extends the model to allow agents to use realized wages and population as signals about local productivity. In the baseline model, I impose that agents do not exploit the wages they receive, or the observed population distribution, to adjust their beliefs about the local productivity. These restrictions deliver tractability, since beliefs about θ_{kt} do not depend on the realization of θ_{kt} . This Section shows that it is possible to allow for wages and the population distribution to perfectly or partially reveal the local value of θ_{kt} or its whole value θ_t , and discusses the implications on the equilibrium mobility patterns.

A.3.1 Complete Revelation of Local Productivity

I begin with an extreme benchmark in which realized wages perfectly reveal the local productivity. Suppose that in each period t , after mobility and production, agents located in region k observe a wage that allows them to infer the true value of θ_{kt} exactly, while they do not use the population distribution to update beliefs. For any worker i with preference shocks ε_{it} who is in k at the end of period t , the posterior belief about θ_{kt} is therefore degenerate:

$$\pi_{ikt,k}(\hat{\theta}_{kt} \mid \theta_{kt}, \varepsilon_{it}) = \mathbf{1}\{\hat{\theta}_{kt} = \theta_{kt}\}.$$

At the beginning of period $t+1$, agents in each location share information using the same rule (13) as in the baseline model. A key modelling assumption is that agents' beliefs feature *imperfect recall*: they treat their and others' current beliefs as primitive objects in the sharing process and do not reconstruct the underlying signals that generated those beliefs. Hence, an individual who observed θ_{kt} at the end of t forms, by the AR(1) process (3), a personal prior for $\theta_{k,t+1}$ of the form

$$\theta_{k,t+1} \mid \theta_{kt} \sim \mathcal{N}(\rho\theta_{kt}, \sigma_{\xi}^2),$$

while an individual who did not observe θ_{kt} carries into period $t+1$ a more diffuse prior for $\theta_{k,t+1}$ based on past information and social learning.

Given the learning rule postulated in Assumption 3, the shared belief in region l about $\theta_{k,t+1}$ at the start of

period $t + 1$ is a nonlinear aggregation of this entire distribution of priors:

$$\log \bar{\pi}_{l,k}(\theta_{k,t+1}) = C_{l,t+1} + \sum_j L_{jlt} \log \pi_{jl,k}(\theta_{k,t+1}),$$

where $\pi_{jl,k}$ denotes the marginal prior about $\theta_{k,t+1}$ held by a representative agent who entered l from origin j at the end of t , and L_{jlt} is the mass of agents who moved to l from j in period t . Because agents who were in k at the end of t hold very precise beliefs centered at $\rho\theta_{kt}$, their presence in l pulls the shared belief in that direction and tightens its dispersion. But unless the entire population of l previously lived in k , the shared belief generally does *not* collapse to the AR(1) prior $\mathcal{N}(\rho\theta_{kt}, \sigma_\xi^2)$: it remains an aggregated object reflecting heterogeneous precisions and heterogeneous past signals across all groups present in l .

A symmetric logic applies to beliefs about $\theta_{j,t+1}$ for $j \neq k$. Whenever productivity in some region j was revealed perfectly in period t , migrants who leave j carry very precise priors for $\theta_{j,t+1}$ to their new locations, where they receive a larger effective weight under the sharing rule and therefore reduce the dispersion of beliefs about region j across space.

Relative to the baseline model, prior beliefs become less biased and more precise. In the baseline, agents hold an optimistic bias about productivity in their current location, which reduces out-migration in levels by acting much like an implicit migration cost. Under perfect revelation of θ_{kt} , this bias is reduced but not eliminated: shared priors at the start of period $t + 1$ aggregate the precise beliefs of former residents of k with the more diffuse beliefs of agents who did not directly observe θ_{kt} .

The main effect on migration *elasticities* operates through the precision of priors. Because the prior for $\theta_{k,t+1}$ is more precise, agents enter period $t + 1$ with more “free” information about local shocks. At the same time, the higher precision of these priors lowers the marginal value of acquiring costly information, so agents optimally invest less in information acquisition. The resulting posterior beliefs therefore combine a more precise prior with a weaker costly-information component, and it is generally ambiguous whether agents end up with more or less total information about contemporaneous shocks compared to the baseline. Consequently, the responsiveness of migration flows to local productivity shocks may increase or decrease, depending on parameter values and the relative strength of these two forces.

Under perfect revelation of local productivity, the realized productivity vector θ_{t-1} becomes an additional state variable. In the baseline model, shared priors at the start of period t depend only the population distribution L_{t-1} . When each θ_{kt-1} is revealed at the end of period $t - 1$, the shared beliefs at the beginning of period t are instead constructed from the AR(1) forecasts $\rho\theta_{t-1}$ combined with the usual information-sharing rule. As a result, the aggregate state at the start of t includes $(\theta_t, \theta_{t-1}, L_{t-1})$ rather than only (θ_t, L_{t-1}) .

Although this added state increases the computational burden, solving the model with perfect revelation is conceptually straightforward for a small number of regions. In practice, however, the partial-revelation case developed in the next subsection provides a cleaner and more general extension, in which wages are informative but noisy about θ_{kt} . The perfect-revelation benchmark can be viewed as the limiting case of that model as the noise variance $\sigma_\tau^2 \rightarrow 0$, and the qualitative insights discussed above follow directly from that limit.

Hence, even under complete revelation of local productivity, information frictions remain economically meaningful. Although agents know θ_{kt} exactly after observing wages, they still face costly information acquisition about other regions, and the diffusion of information across locations remains incomplete. Migration behavior continues to reflect heterogeneous information across space, and the main qualitative insights of the model—that incomplete and spatially asymmetric information shapes mobility—remain intact.

A.3.2 Partial Revelation of Local Productivity

I now write an extension of the model in which workers update their beliefs upon observing the wage they obtain in the location they chose in period t . I still maintain the assumption that agents do not invert the law of motion of population (12) to learn about θ_t . I assume the existence of an additional unobserved local shock τ_{kt} realized in each location k and period t , after mobility has happened, before production occurs. If this shock has a variance equal to zero, wages reveal local productivity perfectly as in the previous case. If this shock τ_{kt} has a positive variance, then even upon observing their wage, agents cannot perfectly infer the realization of the local productivity θ_{kt} .

Each period, agents who move from location j to k receive the flow of payoff u_{ijkt} , not inclusive of information costs, given by (5). The structure of production is similar to the baseline model, with a unique freely traded homogeneous good chosen as numeraire. The production function takes a Cobb-Douglas form using labor as the

single input, so that output in region k at t is given by:

$$y_{kt} = \exp(A_k + \theta_{kt} + \tau_{kt}) L_{kt}^{1-\alpha}, \quad \alpha > 0, \quad (\text{A.20})$$

where θ_{kt} follows the AR(1) process described in (3), and τ_{kt} are iid across locations and time, and drawn from a normal distribution $\mathcal{N}(0, \sigma_\tau^2)$. I assume that τ_{kt} is unobserved. However, contrary to θ_{kt} , agents have no incentive to acquire information about τ_{kt} . This is first because τ_{kt} is realized only after mobility has occurred, so that no information about it can be collected to adjust mobility decisions at t . Second, τ_{kt} is iid over time, so any information about τ_{kt} has no predictive power about its future value.

The first order condition arising from profit maximization yields the following expression for the wage in region k at t :

$$\log w_{kt} = A_k + \theta_{kt} + \tau_{kt} - \alpha \log L_{kt} + \log(1 - \alpha). \quad (\text{A.21})$$

As a result, indirect utility, net of information costs introduced below, can be expressed as

$$u_{ijkt} = A_k + \theta_{kt} + \tau_{kt} - \alpha \log L_{kt} + B_k - \kappa_{jk} + \nu \varepsilon_{ikt}. \quad (\text{A.22})$$

Given that local productivity shocks are the only variables that agents are trying to observe, agents still hold beliefs about the cross-sectional distribution of productivity:

$$\pi_{it}(\theta) = \Pr(\theta_t = \theta | i, t), \quad \forall \theta \in \mathbb{R}^J. \quad (\text{A.23})$$

Figure A.2 represents the timeline of events. In addition to the baseline model, the local shock τ_{kt} is realized once agents have made location decisions. Production occurs according to (A.20). The wage is then observed, and agents in k update their beliefs about θ_{kt} according to Bayes' rule. Note that the observation of w_{kt} can only be used by agents to update their beliefs about productivity in the current location k , since productivity shocks are independent across locations. Denoting by $\pi_{jl,k}(\theta_{kt})$ the marginal belief about θ_{kt} held by agents who moved from j to l during period t , the updating for any possible $\theta_{kt} = \hat{\theta}_{kt}$ upon observing a realization of wages $w_{kt} = \hat{w}_{kt}$ is given by

$$\Pr(\theta_{kt} = \hat{\theta}_{kt} | w_{kt} = \hat{w}_{kt}, j \rightarrow l) = \frac{\Pr(w_{kt} = \hat{w}_{kt} | \theta_{kt} = \hat{\theta}_{kt}, j \rightarrow l) \Pr(\theta_{kt} = \hat{\theta}_{kt}, j \rightarrow l)}{\Pr(w_{kt} = \hat{w}_{kt} | j \rightarrow l)}. \quad (\text{A.24})$$

The resulting belief distribution after this round of updating naturally depends on the equilibrium beliefs $\pi_{jl,k}(\theta_{kt})$ held by agents who moved from j to l . This distribution is known from the updating in the rational inattention block, for which agents pay the information costs (9). The beliefs about the vector θ_t held by agents who moved from j to l are given by

$$\pi_{jl}(\theta_t) = \frac{\bar{p}_{jl}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) \bar{\pi}_j(\theta_t | L_{t-1}, \theta_{jt-1}, \tau_{jt-1})}{q_{jl}(L_{t-1}, \theta_{jt-1}, \tau_{jt-1})}. \quad (\text{A.25})$$

Since the distribution of shared beliefs is set in Assumption 3, and since the mobility rules are given by the equilibrium rule, the distribution of beliefs after observing wages is known and now depends on σ_τ . For example, if the marginal beliefs $\pi_{jl,k}(\theta_{kt})$ were Normally distributed with some mean $\mu_{jl,k}$ and variance $\sigma_{jl,k}^2$, the resulting belief would be expressed as

$$\theta_{kt} | w_{kt} = \hat{w}_{kt}, j \rightarrow l \sim \mathcal{N} \left(\frac{\sigma_{jl,k}^2 (\hat{w}_{kt} - A_k + \alpha \log L_{kt} - \log(1 - \alpha)) + \sigma_\tau^2 \mu_{jl,k}}{\sigma_{jl,k}^2 + \sigma_\tau^2}, \left(\sigma_{jl,k}^{-2} + \sigma_\tau^{-2} \right)^{-1} \right). \quad (\text{A.26})$$

In contrast to the baseline model, the end-of-period posterior beliefs about θ_{kt} depend on the actual realization of w_{kt} , which itself depends on the actual realization of (θ_{kt}, τ_{kt}) . We can rewrite the final posterior beliefs about the vector θ_t , after observation of local wages, as

$$\pi_{jl}(\cdot | L_{t-1}, \theta_{jt-1}, \tau_{jt-1}).$$

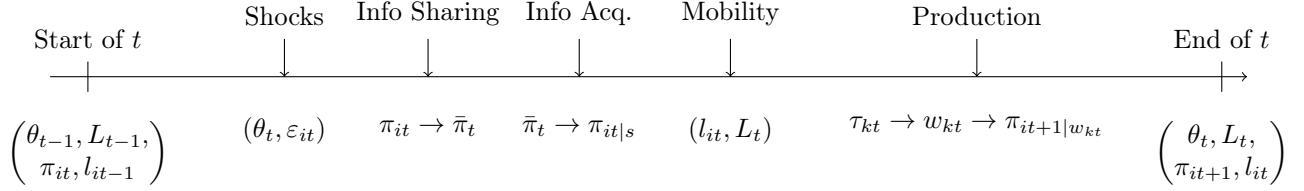


Figure A.2: Sequence of actions during any period t with partial revelation of local productivity

The population in any location k after mobility has occurred is then $L_{kt} = \sum_j L_{jkt}$, where

$$L_{jkt} = L_{jt-1} \bar{p}_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}), \quad (\text{A.27})$$

We can then write the analogous to Proposition 2, expressing the mobility rule for each preference shock ε_{it} , now as a function of the previous local productivity shocks θ_{jt-1} , and τ_{jt-1} .

Proposition 5. *In the stochastic steady state, for each agent located in j at $t-1$, the optimal mobility rule $p_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}, \varepsilon_{it})$ can be expressed as*

$$p_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}, \varepsilon_{it}) = \frac{q_{jk}(L_{t-1}, \theta_{jt-1}, \tau_{jt-1}, \varepsilon_{it}) \exp(u_{jk}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_k(\theta_t, L_{t-1}))^{1/\lambda_j}}{\sum_l q_{jl}(L_{t-1}, \theta_{jt-1}, \tau_{jt-1}, \varepsilon_{it}) \exp(u_{jl}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_l(\theta_t, L_{t-1}))^{1/\lambda_j}} \quad (\text{A.28})$$

where $q_{jk}(L_{t-1}, \theta_{jt-1}, \tau_{jt-1}, \varepsilon_{it}) = \int_{\theta} p_{jk}(\theta, \theta_{jt-1}, \tau_{jt-1}, L_{t-1}, \varepsilon_{it}) \bar{\pi}_j(\theta | L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) d\theta$ and we define the expected future value as $\bar{V}_k(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) = \mathbb{E}[V_k(\theta_{t+1}, L_t, \varepsilon_{it+1}, \theta_{jt}, \tau_{jt}) | \theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}]$, which is independent of θ_{t-1} and τ_{t-1} given distributional assumptions placed on θ_t and τ_t . The continuation payoffs solve

$$V_j(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}, \varepsilon_{it}) = \lambda_j \log \left(\sum_l q_{jl}(L_{t-1}, \theta_{jt-1}, \tau_{jt-1}, \varepsilon_{it}) \exp(u_{jl}(\theta_t, L_{t-1}, \varepsilon_{it}) + \delta \bar{V}_l(\theta_t, L_{t-1}))^{1/\lambda_j} \right), \quad (\text{A.29})$$

and population and beliefs follow the laws of motion (12) - (14).

Proof. The proof follows from the proof of Proposition 2 in Appendix A.2.3, after redefining the state space to include θ_{t-1} and τ_{t-1} . $\square$

After imposing Assumptions 4 and 5 on the equilibrium distribution of beliefs and preference shocks, we can write the equilibrium aggregate mobility rule in a similar fashion to Proposition 3.

Proposition 6. *Under Assumptions 4 and 5, the average mobility rule in the presence of information frictions and preference heterogeneity is given by*

$$\bar{p}_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) = \frac{\exp(\eta_j \chi_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) + \bar{v}_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}))^{\frac{1}{\phi_j}}}{\sum_l \exp(\eta_j \chi_{jl}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) + \bar{v}_{jl}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}))^{\frac{1}{\phi_j}}}, \quad (\text{A.30})$$

where $\chi_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) = \bar{\mu}_{jk}(L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) - \theta_{kt}$ is the expectational error made by agents in j about θ_{kt} , while the continuation payoffs solve

$$V_j(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) = \phi_j \log \left(\sum_l \exp(\eta_j \chi_{jl}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) + \bar{v}_{jl}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}))^{\frac{1}{\phi_j}} \right), \quad (\text{A.31})$$

with $\phi_j = \nu \left(1 + \lambda_j^2 (1 - \eta_j)^2 \right)^{1/2}$ and $\eta_j = \left(1 + \frac{6\sigma_j^2}{\pi^2 \lambda_j^2 \nu^2} \right)^{-1/2} \in (0, 1)$, and π is the constant $\pi = 3.1415...$

When the information cost tends to zero, $\bar{\mu}_{jk}(L_{t-1}) \rightarrow \theta_{kt}$ and $\phi_j \rightarrow \nu$ so that the model reduces to a preference-based migration model:

$$\bar{p}_{jk}(\theta_t, L_{t-1}) = \frac{\exp(\bar{v}_{jk}(\theta_t, L_{t-1}))^{1/\nu}}{\sum_l \exp(\bar{v}_{jl}(\theta_t, L_{t-1}))^{1/\nu}}. \quad (\text{A.32})$$

When the dispersion of preferences ν tends to zero, the solution becomes

$$\bar{p}_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) = \frac{\exp(\rho_j \chi_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) + \bar{v}_{jk}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}))^{\frac{1}{\psi_j}}}{\sum_l \exp(\rho_j \chi_{jl}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}) + \bar{v}_{jl}(\theta_t, L_{t-1}, \theta_{jt-1}, \tau_{jt-1}))^{\frac{1}{\psi_j}}}, \quad (\text{A.33})$$

where $\psi_j = \lambda_j (1 - \rho_j)$ and $\rho_j = \left(1 + \frac{6\sigma_j^2}{\pi^2 \lambda_j^2}\right)^{-1/2} \in (0, 1)$.

The implications of letting workers update their beliefs after observing wages depends on the parameters of the AR(1) process and the variance of the idiosyncratic shocks τ_{kt} . If the persistence ρ is small, then any information about θ_{kt} obtained after migration at t has little value for future mobility choices. At the limit where $\rho = 0$, updating beliefs after observing wages or not does not affect the equilibrium behavior. If $\rho > 0$ but σ_τ^2 is large in comparison to σ_ξ^2 , then wages w_{kt} contain little information about θ_{kt} . As shown by the illustrative case of normal beliefs in (A.26), as the variance σ_{jk}^2 of agents' beliefs about θ_{kt} becomes small relative to σ_τ^2 , the beliefs are almost unchanged after observing the wages. For any intermediate value of ρ and σ_τ^2 , however, observing wages will make local beliefs more precise and influence the equilibrium behavior.

Identifying the variance of the shock τ_{kt} , σ_τ^2 , along with the persistence, ρ , and variance, σ_ξ^2 , of the innovation of the AR(1) process, is possible by computing the variance and autocovariances of the adjusted wages,

$$\tilde{w}_{kt} = \log w_{kt} + \alpha \log L_{kt} - \log(1 - \alpha) = A_k + \theta_{kt} + \tau_{kt}.$$

It is easy to show that the parameters are identified by the following expressions,

$$\begin{aligned} \frac{\text{Cov}(\tilde{w}_{kt}, \tilde{w}_{kt-2})}{\text{Cov}(\tilde{w}_{kt}, \tilde{w}_{kt-1})} &= \rho, \\ \frac{1 - \rho^2}{\rho} \text{Cov}(\tilde{w}_{kt}, \tilde{w}_{kt-1}) &= \sigma_\xi^2, \\ \text{Var}(\tilde{w}_{kt}) - \frac{\sigma_\xi^2}{1 - \rho^2} &= \sigma_\tau^2. \end{aligned}$$

A.3.3 Complete Revelation of the Productivity Vector

Now turning to the case that allows workers to exploit the complete structure of the model to infer information about the productivity vector θ_t . As I discuss in Section 2.5, agents do use the mobility rule $\bar{p}_{jk}(\theta_t, L_{t-1})$ to compute the wages in every location. After mobility has occurred, agents observe the population distribution L_t . Recall that the law of motion of population is given by

$$L_{jkt} = L_{jt-1} \bar{p}_{jk}(\theta_t, L_{t-1}),$$

so that upon observing L_{jkt} for all locations j, k , and with the knowledge of L_{t-1} and the mobility rule $\bar{p}$, agents can in principle invert this equation to recover θ_t . Their ability to exactly infer θ_t depends on whether the function $\theta_t \mapsto \bar{p}(\theta_t, L_{t-1})$ associating to each productivity vector, a vector of moving probabilities, conditional on a population distribution, is injective. If it is not, this inversion is complex. For the sake of this argument, let's consider the case in which it is, so that θ_t is completely revealed to all agents at the end of period t .

This implies that the prior beliefs $\pi_j(\theta_t)$, in period t in every location j , are identical and reflect the AR(1) process governing the evolution of θ_t ,

$$\theta_t \sim \mathcal{N}(\rho \theta_{t-1}, \sigma_\xi^2).$$

This makes social learning nullified in this context, since all information is revealed to all agents. Despite having the same prior beliefs, agents however decide to acquire different information depending on their location

and on their idiosyncratic shocks. There is therefore still a meaningful role for information frictions in this context. In particular, agents will acquire less information about distant locations, leading to the heterogeneous migration elasticity results described in Section 4.1.

A.4 Solution Algorithm

We assume that the stochastic productivity component in each region follows an AR(1) process:

$$\theta_{kt} = \rho\theta_{kt-1} + \varepsilon_{kt}, \quad \varepsilon_{kt} \sim \mathcal{N}(0, \sigma_\varepsilon),$$

with all ε_{kt} are iid across time periods and regions. With the current specification, the ergodic distribution of θ_{kt} is a normal distribution with mean 0 and variance $\sigma_\varepsilon/(1 - \rho^2)$. We make the approximation that p_{jk} , π_{jk} and V_j take the following forms:

$$p_{jk}(\theta'^i, L^i) = g_{jk,L}(L^i) \prod_l g_{jk,l}(\theta'_l{}^i)$$

$$\pi_{jk}(\theta'^i) = \prod_l \pi_{jk,l}(\theta'_l{}^i)$$

$$V_j(\theta'^i, L^i) = V_{j,L}(L^i) \prod_l V_{j,l}(\theta'_l{}^i)$$

Moreover, we assume that the l -partial beliefs $\pi_{jk,l}(\theta'_l{}^i)$ are pdfs of normal distribution:

$$\pi_{jk,l}(\theta'_l{}^i) = \frac{1}{\sqrt{2\pi\sigma_{jk,l}^2}} \exp\left(-\frac{(\theta'_l{}^i - \mu_{jk,l})^2}{2\sigma_{jk,l}^2}\right)$$

and define:

$$V_{k,L}(L'^i) = \sum_{m,j} \beta_{k,mj}^{V,L} \log L_{mj}^i, \quad V_{k,l}(\theta'_l{}^i) = \beta_{1k,l}^V \theta'_l{}^i,$$

$$g_{jk,L}(L^i) = \exp\left(\sum_{m,j} \beta_{jk,mj}^{g,L} \log L_{mj}^i\right), \quad g_{jk,l}(\theta'_l{}^i) = \exp\left(\beta_{1jk,l}^g \theta'_l{}^i + \beta_{2jk,l}^g (\theta'_l{}^i)^2\right).$$

The sample of states consists of N_l population vectors (each of length J^2), and N_s productivity vectors (each of length J). The number of possible values for θ_k is set to $n_s = 2$. There are therefore $N = N_s N_l$ states in total, and we denote by i the index of one such state.

1. Guess partial value functions $V_{j,L}^{(0)}$, $V_{j,l}^{(0)}$, a partial choice rule $g_{jk,L}^{(0)}$, $g_{jk,l}^{(0)}$, and partial beliefs $\pi_{jk,l}^{(0)}$. Use the result of last loop if there is one. Compute the next period population that would arise from any pair (θ'^i, L^i) :

$$L'_{jk}{}^i = p_{jk}(\theta'^i, L^i) L_j^i.$$

Define the gross payoff, exponential total payoff, and shifted exponential payoff of action $j \rightarrow k$:

$$u_{jk}(\theta'^i, L^i) = A_k + \theta'_k{}^i - \alpha \log L_k^i + B_k - \kappa_{jk}$$

2. Compute the end-of period homogenized beliefs for people in j that the current state is θ'^i if the end-of-period population is L^i :

$$\bar{\pi}_j(\theta'^i | L^i) = \prod_{l=1}^J \bar{\pi}_{j,l}(\theta'_l{}^i | L^i),$$

and the l -partial homogenized belief takes the form:

$$\bar{\pi}_{j,l}(\theta'_l{}^i | L^i) = \frac{1}{C_{j,l}(L^i)} \prod_{m=1}^J \pi_{mj,l}(\theta'_l{}^i)^{\frac{L_{mj}^i}{L_j^i}}$$

$$= \frac{1}{\sqrt{2\pi\bar{\sigma}_{j,l}(L^i)^2}} \exp \left(-\frac{(\theta_l'^i - \bar{\mu}_{j,l}(L^i))^2}{2\bar{\sigma}_{j,l}(L^i)^2} \right)$$

with

$$\bar{\sigma}_{j,l}(L^i)^2 = \left(\frac{1}{L_j^i} \sum_{m=1}^J \frac{L_{mj}^i}{\sigma_{mj,l}^2} \right)^{-1}, \quad \bar{\mu}_{j,l}(L^i) = \bar{\sigma}_{j,l}(L^i)^2 \frac{1}{L_j^i} \sum_{m=1}^J \frac{L_{mj}^i}{\sigma_{mj,l}^2} \mu_{mj,l}.$$

Then compute the belief for people in j that the *next* period state is θ'^i if the end-of-period population is L^i :

$$E\bar{\pi}_j(\theta'^i|L^i) = \prod_{l=1}^J E\bar{\pi}_{j,l}(\theta_l'^i|L^i),$$

and the l -partial homogenized belief takes the form:

$$\begin{aligned} E\bar{\pi}_{j,l}(\theta_l'^i|L^i) &= \int_{-\infty}^{\infty} \phi_{\varepsilon}(\theta_l'^i - \rho\theta_l) \bar{\pi}_{j,l}(\theta_l|L^i) d\theta_l \\ &= \frac{1}{\sqrt{2\pi\hat{\sigma}_{j,l}(L^i)^2}} \exp \left(-\frac{(\theta_l'^i - \hat{\mu}_{j,l}(L^i))^2}{2\hat{\sigma}_{j,l}(L^i)^2} \right) \end{aligned}$$

with

$$\hat{\sigma}_{j,l}(L^i)^2 = \rho^2 \bar{\sigma}_{j,l}(L^i)^2 + \sigma_{\varepsilon}^2, \quad \hat{\mu}_{j,l}(L^i) = \rho \bar{\mu}_{j,l}(L^i)$$

3. Solve for the unconditional moving probability $q_{jk}(L^i)$:

$$\begin{aligned} q_{jk}(L^i) &= \mathbb{E}_{\theta'} [p_{jk}(\theta', L^i) E\bar{\pi}_j(\theta'|L^i)] \\ &= g_{jk,L}(L^i) \prod_{l=1}^J \int_{-\infty}^{\infty} g_{jk,l}(\theta_l) E\bar{\pi}_{j,l}(\theta_l|L^i) d\theta_l \\ &= C_{qj}(L^i) g_{jk,L}(L^i) \exp \left(\sum_l \frac{\beta_{1jk,l}^g (\hat{\mu}_{jl}(L^i) + \frac{1}{2}\hat{\sigma}_{jl}(L^i)^2 \beta_{1jk,l}^g) + \beta_{2jk,l}^g \hat{\mu}_{jl}(L^i)^2}{1 - 2\hat{\sigma}_{j,l}(L^i)^2 \beta_{2jk,l}^g} \right) \end{aligned}$$

where $C_{qj}(L^i)$ is a normalizing constant to ensure that $\sum_k q_{jk}(L^i) = 1$.

4. Update the value function by iterating on the Bellman equation. First, compute the expected value in the next period if the current productivity and previous population is (θ'^i, L^i) :

$$\begin{aligned} EV_k(\theta'^i, L^i) &= \mathbb{E}_{\theta''} [V_k(\theta'', L^i) | \theta'^i] \\ &= V_{k,L}(L^i) + \sum_l \beta_{1k,l}^V \int_{-\infty}^{\infty} \theta_l \phi_{\varepsilon}(\theta_l - \rho\theta_l'^i) d\theta_l \\ &= V_{k,L}(L^i) + \rho \sum_l \beta_{1k,l}^V \theta_l'^i \end{aligned}$$

Define the convenient transformed payoff $z_{jk}(\theta'^i, L^i)$:

$$z_{jk}(\theta'^i, L^i) = \exp \left(u_{jk}(\theta'^i, L^i) + \delta EV_k(\theta'^i, L^i) \right)^{1/\lambda_j}$$

Then use the Bellman equation:

$$V_j(\theta'^i, L^i) = \bar{\gamma} + \lambda_j \log \left(\sum_k q_{jk}(L^i) z_{jk}(\theta'^i, L^i) \right).$$

To recover the partial values, run the following regression for each k :

$$V_k(\theta'^i, L^i) = \sum_l \left(\beta_{1k,l}^V \theta_l'^i \right) + \sum_{m,j} \beta_{k,mj}^{V,L} \log L_{mj}^i + \varepsilon_k^i,$$

and update:

$$V_{k,L}(L'^i) = \sum_{m,j} \beta_{k,mj}^{V,L} \log L_{mj}^i, \quad V_{k,l}(\theta_l'^i) = \beta_{1k,l}^V \theta_l'^i.$$

Return to the definition of $EV_k(\theta'^i, L^i)$ with these updated $V_{k,L}$ and $V_{k,l}$ and keep looping until $V_k(\theta'^i, L^i)$ has converged.

5. Update the decision rule $p_{jk}(\theta'^i, L^i)$:

$$p_{jk}(\theta'^i, L^i) = \frac{q_{jk}(L^i) z_{jk}(\theta'^i, L^i)}{\sum_l q_{jl}(L^i) z_{jl}(\theta'^i, L^i)}$$

To recover the partial choice rule, run the following regression for each jk :

$$\log p_{jk}(\theta'^i, L^i) = \sum_l \left(\beta_{1jk,l}^g \theta_l'^i + \beta_{2jk,l}^g (\theta_l'^i)^2 \right) + \sum_{m,j} \beta_{jk,mj}^{g,L} \log L_{mj}^i + \varepsilon_{jk}^i,$$

and update:

$$g_{jk,L}(L^i) = \exp \left(\sum_{m,j} \beta_{jk,mj}^{g,L} \log L_{mj}^i \right), \quad g_{jk,l}(\theta_l'^i) = \exp \left(\beta_{1jk,l}^g \theta_l'^i + \beta_{2jk,l}^g (\theta_l'^i)^2 \right).$$

6. Update the partial beliefs

$$\begin{aligned} \pi_{jk,l}(\theta_l'^i) &= g_{jk,l}(\theta_l'^i) \prod_{i=1}^{N_l} E \pi_{j,l}(\theta_l'^i | L^i)^{\frac{1}{N_l}} \\ &= \frac{1}{\sqrt{2\pi\sigma_{jk,l}'^2}} \exp \left(-\frac{(\theta_l'^i - \mu_{jk,l}')^2}{2\sigma_{jk,l}'^2} \right) \end{aligned}$$

with the updated mean and variances

$$\sigma_{jk,l}'^2 = \left(\frac{1}{N_l} \sum_{i=1}^{N_l} \frac{1}{\hat{\sigma}_{jl}(L^i)^2} - 2\beta_{2jk,l}^g \right)^{-1} \quad \mu_{jk,l}' = \sigma_{jk,l}'^2 \left(\beta_{1jk,l}^g + \frac{1}{N_l} \sum_{i=1}^{N_l} \frac{\hat{\mu}_{j,l}(L^i)}{\hat{\sigma}_{jl}(L^i)^2} \right)$$

7. Compare the updated values of $p_{jk}(\theta'^i, L^i)$ and $\pi_{jk}(\theta'^i)$ to their previous ones, and if they are not very similar, go back to step 1 with the new values of $V_{j,L}$, $V_{j,l}$, partial choice rule $g_{jk,L}$, $g_{jk,l}$, and partial beliefs $\pi_{jk,l}$.

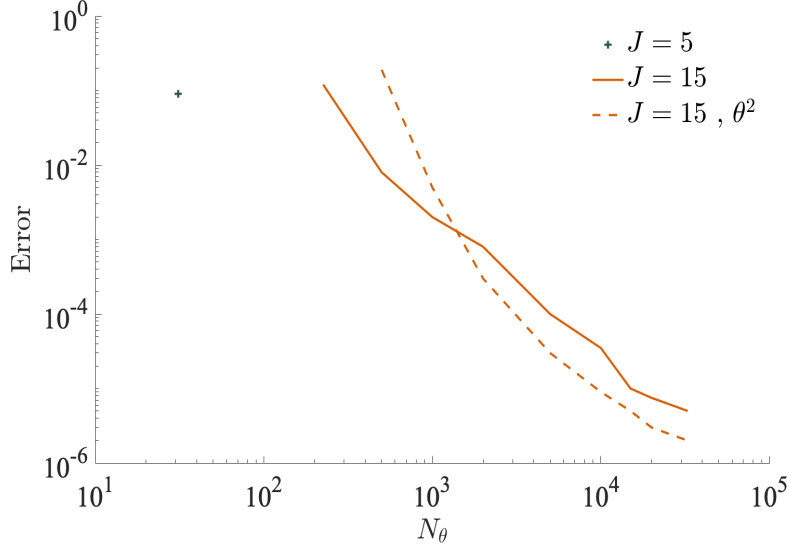
A.5 Accuracy of the Solution Algorithm

A.5.1 Binary productivity process

For few regions and a simple productivity process, it is possible to compute an “exact” solution of the model.

Figure A.3 shows that the simulation error decreases fast with the number N_θ of states (θ, L) drawn for the solution of the model.

Figure A.3: Simulation Error



To compute the exact solution of the model, assume that instead of following an AR(1) process, the productivity is discrete and binary, so that $\theta_{jt} \in \{0, 1\}$, and restrict the number of regions to be at most 15, $J < 15$. Draw $N_L = 1000$ population samples. I then simulate the “almost exact” model x^* , where “almost exact” indicates that the solution is still obtained with a sample of population values. Then compare to the approximation $\hat{x}$, and compute the error as the mean of $|(x^* - \hat{x})/x^*|$. Figure A.3 indicates that the error is lower than 0.1% as soon as $N_\theta = 1000$. The comparison of the solid and dashed line illustrates that allowing for higher order terms in the approximation of the value function have little effect on the overall precision.

A.5.2 Discretized AR(1) Process

In this section, I discuss the consequence of approximating beliefs about the vector of productivity θ_t by independent distributions of θ_{kt} , $k = 1, \dots, J$. I consider the case of normally distributed beliefs

To compare the approximated solution to the exact one, I restrict the analysis to a small number of regions, namely $J = 4$, and assume that preferences are homogeneous across agents, so that $\nu = 0$. I discretize the AR(1) process into $n_\theta = 20$ possible values according to Tauchen’s method. This implies that the total number of possible productivity combinations is $N_\theta = n_\theta^J = 160,000$. I then draw $N_L = 1000$ population samples $L^i = (L_{jk}^i)_{j,k=1,\dots,J}$ for $i = 1, \dots, N_L$, such that $L_{jk} \geq 0$ and $\sum_{j,k} L_{jk}^i = \bar{L}$.

The algorithm for finding the exact solution of the model in this configuration is as follows.

1. Guess an expression $p_{jk}^{(0)}$ for the moving probability between every pair of locations j, k , and for each state $(\theta^{i'}, L^i)$. One simple guess is:

$$p_{jk}^{(0)}(\theta^{i'}, L^i) = \frac{\exp(u_{jk}(\theta^{i'}, L^i))^{1/\lambda_j}}{\sum_l \exp(u_{jl}(\theta^{i'}, L^i))^{1/\lambda_j}},$$

where the flow of utility is determined in general by

$$u_{jk}(\theta^{i'}, L^i) = \theta_k^{i'} - \alpha \log L_k^i + A_k + B_k - \kappa_{jk},$$

and the end-of-period population is given by $L_k^i = \sum_j L_{jk}^i p_{jk}(\theta^{i'}, L^i)$. For the first guess, I set $L^i = L^i$, since no expression of p_{jk} is available yet.

2. Guess an expression $\pi_j^{(0)}$ for the prior beliefs in each region about the productivity vector, conditional on the population vector. I set the belief about $\theta^{i'}$ to be the product of the marginal probabilities

$$\pi_j^{(0)}(\theta^{i'} | L^i) = \prod_k P^*(\theta_k^{i'}),$$

where P^* is the steady state distribution of the discretized AR(1) process, satisfying $P^*\Gamma = P^*$, with $\Gamma(\cdot)$ the transition matrix of the Markov Chain representing the discretized AR(1) process.

3. Using the current guesses of moving probabilities $p_{jk}^{(n)}$ and beliefs $\pi_j^{(n)}$, construct the ex-ante moving probabilities

$$q_{jk}^{(n)}(L^i) = \sum_{i'} p_{jk}^{(n)}(\theta^{i'}, L^i) \sum_{i''} \pi_j^{(n)}(\theta^{i''} | L^i) \Gamma(\theta^{i'} | \theta^{i''})$$

4. Solve for the value function $V_j^{(n)}(\theta^i, L^i)$ as a fixed point of the Bellman equation

$$V_j^{(n)}(\theta^{i'}, L^i) = \lambda_j \log \sum_k q_{jk}^{(n)}(L^i) \exp \left(u_{jk}(\theta^{i'}, L^i) + \delta \bar{V}_k(\theta^{i'}, L^i) \right)^{1/\lambda_j}$$

where the expected value is

$$\bar{V}_k(\theta^{i'}, L^i) = \sum_{i''} V_k(\theta^{i''}, \tilde{L}^i) \Gamma(\theta^{i'} | \theta^{i''}),$$

where $\tilde{L}^i$ is the closest to L^i within the sample of population vectors.

5. Update the moving probability

$$p_{jk}^{(n+1)}(\theta^{i'}, L^i) = \frac{q_{jk}^{(n)}(L^i) \exp(u_{jk}(\theta^{i'}, L^i) + \delta \bar{V}_k^{(n)}(\theta^{i'}, L^i))^{1/\lambda_j}}{\sum_l q_{jl}^{(n)}(L^i) \exp(u_{jl}(\theta^{i'}, L^i) + \delta \bar{V}_l^{(n)}(\theta^{i'}, L^i))^{1/\lambda_j}}.$$

6. Update posterior beliefs

$$\pi_{jk}^{(n+1)}(\theta^{i'} | L^i) = \frac{p_{jk}^{(n+1)}(\theta^{i'}, L^i) \sum_{i''} \pi_j^{(n)}(\theta^{i''} | L^i) \Gamma(\theta^{i'} | \theta^{i''})}{q_{jk}^{(n)}(L^i)}$$

7. Update shared beliefs

$$\log \bar{\pi}_k^{(n+1)}(\theta^{i'} | L^i) = C_k(L^i) + \sum_j L_{jk}^i \log \pi_{jk}^{(n+1)}(\theta^{i'} | L^i).$$

B Appendix to Section 3.1

B.1 Construction of the Main Sample

For every formal job and year, I exploit information on the day of accession into the job and the day of separation (if either of them took place during the corresponding year), the average monthly wage, the number of hours stipulated in the contract, the 2-digit occupation (according to the *Classificação Brasileira de Ocupações*, CBO), and certain characteristics of both the plant at which the worker is employed and of the worker herself. Specifically, I use information on the micro and mesoregion in which the plant is located and its main 2-digit industry of production (according to the *Classificação Nacional de Atividades Econômicas*, CNAE), as well as information on the workers' gender, age, and level of education.⁴⁰

Because RAIS only contains information on the formal employment of workers in Brazil, I have no information on the location of workers that do not hold a formal job in a given year. These workers may be employed in the informal sector, self-employed, unemployed, or out of the labor force. Given that my results will naturally capture only the incidence of informational frictions for migration decisions of workers employed in the formal sector, I limit the analysis to workers that have a sufficiently close labor relationship with the formal sector; specifically, I limit the sample to workers appearing in our sample for at least 5 years between 2000 and 2014.

It is not infrequent that workers in the sample will appear as performing multiple different jobs in the same year. In order to obtain a dataset in which each unit of observation corresponds to a worker and a year, I assign to each worker-year specific pair the location, sector and occupation corresponding to the job that the worker hold for the longest period of time during the corresponding year. However, to determine the total labor income of a worker in a year, I add the labor income earned by the worker in every job in which, according to the data, this worker has been employed in the corresponding year.⁴¹

B.2 Additional Descriptive Statistics on Migration Patterns

In this section, I describe additional statistics about the sample. As illustrated by Figure B.1, the overall migration rate increase by about 1.5-2 percentage points between 2000 and 2014. I then describe which demographic groups are the most mobile. Figure B.2 illustrates a steep decrease of the migration rate with age. Figure B.3 shows that females in the sample are about half as mobile as males. Figure B.5 illustrates that the migration rate is increasing for highly educated, decreasing for low-educated.

Table B.1: Mean and Quantiles of Main Variables

Year	Variable	Mean	Std. Dev.	Min	Max	p10	p50	p90
2000	Age	37.22	8.91	25.00	99.00	27.00	36.00	50.00
	Schooling	5.78	2.17	-1.00	11.00	3.00	6.00	9.00
	Total Income	1,902	12,790	0.00	871,313	342	1,431	4,746
2014	Age	40.43	10.20	25.00	113.00	28.00	39.00	55.00
	Schooling	6.28	1.86	-1.00	11.00	4.00	7.00	9.00
	Total Income	22,976	64,657	0.00	6,068,441	746	5,443	51,529
Average	Age	39.40	9.92	25.00	113.00	28.00	38.00	54.00
	Schooling	6.12	1.98	-1.00	11.00	3.00	7.00	9.00
	Total Income	18,822	54,226	0.00	6,068,441	425	3,681	45,696

⁴⁰Brazilian microregions are groups of municipalities that span the entirety of the Brazilian territory. During our sample period, there were 557 microregions which may themselves grouped into 136 mesoregions.

⁴¹To compute the total labor income of a worker associated to each job this worker has held, I transform the average monthly earnings reported for each job into a measure of average daily wages, and multiply this one by the total number of days between the day of accession and the day of separation into the job reported in the data. If no information on the day of accession or separation is reported, I assume that these ones are January 1 and December 31, respectively.

Figure B.1: Migration rates across mesoregions by Year over 2000-2014

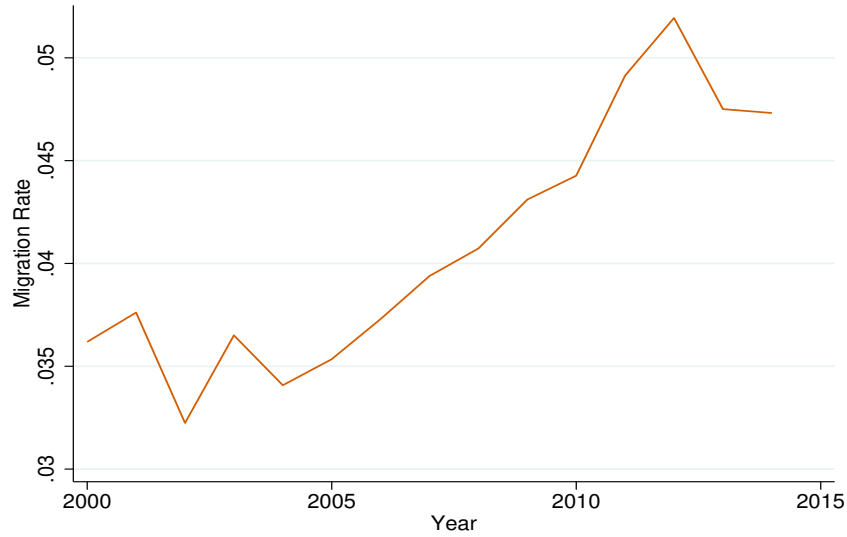


Figure B.2: Migration rates across mesoregions by Year, by age, over 2000-2014

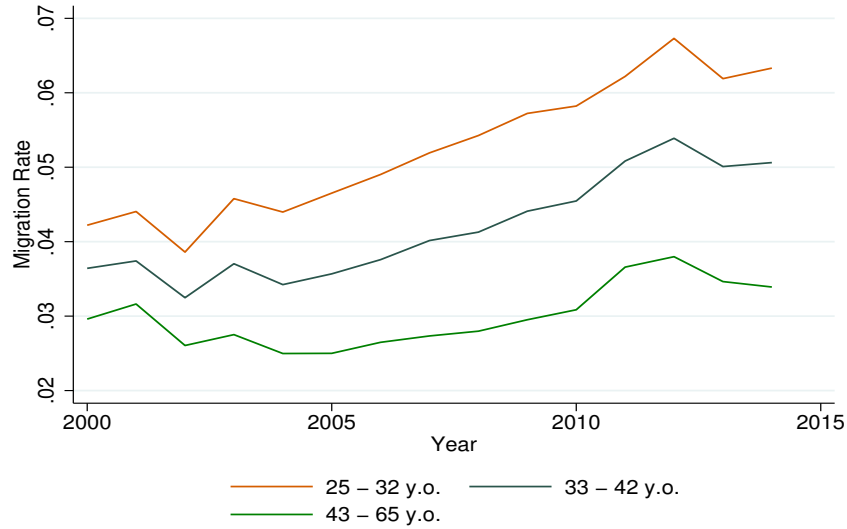


Table B.1 provides descriptive information on the set of workers who appear at least once in the RAIS data between January 2000 and December 2014. The average worker in our sample (weighted by months in sample) is 39.4 years old and has annualized earnings of R\$ 18,822 (in 2014 R\$), before taxes. Over the sample period, the average annual rate of migration across mesoregions is 7.3%. For males, the average yearly migration rate is 8.1%, whereas it is 4.9% for females. Older workers have a smaller propensity to migrate, with a migration rate of 4.2% for workers between 45 and 64 years old, 7.4% for workers between 25 and 44. Workers with a high school degree or above have an average migration rate of 8.1%, while it is 5.1% for workers below high school completion.

Figure B.3: Migration rates across mesoregions by Year, by gender, over 2000-2014



Figure B.4: Migration rates across mesoregions by Year, by race, over 2000-2014

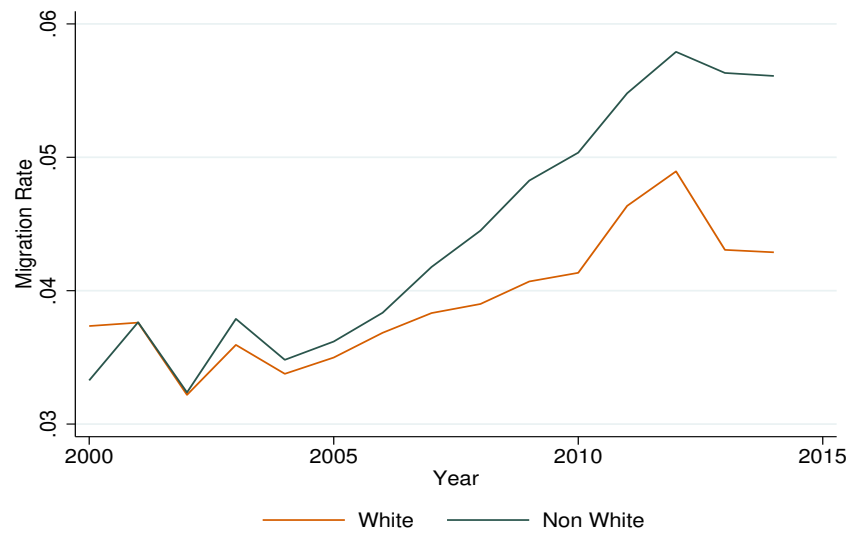
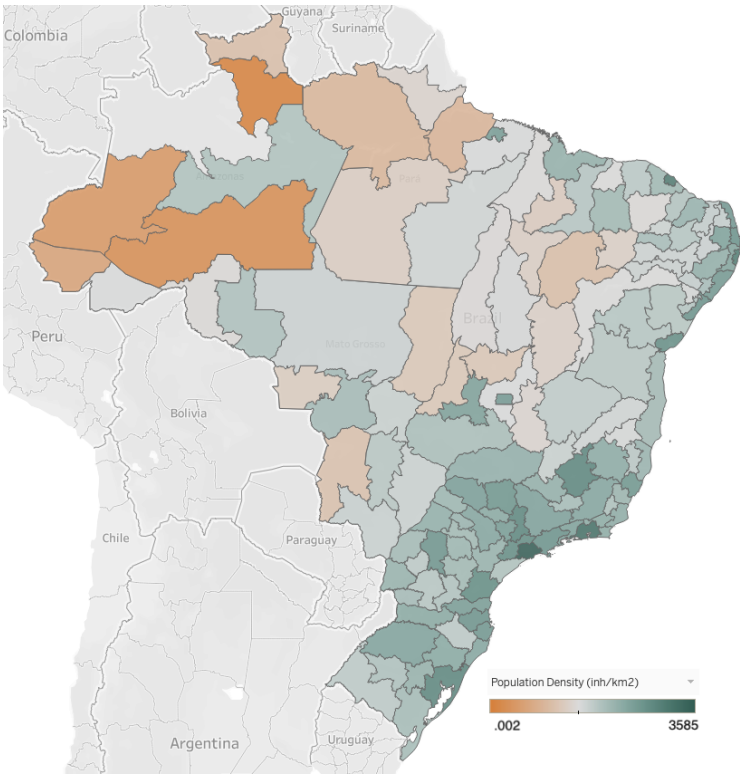


Figure B.5: Migration rates across mesoregions by Year, by educational attainment, over 2000-2014



Figure B.6: Population Density by Mesoregion - Average over 2000-2014



C Appendix to Section 3.2

C.1 Algorithm for Simulated Method of Moments

- Parameters to estimate:
 - Information cost (λ_l, λ_h) for low/high internet regions respectively
 - Role of distance (ϕ_0, ϕ_1) , so that $\kappa_{jk} = \phi_0 \mathbb{1}_{j \neq k} + \phi_1 \text{dist}_{jk}$
 - Decreasing returns in labor in production α
 - Persistence of regional stochastic productivity γ
 - Regional constant productivities $\{A_k\}_{k=1, \dots, J}$
 - Regional constant amenities $\{B_k\}_{k=1, \dots, J}$
- The discount rate is set to $\delta = 0.96$.
- From the wage equation, we can estimate $\{A_k\}_{k=1, \dots, J}$, α and γ :

$$\log w_{kt} = A_k + \theta_{kt} - \alpha \log L_{kt}.$$

From the OLS regression, treating θ_{kt} as the error term, we get estimates of A_k and α . We can then estimate the persistence parameter γ by fitting a 2-state Markov chain on the sequence of residuals $\hat{\theta}_{kt} = \mathbb{1}\{\theta_{kt} > 0\}$ obtained from the regression. Note that θ_{kt} is correlated with L_{kt} so I will need an instrument to estimate α .

- To estimate the remaining parameters, namely amenities, information cost and the role of distance, we use a simulated method of moments (SMM), based on the gravity equation for migration flows predicted by the model:

$$\Lambda_{jklt} = Q_{jklt} + \frac{1}{\lambda_j} \Delta w_{jkt} - \frac{1}{\lambda_j} K_{jkl} + \frac{1}{\lambda_j} (B_k - B_j) + \frac{\delta}{\lambda_j} \chi_{jkt},$$

where

$$\Lambda_{jklt} = \log \left(\frac{p_{jkt} p_{klt+1}^\delta}{p_{jjt} p_{jlt+1}^\delta} \right), \quad Q_{jklt} = \log \left(\frac{q_{jkt} q_{klt+1}^\delta}{q_{jjt} q_{jlt+1}^\delta} \right)$$

$$\Delta w_{jkt} = w_{kt} - w_{jt},$$

$$K_{jkl} = \kappa_{jk} + \delta (\kappa_{kl} - \kappa_{jl})$$

$$\chi_{jkt} = EV_{kt,t+1} - V_{kt+1} - (EV_{jt,t+1} - V_{jt+1})$$

$$EV_{kt,t+1} = \mathbb{E}[V_k(\theta_{t+1}, L_t) | \theta_t, L_{t-1}]$$

- In the data, I observe Λ_{jklt} , Δw_{jkt} and dist_{jk} for all regions and years, as well as the population distribution $\{L_{jkt}\}_{j,k=1, \dots, J}$. From the wage regression, I also know the current estimated productivity state θ_{kt} in every region-year.
- I employ the following iterative procedure:
 - Guess a set of parameters $(\lambda_l^0, \lambda_h^0)$, (ϕ_0^0, ϕ_1^0) and $\{B_k^0\}_{k=1, \dots, J}$.
 - Simulate the model using these parameters and the previously estimated parameters.
 - For any triplet-year $jklt$, compute the simulated Q_{jklt}^0 , K_{jkl}^0 and χ_{jkt}^0 , and run the regression:

$$\Lambda_{jklt} - Q_{jklt}^0 = \beta_1 (\Delta w_{jkt} + \delta \chi_{jkt}^0) + \beta_2 K_{jkl}^0 + D_k - D_j$$

- Update $\lambda^1 = \beta_1^{-1}$, $\phi^1 = \beta_2 / \beta_1$, $B_k^1 = D_k$, and return to step 1 until convergence.

- Joint estimation:

- For any $(\lambda_l^0, \lambda_h^0)$, (ϕ_0^0, ϕ_1^0) and $\{B_k^0\}_{k=1, \dots, J}$, define

$$\varepsilon_{jkl t} = \Lambda_{jkl t} - \left(Q_{jkl t} + \frac{1}{\lambda_j} \Delta w_{jkl t} - \frac{1}{\lambda_j} K_{jkl} + \frac{1}{\lambda_j} (B_k - B_j) + \frac{\delta}{\lambda_j} \chi_{jkl t} \right).$$

where $\Lambda_{jkl t}$ and $\Delta w_{jkl t}$ are the observed migration flows and wage gaps, and the remaining terms are simulated using the parameters and the observed states (θ_t, L_{t-1}) .

- The model predicts the following moment conditions must hold:

$$\mathbb{E}[\varepsilon_{jkl t}] = 0,$$

$$\mathbb{E}[\varepsilon_{jkl t} Q_{jkl t}] = 0,$$

$$\mathbb{E}[\varepsilon_{jkl t} \Delta w_{jkl t}] = 0,$$

$$\mathbb{E}[\varepsilon_{jkl t} K_{jkl}] = 0,$$

$$\mathbb{E}[\varepsilon_{jkl t} (B_k - B_j)] = 0,$$

$$\mathbb{E}[\varepsilon_{jkl t} \chi_{jkl t}] = 0,$$

Using these moment conditions, I then look for the set of parameters (λ_l, λ_h) , (ϕ_0, ϕ_1) and $\{B_k\}_{k=1, \dots, J}$ that minimize an objective function constructed from these moment conditions.

- Data collection:

- For the 100 most populated microregions:
 - * Average Wages across all individuals (for now), each year
 - * Number of workers per municipality and year
 - * Distance between all regions
 - * Bilateral annual migration flows between each microregion

C.2 Construction of the Instrument for Internet Access

To instrument for the share of residents in a mesoregion with an active internet connection, I rely on the expansion of the internet infrastructure over the time period of the sample. In particular, as documented by Tian (2021), Brazil witnessed a rapid expansion of its internet network during the period between 2008 and 2014. Figure C.1 illustrates the change in the fraction of residents with an internet connection between 2000-2007 and 2008-2014 in each mesoregion. To the map is superimposed the grid of internet infrastructure that was implemented during the 2008-2014 period. These additional elements are backbones of the internet network. These backbone cables are essential parts of the internet infrastructure. From these main cables, smaller ones fan out and provide broadband connections. As discussed by Tian (2021), the strength of the signal decays with distance to the backbone cable, so that providing a connection of satisfactory quality is typically difficult beyond 250 km from the cable.

Motivated by the technical role played by these cables in providing access to internet, I construct, for each mesoregion, a dummy variable equal to 1 if it is on the path of a backbone cable that was added during the period. Since mesoregions can be quite large, it is typically verified that a mesoregion with no backbone cable passing through it will have most of its population centers located at more than 250 km from the backbone cable.

Denoting by Δz_{jt} the indicator variable equal to 1 if mesoregion j is on the path of a backbone cable that was added during the 2008-2014 period, I run the following first stage regression:

$$\Delta int_{jt} = \delta \Delta z_{jt} + \nu_{jt}.$$

As discussed by Akerman et al. (2022), the exclusion restriction at the basis of the identification is that timing of the roll-out of the backbone cables is unrelated to changes in confounding factors that occurred over the same period.

The identifying assumption is that mesoregions close to and farther away from new broadband backbones were on parallel trends in the outcome of interest prior to the completion of the new backbones, and did not experience systematically different idiosyncratic shocks after the new backbones arrived.

As first argued by Tian (2021), there are two main reasons why this assumption is plausible in the context of the internet expansion in Brazil over this period. First, alignment of the backbones was announced at the beginning of the 2008-2014 period and followed other infrastructures that had existed long before 2008, making it harder for policymakers to align the broadband cables in anticipation of economic changes in certain areas. Second, the order in which municipalities are connected is approximately geographically determined, according to their distances to the submarine cable landing points along the coast. It is thus a priori unlikely that the availability of the new backbones across different municipalities correlates with the temporal variation in the extents of firms' division of labor of areas on and off the new backbone cables in Brazil.

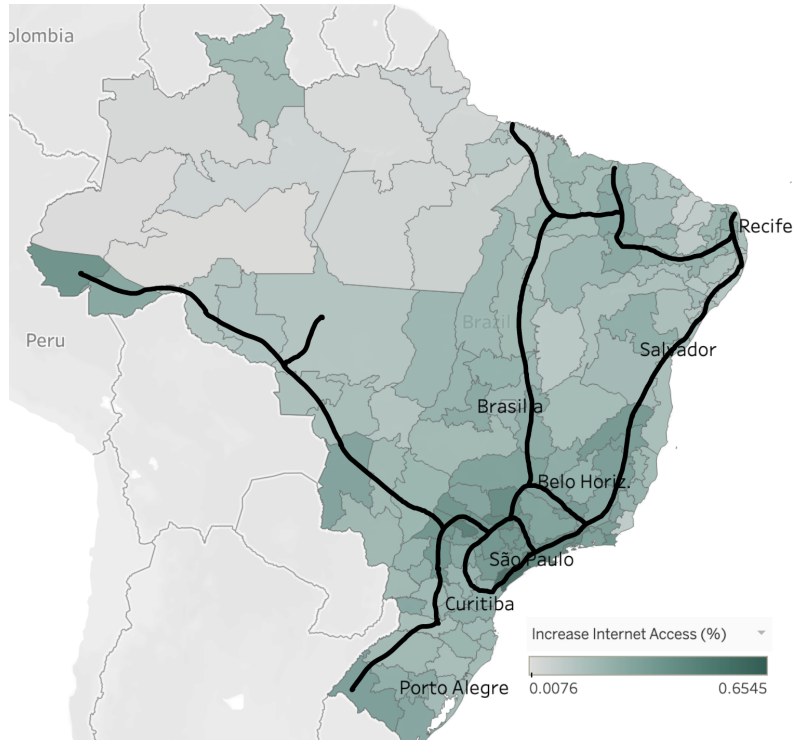
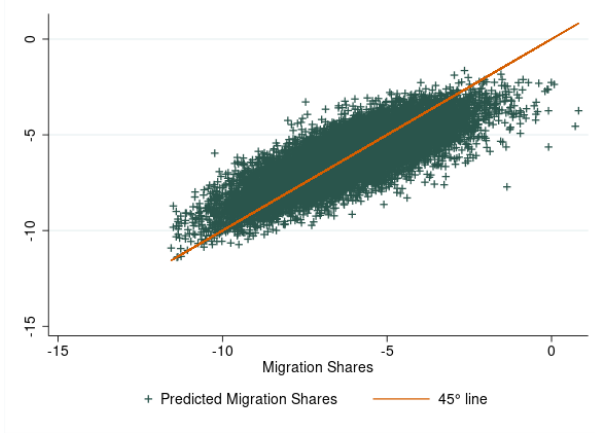


Figure C.1: Change in internet access from 2000-2007 to 2008-2014 and new backbone cables

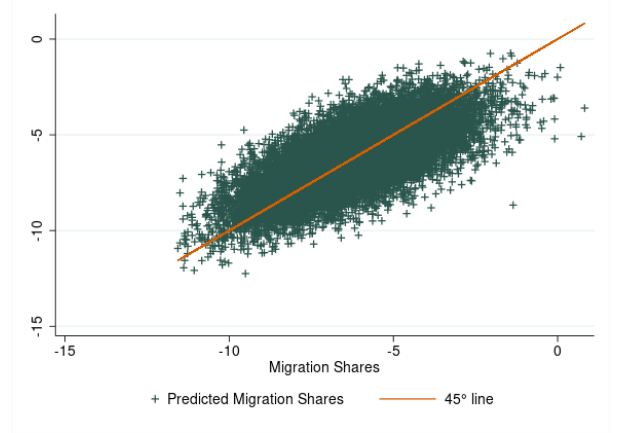
D Appendix to Section 4

D.1 Fit of Migration Flows

In Figure D.1a and Figure D.1b, I report the scatter plot of the log of observed migration shares between every pair of mesoregions for all years, against their predicted values according to the model with information frictions, and without, respectively. When information frictions are allowed, the R^2 of the regression of predicted on actual migration costs is 0.70, whereas it is only 0.61 in the model with no information frictions.



(a) Predicted vs. Actual Migration Flows: with Information Frictions



(b) Predicted vs. Actual Migration Flows: No Information Frictions

D.2 Heterogeneous Migration Elasticities in the Model with No Information Frictions

D.3 List of Local Shocks

In order to empirically identify a mesoregion as having experienced a positive labor market demand shock, I require this mesoregion to verify three criteria. First, there must exist a year such that the average immigration rate in the next four years is at least 50% larger than the average immigration rate in the previous four years; this year is defined as the year of at which the positive labor demand shock took place. Second, the average population in the 4 years preceding the year of the shock is at least 20,000 workers. Third, indexing the year at which the shock took place as t , the total number of immigrants to the shocked region over the period $t - 2$ to $t + 4$ must be at least 20,000 workers. The first criterium uses a discontinuity in immigration rates to identify regions experiencing a positive labor demand shock. The second and third criterium restrict the shocks of interest to those taking place in mesoregions that are sufficiently large and that drove a sufficiently large number of workers into the shocked region.

Once I have identified a mesoregion that experienced a positive labor demand shock according to the criteria outlined in the previous paragraph, I focus on the workers that migrated to the shocked region during a period encompassing the two years prior and the four years subsequent to the shock, and relate the timing of their migration decision to different characteristics of both the migrants themselves and of the mesoregion from which they migrated.

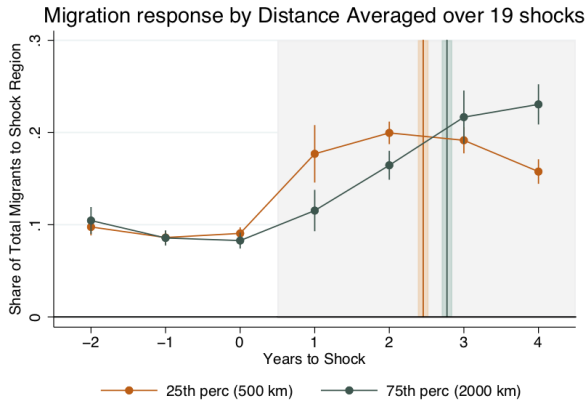
Given that the last year for which I observe migration decisions in the data is 2014 and that I want to study the evolution of the migration flows to a region experiencing a positive labor demand shock in the four years subsequent to the shock, I only search for regions that experienced positive labor demand shocks prior to 2010. This procedure to identify such regions identified a total of ten mesoregions which experience positive local labor demand shocks. As summarized in Table D.1, these are diverse in their location, the year in which they took place, the total number of workers they draw into the shocked regions and the underlying cause of the shock.

Table D.1: Description of Positive Local Labor Demand Shocks

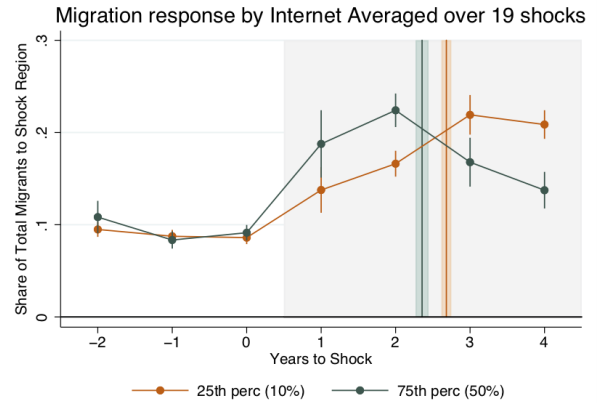
Municipality	State	Year Shock	Number Migrants	Source Labor Shock
Ipojuca	Pernambuco	2009	89,067	Refinery Construction
Natal	R. G. do Norte	2008	74,294	Oil Boom
Santo Antonio	Rondonia	2008	65,746	Dam Construction
Maceio	Alagoas	2008	49,943	Tourism Boom
Belo Monte	Para	2010	40,644	Dam Construction
Uberaba	Minas Gerais	2008	40,301	Sugar Cane Energy
Araucaria	Parana	2006	37,684	Tourism Boom
Cidelandia	Maranhao	2010	37,216	Palm Oil Boom
Suape	Pernambuco	2008	32,752	Refinery Construction
Itabira	Minas Gerais	2010	21,115	Mining Boom

D.4 Replicating Local Shocks in the Model

Figure D.2: Average Delay in the Migration Response to the 10 Local Shocks



(a) Distance: Data



(b) Internet: Data

Figure D.2a and Figure D.2b illustrate the estimates that arise from pooling the data across the ten shock listed in Table D.1. They illustrate, for each year between $t - 2$ and $t + 4$, the predicted migration probability when the corresponding covariate X_{it+s}^k is set to its 25% percentile (labeled as “Low” and painted in orange) or to its 75% percentile (labeled as “High” and painted in green) are set to their mean values. The whiskers attached to each dot represent the 95% confidence interval for each predicted migration probability. The dark thin vertical lines indicate the estimated expected number of years of delay implied by the expected probabilities, and the light-colored thick vertical lines illustrate the corresponding 95% confidence intervals. The standard errors of the estimates of the implied expected probabilities and the expected number of years of delay are computed using the Delta method and standard errors for the estimates of the parameter $\{\gamma_{it+s}^k; k = 1, \dots, K\}$ clustered by year and mesoregion of origin of the migrant (i.e. by the mesoregion in which the migrant was located at period $t + s - 1$).

The results in Figures D.2a and D.2b illustrate that everything else equal, workers living in geographically close mesoregions or in mesoregions in higher broadband internet penetration tend to react faster to positive labor demand shocks happening in mesoregions other than their location of residence. I interpret these estimates as suggestive of the hypotheses that, everything else equal: (a) workers tend to have better information about labor demand shocks taking place in markets that are geographically close to their location of residence; and (b) workers

located in areas with higher internet penetration have better information about every labor demand shocks, no matter where this one took place.

E Additional Figures

E.1 Cardell Distribution

$$g_{\beta}(z) = \frac{1}{\beta} \sum_{n=0}^{\infty} \frac{(-1)^n e^{-nz}}{n! \Gamma(-\beta n)}.$$

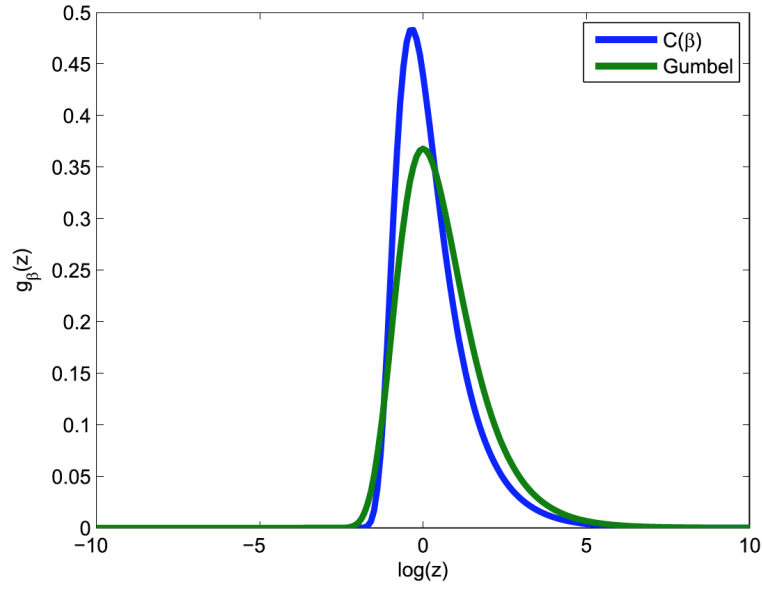


Figure E.1: Distribution Cardell with dispersion $\beta = 0.5$, vs. Gumbel. Source: Dasgupta and Mondria (2018).